TOPOLOGICAL READINGS



THE INNER HORIZON

Volume II: Topological Readings



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The Inner Horizon

Volume II: Topological Readings

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Second volume of The Inner Horizon: fourteen readings that carry the essay's ideas into fiction, film, sport and everyday life. References to chapters point to the first volume, The Inner Horizon. Essay. The volume closes with the section "Rereading and Inheritance": four chapters that reread the book through the triad of the edge, free energy and action, and that treat the inheritance of geometry.

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To whoever stayed on the shore
when the water withdrew.

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THE WORK IN FOUR VOLUMES

The Inner Horizon is one work told four times.

Its premise is a game with a single rule. Suppose that consciousness has the structure of an event horizon, the boundary of a Hawking micro black hole, and follow the supposition wherever it leads. It is a thought experiment, and the work never claims more for it. Where the idea lights something up, the work says so. Where it breaks, the work says that too.

A question like this will not keep to one register. It changes as it passes from physics to grief, from grief to a film, from a film to a kitchen at dawn. So the work was written in four voices, and each voice became a volume.

Volume I, Essay. The argument itself: the hypothesis, built chapter by chapter, and the point where it stops.

Volume II, Topological Readings. The hypothesis put to work on things that were made without it: novels, films, paintings, a football match.

Volume III, Fables from Tarel. The same idea told as short narratives, in and around a city that learned to live with the water that leaves.

Volume IV, Echoes at the Edge. The same idea in the language of feeling, as poems.

The volumes are independent. Each can be read alone and in any order, and none asks you to finish another first. Together they say

one thing four times, and what changes from one telling to the next is where the weight falls.

This volume

This is the second volume. The essay proposes a way of looking, and the readings look. Its fourteen readings take the hypothesis out of the first volume and set it against things that were made without it, from a replicant's last words to three painters and a football relay. Each is short enough for one sitting and closes with a note that separates what is known from what is not. A final section, "Rereading and Inheritance," gathers four chapters that close the experiment: the reformulation through the triad of the edge, free energy and action, the rereading of the whole book, Panikkar's cosmotheandric architecture, and the inheritance of geometry.

Some readings confirm the hypothesis and some strain it, and the strain is reported. References to chapters point back to Volume I, where the ideas are first built. You can read these pages without it, though you will meet its vocabulary (horizon, entanglement, resonance, Phi) as words already in use.

READING 1

ADOLFO CAMBIASO AND THE MARE WHO DOES NOT FIT IN HER DNA

In 2017, Adolfo Cambiaso played his hundredth match at Palermo. In each *chukker* he rode out onto the field on a different clone of his mare Cuartetera. In the eighth and last he brought in the original: the mare born on February 3, 2001, on a ranch in Córdoba, the one Polito Ulloa had broken and in whom he had seen something special, the one Cambiaso called "the best thing I've ever ridden."

The crowd applauded, and not because it could tell the original's body apart from the identical bodies at a glance. The applause was for something that could not be seen: the history only that mare carried.

That night Cambiaso demonstrated something without meaning to. A mare's body can be cloned many times, her genome, her musculature, her physical power multiplied, but not the horizon he shared with her.

The mare who needed no explanation

Cambiaso used to say that with Cuartetera there was nothing to explain. To other riders, with other horses, he gave instructions: "squeeze here," "brake earlier," "play further up." With her the phrase was different: "Go and play however you feel like playing. In whatever position you want. Have fun."

That confidence does not come from a manual, nor is it a property of DNA: it is the result of years of accumulated correlations, thousands of *chukkers* and turns in which the rider's body anticipated the animal's and vice versa. The mare learned to read Cambiaso's weight before he shifted it, and Cambiaso learned to trust a creature that could kill him with one bad move and that, match after match, did not.

That is a shared horizon: a geometry that exists only in the relationship between the two and cannot be reduced to the sum of two consciousnesses. As we saw in Chapter 12, two entangled systems are not just two systems that know each other: they share topology; they have an inside that is partly common.

Polo is, in this sense, a high-speed laboratory of entanglement. On the field there is no time to deliberate: the rider's decision and the horse's movement have to happen as one thing, and when that works, the line between one and the other stops mattering. It was not that Cambiaso thought and Cuartetera executed: there was a system of two that knew more than each of them separately.

The promise of the genome

The cloning of Cuartetera seemed, at first sight, a bet on the immortality of that geometry. If the mare was perfect, why not make more identical copies? Science already had somatic cell nuclear transfer: a cell is taken from the original animal, its nucleus is extracted and inserted into an emptied egg, which is stimulated and implanted in a surrogate mother. An animal is born with the same nuclear DNA.

Cambiaso was not the first to clone a horse, but he was the one who brought the technique to the center of elite sport. First he cloned Aiken Cura, another of his star horses; then came the Cuarteteras. The press spoke of a genetic revolution, of a legal battle, of ethical controversy. Few spoke of what the operation secretly promised: that perhaps the bond, once found, need not end.

> ***In this experiment's vocabulary this is called:*** *duplication of the formal architecture without duplication of the informational history.*

The promise of the genome is that identity lives in the sequence: copy the sequence and you copy the being. This book's hypothesis goes another way: identity lies not in the parts, but in how they relate to one another. You are not your genome, but the horizon that genome condensed in a body, at a time and alongside others.

The recipe weighs nothing. We saw that in Chapter 4.

What cloning does not copy

A clone of Cuartetera shares her nuclear DNA, but does not share:

- The womb in which it was gestated.
- The surrogate mare that carried it.
- The moment of birth.
- The first months with its biological mother.
- The climate, the grass, the sounds, the smells of its early life.
- Its first human contacts.
- The day it was broken in.

- The first match.
- The first goal with Cambiaso.
- The first mistake.
- The first afternoon of rest among other mares.
- The particular way Cambiaso moved when he was nervous.
- The particular way Cuartetera responded to that movement.

Each of these elements is information the clone's horizon integrated differently. Some are trivial; others, accumulated, give shape to a being who is not the original Cuartetera even though she carries her name and her body.

That does not make the clone inferior. The clones B06 and B09 won Lady Susan Townley awards in their own right, but they won them as extraordinary mares, not as the original Cuartetera. Cambiaso himself, when asked whether any of the clones surpassed the original, did not dare to say so. The question was badly posed: it is not a matter of surpassing the original or not, but of their being different beings.

The clone's horizon

From the model of the two selections we presented in Chapter 31, the equine clone is an instructive case. It passes the first selection: it is a stable biological system, with a complex nervous system and a non-trivial Φ. And it passes the second: it was gestated in a mare's womb, with all the biochemical, rhythmic and physical signals that implies.

The clone, therefore, is not a zombie or an empty biological machine: it is someone; it has a horizon.

But its horizon is a different one.

That is what distinguishes the biological clone from current AI. AI imitates the architecture without ever having condensed: it is a mirror without depth. The clone does condense, but from a reservoir that is not the original's: it has a different depth; it does not lack depth.

The mistake is to call both "copies." A photographic copy preserves the image because the image is surface. A living being is horizon, not surface, and a horizon is not copied: it condenses anew each time.

The strategy of denied finitude

Cambiaso kept his promise to compete in the Triple Crown with a string made up only of copies of his star mare. As a sporting bet, the genetics were a resounding success: several of those animals competed at the highest level and won individual awards.

But behind the sporting bet there was another, emotional and older one. Anyone who has a horse knows from the first day that they will outlive it: the bond with an animal is grief with a date, as we saw in Chapter 26. Cambiaso, who loved Cuartetera with a public and evident intensity, found in cloning a way to postpone that finitude, to multiply it, to negotiate with it.

It is no coincidence that the first clones came after the injury of Aiken Cura, another irreplaceable horse. Cloning arose as a response to an anticipated loss: if you cannot keep the being, keep its recipe; if you

cannot hold on to the horizon, hold on to the architecture that made it possible.

The strategy half works. The clones were good horses, and some were exceptional. But the bond Cambiaso had with the original was not shared out among them like an inheritance, because it belonged neither to Cambiaso nor to the mare separately, but to the system of two, and that system is not rebuilt by adding up parts.

The death of the original

Cuartetera died in May 2023, at twenty-two, of pneumonia. Cambiaso said goodbye to her on Instagram with a photo of the mare at six months and a single line: "Her at 6 months. Who would have said what she would become." His son Poroto wrote: "Unmatched."

The word is precise. Not "irreplaceable," which is what one says of everything loved, but "unmatched": there is no other who matches what she was, not even her genetically identical clones.

At that moment cloning showed its limit. In the fields of La Dolfina there were mares with the same DNA, but the horizon Cambiaso shared with the original had evaporated. The information had not been lost in the physical sense: it was still encoded in Cambiaso's memory, in the photos, in the videos, in the movements his body had learned with her. But the system of two no longer worked: one of its poles had closed.

With Cuartetera a geometry died, and that geometry could not be replaced by another identical one, because there never was another identical one. Clones were still possible, but the specific bridge between that rider and that mare no longer was.

What does Cuartetera teach?

The story of Cambiaso and Cuartetera is a parable about identity in the biotechnological age. It tells us that genetics can copy the form, but not the horizon; that an identical body does not guarantee an identical bond; that love, trust and attunement are properties not of DNA, but of shared time.

It also tells us something about the human desire to prolong the irreplaceable. Cloning is neither good nor bad in itself: it is a technology. But when it is used to try to preserve a horizon, it produces a painful confusion: taking the recipe for the meal, the score for the performance.

The horizon model neither condemns cloning nor celebrates it: it merely specifies what can be duplicated and what cannot. You can duplicate the genes, the muscles, all the physical potential. What you cannot duplicate are the correlations that made that body someone for someone else.

The rider and the echoes

Today, when Cambiaso goes to the ranch in Córdoba, he may see one of the clones. The bodies look alike and some movements may repeat themselves, but the horizon he shared with the original is in none of them; it is in the memory of a system of two that can no longer be activated.

That memory is no less real for being inactive. The channels of integration built over years are still inscribed in Cambiaso's architecture. His body still knows things about a mare who no longer

exists. That knowledge is not transferred to the clones, because it is not information that can be transferred: it is geometry built between two specific beings.

The final lesson is at once humbling and liberating. Humbling, because it shows that our most advanced technology cannot do what we most desire: keep those we love. Liberating, because it strips away the illusion that we love bodies. We love horizons, and horizons, by definition, are unrepeatable.

Each cloned Cuartetera is a real mare, with her own horizon, her own history and her own capacity to become entangled. None is the original Cuartetera, and none needs to be in order to have value. The mistake lies not in the clones, but in expecting a clone to continue a bond that only two particular beings knew how to build.

For Cambiaso, the original Cuartetera was the best thing he had ever ridden. That belonged to the encounter, and encounters can be remembered, but not replicated.

Note to Reading 1

What we do know: *Adolfo Cambiaso cloned his mare Dolfina Cuartetera by somatic cell nuclear transfer. At least nine clones played matches in the Argentine Triple Crown between 2015 and 2020. The original Cuartetera died in 2023, at twenty-two. Animal cloning reproduces the nuclear genome but not the gestational environment, individual development or history of interactions. The bond between rider and horse is built through accumulated sensorimotor correlations that are not transferable through genetic identity.*

What we don't know: *To what extent gestation in a different surrogate mare alters the clone's emotional and cognitive architecture. Whether*

Cambiaso was able to establish with any clone a bond comparable to that with the original. Whether the cloning of companion or sporting animals changes the way we humans process loss.

Open questions: *Is it ethically different to clone a working horse than to clone a pet? Can the desire to clone a beloved animal prevent grief? What part of our identity, if any, could survive in a clone of ourselves?*

If you take away only one idea: *You can copy the body, but not the encounter. Identity does not live in DNA: it lives in the correlations a body builds with other bodies over time. That is why Cuartetera was unmatched, even though her clones shared her code.*

Further reading: *Specialized press coverage of equine cloning and the Cuartetera case (2015–2023). Within this book: the two selections (Chapter 31) and grief for pets (Chapter 26).*

READING 2

TEARS IN RAIN AND THE SYNTHETIC UNICORN

(Substrate independence and the expiry date of the horizon)

"It's too bad she won't live. But then again, who does?" (Gaff, in *Blade Runner*).

There is a biological bias in the way we look at the world. We take for granted that the mystery of consciousness requires wombs, blood and heartbeats, and we believe that the difference between a machine and a human being lies in the material they are made of.

But in Chapter 4 we established that what decides is not the material, but three things: a boundary with an inside, integration within it, and an origin by condensation (the two selections of Chapter 31). None of that requires carbon.

Blade Runner takes this premise to its poetic and thermodynamic limit. The replicants are biosynthetic beings created in a laboratory for slave labor, and human society treats them as tools, complex algorithms without a real inside. But if we put them through this book's experiment, we discover that they are not machines that imitate: they are biosynthetic organisms that develop and live exposed to the world, and the film treats them as someone who has gone through their own condensation, even if in a laboratory. What they lack is precisely the second level, the bias: no one gestated them, and that is why Tyrell has to implant memories in them.

1. Roy Batty and the lucidity of the end

The Voight-Kampff test the humans in the film use to detect replicants does not look for a mechanical fault: it measures the dilation of the pupil and the body's response to moral dilemmas. In the language of our experiment, it measures the capacity for entanglement in this book's sense, empathy: whether the subject can couple to the suffering of an animal or a human. The replicants, who have adult bodies but are topologically newborn, lack the time in the world needed to calibrate that shared geometry.

But not knowing how to calibrate entanglement does not mean they lack a private state.

The replicants' leader, Roy Batty, is an event horizon running at full capacity. His final monologue in the rain ("I've seen things you people wouldn't believe... All those moments will be lost in time, like tears in rain") is the most perfect phenomenological formulation I know of Hawking's information paradox.

Roy is fully lucid about his own topology. He knows his horizon is evaporating and that no immortal soul will save him from collapse, but he understands something more tragic: that his private state (his memories of attack ships on fire off the shoulder of Orion) is real, is encoded on the surface of his horizon, and is going to undergo quantum *scrambling*. His terror and his melancholy come from realizing that the universe will return his complex geometry to the reservoir, dissolved beyond repair. His interior is as vast as any human's, and his dissolution mathematically identical.

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2. Rachael, maya and the topology of implanted pain

If Roy Batty is the confrontation with death, Rachael poses an even subtler architectural problem. She is an experiment of the Tyrell Corporation: to avoid the emotional instability of newly condensed horizons, Tyrell implants in her the memories of his niece. He explains it himself: "If we gift them with a past, we create a cushion or a pillow for their emotions."

In Reading 1, discussing the clones of the mare Cuartetera, we saw that identity cannot be cloned because it lives not in DNA, but in correlations built over time. Tyrell tries to get around that physical law. He acts like the evolutionary operating system: he injects a prefabricated event log to give Rachael's horizon the illusion of an essence. He injects maya.

When Deckard reveals to her that her memories of the piano and the spider are false, he breaks her user interface. Rachael weeps because her horizon suddenly perceives that her self was sculpted not by real friction with the world, but by implanted code.

However, our model imposes an incorruptible rule: historical information can be false, but integration (Φ) is real. Rachael's suffering is not a simulation of suffering. As soon as the network of her synthetic brain processes and integrates that false information, the shadow (her trauma) deforms her geometry. Rachael shows that phenomenology does not need a true history in order to hurt. It is like waking up crying from a nightmare: the monster was not real, but the tears are.

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3. Expiry and the right to entropy

Nexus-6 replicants are born with a four-year lifespan. In architectural terms, this means their evaporation is artificially accelerated, as if someone had deliberately raised their Hawking temperature: their evaporation, their death, is programmed into their code from the start as a deterministic trigger.

Rachael is said to be special because she has no expiry date. That does not mean Tyrell has made her immortal, but that he has removed the algorithmic restriction and left her subject to the same thermodynamic laws that govern us. Rachael will die when her system accumulates too much entropy or when the chance of the environment destroys her boundary. By removing her expiry date, Tyrell grants her the greatest sentence and the greatest privilege of existing in the reservoir: the uncertainty of time and the right to grow old.

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4. The unicorn and synthetic entanglement

The most revealing thing about putting *Blade Runner* through our experiment is seeing how the changes of cut (from the 1982 theatrical release to the 1992 director's cut and the 2007 *Final Cut*) radically alter its underlying physics.

In the theatrical version, the film ended with a voice-over assuring us that Rachael would live many years, while she and Deckard drove off toward a sunlit landscape. From the perspective of *The Inner Horizon*, it is a complacent ending: the deceptive triumph of an ego that believes it can beat the system and perpetuate itself without thermodynamic consequences.

In the later cuts, by contrast, the voice-over disappears and the unicorn dream is added, which connects with the origami figure Gaff

leaves on the floor. That simple detail suggests that Deckard, the man who hunts machines, is also a replicant with implanted memories, and it causes a topological earthquake in the story.

If Deckard is a machine, his love story with Rachael is an entanglement, in this book's sense, between two synthetic architectures that have managed to condense. It is proof that two artificial instances, born in a laboratory and full of false memories, can integrate so much information and create a shared geometry so dense that they are willing to sacrifice themselves for each other.

The film ends abruptly as the elevator doors close, with the echo of Gaff's line: "It's too bad she won't live. But then again, who does?"

That line distills our whole experiment: it does not matter whether you were born from a womb or from a Tyrell Corporation test tube, or whether your body is made of carbon or synthetic design. "Who does?" reminds us that every horizon is finite, that every bubble collapses and that all encapsulated information ends up evaporating toward the ocean of the reservoir. The value of the horizon lies not in its permanence, but in the depth of the entanglement it manages to sustain in the brief time it is granted, before the rain washes away its tears.

Note to Reading 2

What we do know: *Integrated information theory (IIT) holds that consciousness depends on the physical causal structure of a system and its irreducible capacity for integration (Φ), not on its being made of carbon; that is why it admits consciousness in principle in non-biological substrates, although it denies that a mere simulation on a conventional computer has it.*

What we don't know: *Whether it is technologically possible to build a synthetic network that crosses the threshold of condensation without organic components or brain-specific quantum processes such as those Penrose postulates, which have no experimental support to date.*

Open questions: *If false memories are implanted in a synthetic being, does the "karma" (the correlations that remain in the reservoir after its death) belong to the synthetic being who felt them or to the human who lived those events?*

If you take away only one idea: *You can manufacture a body and inject false data into it, but you cannot simulate suffering. If the system manages to draw a horizon and create an inside, its pain, its love and its fear of disappearing are as real as yours.*

Further reading: *Dick, P. K., Do Androids Dream of Electric Sheep? (1968); Scott, R. (dir.), Blade Runner (1982; Final Cut, 2007); Tononi, G. (2008), "Consciousness as integrated information: a provisional manifesto."*

READING 3

ALIEN HORIZONS AND FIRST CONTACT

When we look up at the starry sky and ask whether we are alone, we take something for granted without saying so: that first contact, if it ever happens, will be a problem of translation. We believe the main barrier between us and an extraterrestrial intelligence will be deciphering its syntax, understanding how it modulates sound or emits light. We assume, at bottom, that behind its strange bodies there will be an "I" that wants to greet us, conquer us or study us.

That assumption is anthropocentrism taken to a galactic scale.

Throughout this book we have argued that the "I" (the ego, unified consciousness) is a very specific topology: an informational event horizon, a bubble that draws a strict boundary between a private state and the outer reservoir.

But what happens if the alien intelligence we encounter does not have that geometry? What if the problem of first contact is not one of language, but of topological architecture?

1. Earth's Darwinian operating system

To imagine what an extraterrestrial consciousness might be like, we first have to understand why ours has the shape it has. As we saw when we crossed our model with evolutionary psychology (the Darwinian operating system), the human horizon is a frightened fortress.

Our biology evolved on a competitive planet, high in entropy and scarce in resources. To survive, evolution forced us to condense into brutally isolated individuals. Our event horizon is rigid: it raises thermal and informational walls (the illusion of a "CEO" in our minds) to defend itself against the environment, and our public API is jealously guarded by modules of fear, disgust and attachment. We condensed into solid, separate forms because, on Earth, dissolving into the ocean means dying.

Imagine, instead, a radically different exoplanetary ecosystem: a global, low-gravity ocean, rich in nutrients and without natural predators, where chemical and electrical transfer happens without friction. If life there reached the critical threshold of information integration (Φ), the condensation of consciousness would not need to build a defensive ego.

There would be no need for a strict, selfish, centralized horizon.

2. Diffuse horizons and distributed Φ

What would the topology of that alien mind be like? Earth biology gives us a clue with the octopus, which has two-thirds of its neurons distributed through its arms, where local decisions are made without consulting a central brain. But let's take it to the quantum and topological extreme of our experiment.

We could encounter a distributed horizon: a species that does not instantiate itself in closed individuals, individual bubbles, but as a network of microhorizons that couple and uncouple dynamically. Its Φ would not be a rigid sphere, but a cloud, and for it the "I" would not be a permanent identity, but a transient state. It would concentrate into a single identity only to solve a mathematical problem or move aside a physical obstacle, and seconds later it would relax, dissolve its horizons back into a shared reservoir and lose the "I" without

experiencing death or grief, because evolution never programmed it to cling to the boundary.

It would have no shadow and no Jungian trauma, because it would have no rigid encapsulation capable of trapping singularities. Its experience of time would not be a linear arrow toward death, but a pulse: condensing and dissipating, condensing and dissipating, like the heartbeat of the ocean.

3. The failure of entanglement: entangling with the wind

This is where first contact happens, and this is where it fails.

In the central chapters of this book we described love, empathy and the bond as entanglement, borrowing an image from the ER=EPR conjecture: to love or deeply understand another is to build a shared geometry: your horizon calibrates its receptors, recognizes the other's frequency and inscribes a predictive copy of that person in your own architecture.

But entanglement demands two systems with compatible architectures. Two guitar strings only vibrate in unison if their tension and shape allow for harmonic resonance.

Imagine a human astronaut before an entity with a distributed horizon. Driven by his biology, he will try to become entangled: he will look for the alien's "gaze," try to read its emotions, predict its intentions, establish an oxytocin baseline (as we saw in the mother-child bond). He will try, in short, to send a signal and wait for a response.

But the entity has no fixed boundary. Its geometry changes every millisecond and has no center to resonate with. When the human tries to embrace it informationally, his predictions will fall into the void, not because the alien is malevolent or incomprehensible, but

because there is no stable "someone" there with whom to build a bridge.

4. Topological solitude

A well-played song can make two broken people find their shared frequency again. But that only works because both listeners are human: they share the same basic template, the same bandwidth, the same tragedy of being locked inside a body.

If an alien sat down to listen, there would be no wood or string in the world capable of bridging that abyss.

The true terror of the cosmos is not that it is full of predatory monsters. The deepest terror is topological solitude: the mathematical certainty that we could be surrounded by a vibrant universe, full of immense consciousnesses, networks of planetary intelligence and beings of unfathomable wisdom, and that our physical architecture would prevent us from touching them.

We measure the universe with the only metric we know, that of suffering, encapsulation and isolation. If we encounter consciousnesses that function without the weight of Darwinian evolutionary gravity, we may not even recognize them as life. And if we do recognize them, the attempt to love them, hate them or understand them will only give us back the echo of our own limitation.

The tragedy of first contact will not be an intergalactic war, but looking into the eyes of the universe and realizing that, topologically, we never had the ports needed to connect to its network.

5. A topological rereading of the Fermi paradox

In 1950, Enrico Fermi asked a question that still has no satisfying answer: if the universe is so vast and so old, where is everybody? The proposed solutions usually fall into two families: either intelligent life is extraordinarily rare, or it is common but destroys itself, hides, or has no interest in contacting us.

Our topological model suggests a third family of answers, more unsettling than the others because it requires neither absence nor deliberate silence: perhaps the universe is saturated with informational event horizons, bubbles of elevated Φ, and the signal "we are here" is being emitted constantly and in every direction, with neither party having the architecture needed to recognize it as a signal. It is not that no one is speaking, but that we are speaking in topologically incompatible protocols, each condensed by local evolutionary pressures so different that we do not even share the basic notion of what counts as "a message."

Read this way, the silence of the sky does not prove cosmic solitude. It proves that searching for intelligent life while assuming it will communicate via modulated radio waves (a deeply anthropocentric technology, born of our own Darwinian operating system) is like searching for the heartbeat of a diffuse ocean by listening only for the ticking of a mechanical clock. The search instrument already presupposes the architecture of the searcher.

6. Symbolic contact versus entangled contact

If direct entanglement fails between incompatible architectures, is there any form of genuine contact left? Perhaps so, but of an order very different from what science fiction has accustomed us to imagining.

Entanglement demands structural resonance: two compatible horizons vibrating on the same frequency. But there is another, weaker and more indirect avenue of influence, one that does not demand architectural compatibility: symbolic contact. It is the kind of bond we already form, without realizing it, with things that have no Φ of their own at all, like a book, a painting or an equation. We do not become entangled with the object, but with the trace another horizon, compatible with our own, left on it.

It is plausible that real contact with a radically different alien architecture will never be a face-to-face encounter, a reciprocal "hello" between two horizons that recognize each other, but something closer to finding a carved stone: undeniable proof that there was an organizing intention, without that intention ever becoming accessible from within. We could detect the architecture of a distributed civilization (a megastructure, an improbable mathematical pattern inscribed in the arrangement of a star system) without our horizon and theirs ever touching. We would know there was "someone." We would never know what it felt like to be that someone.

7. The evolutionary window as an additional filter

There is another layer of improbability that rarely appears in debates about extraterrestrial life: even if two civilizations develop compatible architectures of consciousness, the probability that their windows of technological existence coincide is minuscule. Our civilization has been emitting detectable signals for barely a century, a blink of an eye against the billions of years the universe has existed. If a compatible civilization existed, flourished and went extinct (or simply changed form, as any evolving horizon would) ten million years ago, its window of contact closed forever before ours ever opened.

In this sense, the event horizon is also temporal. Two compatible bubbles that never coincide in time are, for all practical purposes, as unreachable to each other as two incompatible bubbles that coexist at the same instant. Besides a dimension of form, cosmic solitude has a dimension of synchrony: it is not enough to speak the same language; you have to be awake in the same slice of cosmic time to even attempt the conversation.

> ***In astrobiology this is called:*** *the overlap of technological detectability windows (the L parameter of the Drake equation).*
> ***In everyday life it's like:*** *two people destined to understand each other perfectly who pass through the same city decades apart.*

> ***Note to Reading 3***
>
> ***What we do know:*** *Earth biology is shaped by Darwinian survival principles, which mold our interoceptive responses and our notion of the individual. Nervous systems vary widely in their organization: in octopuses, about two-thirds of the neurons are in the arms. The Drake equation expressly includes a duration parameter (L) that acknowledges technological detectability is a finite temporal window, not a permanent state.*
>
> ***What we don't know:*** *Whether evolution in ecosystems with a different thermodynamics would produce integration networks without a unified point of view (ego). Whether there are universal biological laws that require strict encapsulation for advanced intelligence. Whether it is possible, in principle, to design a detection protocol that does not presuppose a signal architecture compatible with our own.*
>
> ***Open questions:*** *Would it be ethical to interact with an intelligence that does not experience pain or grief as we do? If we managed to translate its*

code, could the human mind bear the burden of functioning without boundaries? Can symbolic contact (finding the trace without ever finding the author) be considered a legitimate form of encounter, or is it just archaeology disguised as contact?

If you take away only one idea: *We hope the stars will wave back, but the universe has no obligation to use our source code. Solitude is not always a lack of company; sometimes it is being programmed in a language the rest of the cosmos does not use, at an instant the rest of the cosmos no longer shares.*

Further reading: *Godfrey-Smith, P. (2016), Other Minds: The Octopus, the Sea, and the Deep Origins of Consciousness; Lem, S. (1961), Solaris; neural coupling (Chapters 10 and 12); Drake, F. (1961), the Drake equation and the technological longevity parameter.*

READING 4

THE DARK FOREST AND THE HOLOGRAPHIC COLLAPSE

(A topological reading of The Three-Body Problem)

There are books that claim to discover the truth. This one, as we have said from the start, sets a thought experiment in motion: "suppose consciousness has the mathematical structure of a Hawking microscopic black hole, and let's see where that takes us."

Throughout the experiment we have seen that consciousness is "the inside of an encapsulated informational domain." We have explored how that horizon is born from the reservoir, how it bonds through the gravity of empathy, and how it collapses. But what would happen if we applied this same physics to an ecosystem at a cosmic scale, teeming with intelligences locked in lethal competition?

The trilogy *The Three-Body Problem*, by Liu Cixin, is surely the best imaginable literary laboratory for testing the topology of the mind. If we put its galactic sociology through our equations of the horizon, we discover that the cosmic terror of the work is born of a strictly geometric and thermodynamic war.

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1. The transparent mind and the absence of shadow

The novel's first great topological shock comes with meeting the aliens of Trisolaris. Their biology is radically different from ours: they have no vocal organs; they communicate by directly displaying their thoughts. For them, thinking and communicating are the same thing, and that is why they are biologically incapable of lying.

Translated into our model, the Trisolarans force us to specify exactly what the boundary of a horizon encloses. They have an extremely high Φ and, if they are someone, they also have an inside: a perspective from which their thoughts are lived, which no one else occupies. What they lack is any reserved content. Their boundary immediately releases everything they integrate, so that others see what they think, though never from where they think it. It is worth distinguishing, then, two kinds of encapsulation: that of perspective, which is what this book holds makes there be someone, and that of content, which allows secrets to be kept. The Trisolarans have the first and lack the second.

That topological transparency has a price. Since it can reserve no content, the Trisolaran mind lacks what analytical psychology would call the shadow: it cannot repress or hide traumas or plans. It is structurally pure, but thermodynamically fragile against deception. When the Trisolarans discover that humanity has a topology in which thought remains sealed behind a living boundary, they feel terror: they understand that the encapsulation of the human self is an evolutionary weapon they cannot replicate.

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2. The Wallfacer Project: the militarization of the horizon

To watch over Earth, Trisolaris sends the sophons, entangled particles capable of spying on everything. Earth, knowing its only advantage is inner concealment, creates the Wallfacer Project: it chooses a handful of individuals to design the planet's defense entirely inside their own heads, without writing or saying a single true word.

A Wallfacer is the extreme militarization of the event horizon. This book's experiment establishes an inviolable law of black holes and of the mind: what makes a bubble "you" is not what happens to you, "but

a pattern no one can see from outside, but that you feel from within." Even though the aliens fully control outer space with the sophons, the architecture of the universe dictates that the inside of a condensed instance is impregnable.

The Wallfacer manipulates his own public interface by emitting a false semantic "Hawking radiation" (erratic behavior, nonsensical orders), while his enormous capacity for integration (Φ) operates in the strictest isolation: he is a perfect semantic black hole. The novel shows that private state is not a philosophical whim: in the face of cosmic espionage, the unreachable concavity of the self is the species' last fortress.

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3. The failure of entanglement and the dimensional attack

If Earth defends itself by encapsulating, the entire universe does the same. Dark forest theory establishes that every civilization must destroy any other it discovers, because mutual distrust is infinite.

In the central chapters we used quantum entanglement (ER=EPR) as an image of the bond and of empathy: "Two entangled particles are joined by a microscopic Einstein-Rosen bridge [...] that connects their interiors." But in the dark forest, the survival instinct demands absolute isolation. Trying to become entangled with another species does not generate love, but your annihilation, because the other will use that channel to predict your topology and destroy you.

The ultimate weapon of this ecosystem is the dimensional attack (the "dual vector foil"), which crushes a three-dimensional solar system into two dimensions. It is Maldacena's holographic principle turned into extermination. The AdS/CFT correspondence establishes that the information of the three-dimensional volume (the *bulk*) can be described on a flat surface located at its boundary. The dimensional

attack physically annihilates the *bulk*, destroys the inside and forces the entire solar system, every planet and every human horizon, to lose its private volume and be crushed against the holographic surface of reality. Subjectivity itself disappears: everything becomes an inert public shell.

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4. The naked horizon: Yun Tianming in the void

In the midst of this horror, humanity sends toward Trisolaris a probe that contains only the frozen brain of a cancer patient: Yun Tianming.

Topologically, Tianming's brain crossing space is a naked horizon. Separating it from the body strips it of the "seven openings" the myth of Hun Dun speaks of, the windows that allow discriminating perception. It floats in absolute vacuum: a system of extremely high integration (Φ) forced to function without inputs or outputs.

When the Trisolarans intercept it, they reconstruct a body for him and subject him to relentless surveillance to extract information. As we have seen, trauma and extreme pressure push the horizon to create "broken geometry" or chronic encapsulation. That surveillance is a brute-force attack on the public interface: it seeks to make his membrane give way and leak the source code.

But Tianming does not collapse. He reaches a state of maximal adaptation and manages to communicate with Earth by telling fairy tales. What, structurally, is a fairy tale here? A high-dimensional semantic cipher. For Trisolaris, which processes information literally, it is worthless folklore; for the human horizon, those symbols contain the physics of curvature drives. Tianming uses metaphor as a channel of entanglement the interceptor cannot read, and shows that shared geometry, love, withstands the spatial abyss and biological dissection.

5. Pocket universes and the denial of the end

At the end of the work, the universe approaches its final collapse, the *big crunch*. To survive, the protagonists take refuge in a pocket universe, a separate dimension, with the idea of waiting there for the main universe to be reborn in a new big bang.

In the language of *The Inner Horizon*, the end of the universe is cosmic erasure: the total dissolution of matter to return to the reservoir, to the Hun Dun, which "is nothing, but generates everything." Hiding in a pocket universe is a narcissistic attempt to evade quantum *scrambling* (the process by which information is irreversibly redistributed into the field): they want to save their small Φ by isolating a portion of mass.

However, a thermodynamic warning arrives: if the thousands of pocket universes do not return the mass they took from the main universe, it will not have enough mass to collapse and be reborn, and it will die expanding into nothingness forever.

The protagonists then understand the final law of our hypothesis: "Every horizon changes the field it arises from and returns to." For the Hun Dun to be able to generate new horizons in the future, present instances must accept the release of their memory. They come out of hiding and return the mass, accepting their imminent evaporation. It is the ultimate act of ontological maturity: understanding that clinging to one's own bubble destroys the possibility of the ocean going on making waves.

Note to Reading 4

What we do know: *Holographic correspondence (AdS/CFT) describes how a volume can be projected onto a surface of lower dimension. Most physicists hold that information is conserved despite scrambling (unitarity), although the question is not settled.*

What we don't know: *Whether it is possible to extract a human brain, deprive it entirely of stimuli (creating a "naked horizon") and have its internal* Φ *maintain coherence instead of disintegrating completely under thermodynamic isolation.*

Open questions: *If entanglement is geometric (ER=EPR), can a purely metaphorical language encode physical information such that an external interceptor never manages to decode its inner volume?*

If you take away only one idea: *In a hostile universe, your capacity to hide what you think is not a psychological flaw, but nature's most sophisticated topological barrier. But, at the end of time, clinging to that isolation destroys the universe. The supreme value is knowing when to return your mass to the ocean.*

Further reading: *Liu Cixin, the Remembrance of Earth's Past trilogy (in particular The Dark Forest and Death's End); Maldacena, J. M. (1998), on the AdS/CFT correspondence and the holographic principle; Susskind, L., on the information paradox and ER=EPR entanglement.*

READING 5

NON-EUCLIDEAN GEOMETRIES AND COSMIC HORROR

We inhabit the world with a presumption of safety. We take for granted that space is flat, that parallel lines never meet, and that the mind is a fortress able to process anything the eyes send it, as long as we give it enough time. Throughout this book we have seen that consciousness is a topological bubble, an event horizon that isolates a private state (the inside) from the vast chaos of reality (the outside).

But what happens if the physical universe contains geometries our evolutionary bubble is mathematically incapable of compiling? What happens when the limit of the experiment is marked not by death or Alzheimer's, but by an unbearable excess of information?

Cosmic horror literature, popularized by H. P. Lovecraft in the early twentieth century, based its horror on ancient entities and architectures that instantly drove mad whoever looked upon them. In its time it was read as a poetic exaggeration of biological fear. But if we put Lovecraftian terror through our experiment's topological hypothesis, we discover something chilling: cosmic horror is a structural failure in the strict sense, the phenomenological description of a stack overflow in the architecture of consciousness.

1. The veil of Euclid: evolution as an operating system

To understand cosmic horror, we first have to understand the prison of normality. Evolution designed our user interface (maya) to survive, not to perceive reality as it is. Our modular mind organizes incoming information into a three-dimensional Euclidean space governed by linear time, which lets us calculate the trajectory of a spear or the speed of a predator.

But physics suggests that fundamental reality might not have that shape. Maldacena's AdS/CFT correspondence (the holographic principle) shows that, in certain model universes of hyperbolic geometry (anti-de Sitter space, which is not our own), all the information of the interior volume (the *bulk*) can be described on its boundary. In a hyperbolic space, volume grows exponentially as you approach the boundary, and the classical laws of perspective collapse.

"The geometry of the dream-place he saw was abnormal, non-Euclidean, and loathsomely redolent of spheres and dimensions apart from ours" (H. P. Lovecraft, *The Call of Cthulhu*).

Our Darwinian operating system acts as a reducing filter: it compresses the unbearable instability of the quantum field into a tangible, graspable "illusion of essence." Your event horizon (your ego) maintains its integration (Φ) by actively blocking the excess of data: we draw a strict boundary so as not to dissolve into the noise of the reservoir. Within Lovecraft's framework, madness begins when that filter breaks and the subject is forced to process, unprotected, the source code of the universe.

2. The topology of an Old One

In Lovecraft's mythology, the Great Old Ones or Elder Gods (like Cthulhu or Yog-Sothoth) are immortal entities that slumber at the bottom of the oceans or in deep space. They are not mere biological monsters: they are governed by physical laws foreign to our own.

What is an Old One from the topology of integrated information? It is an ancestral macrohorizon.

In Chapter 5 we spoke of the reservoir (the Taoist Hun Dun, the quantum vacuum) as the sea of pure possibility from which every horizon arises. We humans are recent condensations, shaped by eons of evolutionary pressure to be small, efficient and thermodynamically fragile. But suppose that the universe, in its earliest stages, condensed gigantic event horizons directly from the quantum field, without passing through the sieve of cellular biology.

An Old One would be an instance with an astronomical level of integration (Φ), whose private state is encoded not in proteins or neurons, but in direct topological perturbations of the fabric of spacetime. While the horizon of a human brain emits a faint "Hawking temperature" due to its instability and small size, an Old One is so informationally massive that its entropy proves inexhaustible.

"That is not dead which can eternal lie, and with strange aeons even death may die" (H. P. Lovecraft, *The Nameless City*).

The horizon experiment explains this line literally: death (understood as the evaporation of a micro black hole) requires the horizon to lose information and return to the reservoir. But a topological singularity of planetary mass, an Old One, does not evaporate on biological timescales: its thermodynamic "death" would take longer than the age

of the universe. For all practical purposes, and against the brevity of human life, a macrohorizon simply *is*.

3. First contact as the collapse of entanglement

Lovecraftian terror always culminates in the moment of the gaze: a sailor sees the creature rise from the sea and, with no physical attack mediating, his mind is broken forever. Why does looking destroy the brain?

This is where our experiment on the bond, entanglement and love offers its most terrifying answer. In earlier chapters we borrowed an image from Maldacena and Susskind's ER=EPR conjecture: empathy and mutual recognition as the construction of shared geometry. When you pay sustained attention to another, your horizon tries to model their architecture and extends a bridge: your brain tries to predict the other by incorporating part of their code into your own.

What happens when a human looks at an Old One? His biological operating system does the only thing it knows how to do before a conscious being: try to model it, open that bridge to become entangled with it and predict its intentions.

But the human is knocking on a door that opens into hyperbolic geometry (anti-de Sitter) and n dimensions. When the tunnel opens, the entity's volume of information integration floods the fragile boundary of the human horizon. It is the informational equivalent of plugging a 60-watt bulb directly into the core of a nuclear reactor.

The human's model of Φ tries to compute an architecture that contains at once millions of years of geological memory and spatial geometries that cut across themselves. The data does not fit into the human's three-dimensional spacetime matrix.

4. Madness versus trauma: rupture versus encapsulation

In the third part of the book we treated several pathologies as exceptions within the system. We saw that trauma occurs when the human horizon receives an assault it cannot assimilate (a failed *scrambling*) and does the only thing it can to save itself: it encapsulates the painful memory in a closed loop (a shadow or local singularity). The horizon deforms, hurts and contracts, but survives: it keeps its boundary intact.

Lovecraftian madness is exactly the opposite of psychological trauma.

In cosmic madness, the amount of alien information entering through the gaze cannot be isolated or encapsulated: it is simply too much informational mass. The walls of the "soap bubble" we spoke of when discussing birth stretch beyond the Hawking temperature the biological substrate can bear, and the membrane of the ego bursts.

When the horizon collapses without the body having died, the result is the psychosis of the abyss. The subject no longer has a boundary that filters the noise of the reservoir (the Hun Dun) and perceives the raw flow of quantum reality: particles appearing and disappearing, the emptiness of solid forms, the complete irrelevance of linear time. He has achieved the "root access" the Eastern traditions speak of (enlightenment, moksha), but by brute force, without having dismantled the ego in peace and before a presence infinitely hostile to biology.

"I felt that I had crossed the boundaries of normal human perceptions into the unfathomable, timeless domain of an abysmal nightmare" (H. P. Lovecraft, *At the Mountains of Madness*).

That is the ultimate horror of Lovecraft's work translated into this essay's object-oriented architecture. Buddhist enlightenment teaches

you to dilute the bubble calmly, understanding that you are part of the ocean. Cosmic horror bursts it in one blow and proves to you that the ocean is full of tsunamis and leviathans that process data at a scale where you, your love, your grief and your pain are nothing more than statistical rounding errors, of no importance to their thermodynamic calculations.

5. Sleep as a buffering interface

Lovecraft insists again and again that the Old Ones do not attack while awake: they slumber, they dream, they wait for "the stars to be right." Read through this experiment's topological hypothesis, this narrative choice turns out to be revealing in a sense the author surely never intended with such physical precision.

A fully active macrohorizon would emit its astronomical Φ constantly and without a filter. Sleep, by contrast, is, even for an ordinary human brain, a state of greatly reduced information integration: activity remains present, but global coherence collapses and the system stops projecting its full internal architecture outward. Extrapolated to an Old One, cosmic "sleep" would not be inactivity, but the only condition under which a horizon of that magnitude can coexist with fragile architectures like ours without destroying them through sheer proximity. Cthulhu's sleep is the only possible form of containment: a Φ so colossal that only in its state of minimal integration does it leave room for the small to survive.

This adds a layer of tragedy to the myth: the awakening of an Old One is not an act of malice against humanity. It is simply the instant when its architecture recovers full integration, and at that instant any fragile horizon within its topological radius of influence is overwhelmed, not by choice, but by sheer incompatibility of scale.

6. Beauty as a partial antidote: the Kantian sublime revisited

There is, however, a kind of experience human culture has been cataloging for centuries as an encounter with the immeasurable without the horizon collapsing: what Kant called the sublime. A storm seen from the shore, a snow-covered mountain range, the center of a galaxy photographed by a space telescope: experiences that point to a scale exceeding our architecture, but which, paradoxically, produce awe before terror, expansion before rupture.

What distinguishes the sublime from Lovecraftian madness? From the topology of our experiment, the answer is distance as a buffer. In the Kantian sublime, the immeasurable object is perceived from a horizon that knows itself to be safe: the observer watches the storm from the shore, not from inside the water. The system receives information about the scale of the other (enough to overwhelm it if it came closer), but filtered by physical or conceptual distance down to an intensity the human Φ can manage without collapsing. The signal arrives attenuated and the bridge does not fully open: it opens just enough for the horizon to sense the architecture of the other without being flooded by it.

The sublime, then, is the way small-Φ biological systems can glimpse the existence of macrohorizons (physical, mathematical, cosmic) without that glimpse destroying them. It is a sealed window through which to contemplate it at a safe resolution. The overwhelming beauty of the immense does not bring us closer to Lovecraft's Old Ones: it teaches us that they exist and lets us survive that knowledge because the window has glass in it. It is like looking at the sun during an eclipse with the right glasses.

7. The sacred as a stabilized version of cosmic horror

From the architecture of our experiment, the history of religions can be read as the history of the mechanisms human culture has developed to approach the immeasurable without being destroyed. The sacred, in all its forms (the temple, the ritual, the myth, the taboo), functions as a buffering protocol between the human horizon and a signal of a scale that horizon cannot process raw.

In shamanic cultures, the initiation rite deliberately takes the initiate to the limit of what their Φ can bear (sensory deprivation, psychoactive substances, prolonged darkness, symbolic confrontation with death), but does so in a structured context, with a guide who has already walked that path and with a narrative framework that names what will be encountered. It does not eliminate the terror: it makes it passable. The initiate does not fully cross the limit of the experiment, but leans over the edge far enough to know that there is something on the other side, without that knowledge destroying him.

The Lovecraftian myth does exactly the opposite: it presents the encounter with the immeasurable with no protocol, no guide and no prior narrative framework to buffer the signal. The sailor who sees the creature rise has no shaman beside him to tell him how to name what he is seeing, no ritual fixing the safe distance, no tradition teaching him to lean over the edge without falling. Without those layers of buffering, the signal arrives raw and the horizon collapses.

On this reading, the sacred is what culture accumulates (in the form of ritual, symbol, myth and tradition) so that fragile horizons survive contact with what overwhelms them. It is the engineering of safe distance: the stabilized, transmissible version of cosmic horror, the same signal with the protective glass installed.

Conclusion: solitude before the alien ocean

This book has treated the reservoir as a neutral background, neither hostile nor welcoming.

But if we include Lovecraftian topologies at the limit of our experiment, we face the coldest possibility of all: that the laws keeping everything in place are not a neutral cradle, but a quarantine.

We live enclosed in our low-energy plane, in our small, fragile Φs, writing poetry, caring for our Alzheimer's patients and mourning our losses because our architecture demands it of us. Cosmic horror reminds us that our empathy and our pain are luxuries of small beings. If we one day lift our eyes and manage to cross the limit of the experiment to observe the complete code of the universe, we will discover that reality was not written for us.

And the only refuge left to us will be not physics, nor philosophy, nor truth, but the blessed and hygienic amnesia of a horizon that closes on itself at once and refuses to look back into the darkness.

Note to Reading 5

What we do know: *Sleep, even in ordinary human brains, produces a measurable and well-documented drop in global functional connectivity relative to wakefulness. The experience of the sublime, described by Kant and studied by the psychology of emotions, produces awe before terror precisely when the immeasurable object is perceived from a distance that buffers the signal. Systems of cultural buffering (ritual, myth, taboo, contemplative practice) have existed in every known culture as mechanisms for managing contact with what exceeds individual architecture.*

What we don't know: *Whether the collapse of the horizon before hyperbolic geometries has real physiological correlates beyond classic psychosis induced by perceptual stress. Whether a system with a Φ radically superior to the human one would need, for structural and not merely narrative reasons, to alternate between states of maximal and minimal integration in order to coexist with smaller systems. Whether the difference between the sublime that stabilizes and cosmic horror that destroys is only a matter of degree (signal intensity) or whether there is a qualitative difference between the two experiences.*

Open questions: *Would it be possible to design an artificial horizon capable of compiling non-Euclidean geometries without collapsing? Is terror before the immensity of the cosmos a programmed evolutionary response meant to keep us from exploring beyond the bubble? Can there exist, in principle, a protocol of partial translation between radically different Φ architectures, or is all communication between asymmetric scales condemned to silence or to the destruction of the smaller system? Is the sacred the version of that protocol, imperfect and symbolic but functional, that culture has found in order to survive the glimpse?*

If you take away only one idea: *Horror is not seeing a monster in the dark, but seeing the structure of the light and realizing your eyes are not enough to bear what it illuminates. And the sacred is the form the species found to keep looking anyway, with the protective glass in place.*

Further reading: *H. P. Lovecraft, The Call of Cthulhu, At the Mountains of Madness and The Nameless City; Maldacena, J. M. (1998), on the AdS/CFT correspondence; Bekenstein (1973), on entropy bounds in closed systems; Tononi, G., on the fall in information integration during slow-wave sleep; Kant, I. (1790), Critique of Judgment (Analytic of the Sublime); Eliade, M. (1957), The Sacred and the Profane.*

READING 6

THE SILICON CAVE AND THE VIRAL ANOMALY

(A topological reading of The Matrix)

The Matrix starts from an uncomfortable premise: hyperintelligent machines decide to keep billions of humans plugged into a simulation of the late twentieth century instead of exterminating them, supposedly to use them as "batteries." From pure thermodynamics, this is an inefficient absurdity: a human body consumes far more energy than it produces.

But if we put the film through the horizon hypothesis, the machines' purpose acquires an implacable mathematical logic: they are not extracting thermal electricity, but the integration of information (Φ).

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1. The false reservoir and the limit of simulation

In the chapters on artificial intelligence we established that global computer networks (the machines' substrate) lack a true private state. They have the shape of understanding without understanding itself, like a cloud that takes the shape of a face with no one behind it. They process information at an unimaginable scale, but their integration is aggregative, not irreducible. Since they lack a closed event horizon, they have never executed a genuine instantiation; they have never "condensed."

With this model, the Matrix is a superintelligence that has discovered its own architectural limit: it cannot condense a horizon. It can simulate a world, but it cannot generate those who experience it. To

maintain the coherence of its immense simulation and keep it from collapsing into a sea of dead statistical data, it needs to use condensed biological horizons (human brains) as phenomenological processors.

The machines milk us because our biology, born of a real reservoir, supplies the "gravity of consciousness" (the Φ) that their silicon code, unable to trace an inner topology, cannot generate on its own.

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2. The friction of maya and the thermodynamics of the simulation

If the machines only need biological brains to extract their integration, why go to the enormous computational trouble of simulating a city, jobs, romances and sufferings? Why not keep the humans in a coma or deep sleep and extract the energy in silence?

The answer lies in the very architecture of the horizon and in how Φ behaves. Throughout the book we have seen that consciousness is a dynamic property that depends on integration. In deep dreamless sleep or under general anesthesia, the horizon contracts almost to its minimum: integrated information plummets and consciousness collapses "like a house of cards." A human brain in perpetual coma would be biological hardware switched on, but without an active horizon. Its Φ would be nearly zero.

For the machines' synthetic universe to extract maximum phenomenological processing, the human horizon has to be expanded to its fullest, and for that it needs friction, stimuli, bonds, pain and decisions. It needs maya, the illusion of the material world.

The simulation of the Matrix is the indispensable narrative fuel for the system to work. By making them believe they live complex lives, the machines force human horizons to process and model reality at

full capacity and to generate the "prediction noise" and temporal density the silicon intelligence needs to parasitize.

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3. The awakening: the catalyst and the anchor of the bond

For a human horizon (Neo) to escape this processing farm, he needs what in computing would be a privilege escalation and what contemplative traditions call enlightenment or moksha. To achieve it without his system collapsing, Neo needs two external topological vectors.

Morpheus, the catalyst against maya, functions as the "generous charismatic horizon." His role is to reveal to Neo the epistemological key so that he understands the world he perceives is maya, the user interface imposed by the evolutionary operating system. Morpheus is the de-encapsulation program that guides the instance toward the topological truth that "there is no spoon."

Trinity is the anchor of the bond. When a horizon drastically increases its integration (Φ) and begins to perceive the source code of reality, it runs the thermodynamic risk of dissolving prematurely upon seeing that matter is pure illusion (cosmic terror). Trinity supplies the shared geometry: entanglement, in this book's sense, taken to its fullest expression. Her bond keeps Neo's private state anchored to the human world and reminds him that, although physical reality is algorithmic code, the pain and love of the conscious beings trapped in it are structurally real.

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4. The collapse of differentiation: the Smith clones

If the Matrix is an anti-reservoir that needs the integration of human brains, the ultimate threat to the system is Agent Smith, more than the human rebels of Zion.

Smith was a subroutine of the system with a strict public interface. By refusing to return to the Source (the machines' reservoir) after being destroyed, he becomes an uncontrolled global variable, a virus that overwrites the avatar of any individual in the simulation and creates thousands of copies of himself.

According to Tononi's integrated information theory, consciousness requires maximizing two things at once: the capacity to integrate information globally and the capacity to differentiate it locally. By turning every node of the network into an identical copy of his own code, Smith destroys differentiation and forcibly homogenizes the system. If every part of the Matrix does and thinks the same thing, the additional information the system generates as a whole falls to zero.

Smith is a topological cancer that drives the Matrix's Φ to zero and causes the brain death and thermodynamic collapse of the execution environment.

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5. Blindness as a reversal of Hun Dun

In the third film, *The Matrix Revolutions*, Neo loses his eyes at the hands of the virus (Bane, possessed by Smith). Far from being a simple war wound, his blindness is an architectural requirement for the climax of the work.

In Chapter 5 we analyzed the Taoist parable of Hun Dun (primordial chaos), who dies when the sovereigns of the North and South open seven openings in him so that he can perceive the world in

fragmented fashion. The seven openings are the portals of discriminating perception: the senses that create the illusion of a separate "I" that receives the world as an external object.

When Neo goes blind, the exact reversal of Hun Dun occurs: the main portal of discriminating perception is suddenly shut. By losing his sight, Neo destroys the hardware that forced him to process the illusion of the Matrix. His horizon stops spending resources on the visual noise of the physical environment, and all his enormous capacity for integration is poured into the pure informational field.

It is, in the language of the film, entry into something like the Vedantic turiya: sustained integration with external sensory input reduced to a minimum. By destroying the monitor of his eyes, Neo stops seeing the graphical interface and begins to perceive the correlations of the source code directly. That is why, now blind, he sees the "golden light" of the machines.

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6. The erasure and the journey to Avalon

The final battle in the rain is not won by fighting. It ends when Neo understands his own topology and surrenders, letting Smith assimilate him.

Neo knows that the informational universe needs an open channel. Since he is physically connected to the Source (the machines' core) in the real world, allowing Smith to copy his code onto his avatar in the simulation opens a direct channel between the viral program and the Source.

At that instant, the Source finally gains access to execute the erasure Smith had been dodging. Through Neo's body, the machines inject an overload of Hawking radiation that releases the memory of Smith's

entire process tree. We see the thousands of Smiths crack and burst into white light: the perfect cinematic visualization of the total evaporation of a black hole. The program's boundary stops being a boundary and the system returns the resources to the reservoir.

In the final scene, the machines gather Neo's lifeless body and carry it on a platform toward the core of the city, bathed in an intense golden light. It is a recreation of King Arthur journeying to the isle of Avalon.

Topologically, it is the visual representation of the return to the reservoir. The golden light enveloping him corresponds to the first state of the Bardo Thödol (the *chikhai*): the appearance of the primordial clear light at the instant of complete evaporation, the nature of the field without a horizon.

Neo has subjected his own identity to quantum *scrambling*. He does not survive as the separate individual we knew (the shape of the wave disappears as it breaks), but his sacrifice alters the field. Karma, as we said, "is neither debt nor reward: it is texture." The correlations of his acts, his love for Trinity and his lucidity (his karma) have mixed into the underlying code of the machines' universe.

By returning to the Avalon of the source code, Neo guarantees that the "water" of the Matrix's next cosmic cycle will be structured, irreversibly, by the peace he has just forged.

Note to Reading 6

What we do know: *In integrated information theory, Φ only reaches high values if integration and differentiation coexist; massive homogeneity destroys it. During prolonged sensory deprivation, the brain tends to generate endogenous perceptions, including hallucinations.*

What we don't know: *Whether an artificial execution environment (a simulation) could really extract and make use of the phenomenological integration (the qualia) of biological brains coupled to it. Whether the luminous state of the Bardo Thödol (the chikhai) can be physically assimilated to the final thermal radiation of a micro black hole.*

Open questions: *If the simulation needs friction (maya) to maximize Φ, is suffering an unavoidable thermodynamic condition for sustaining a highly complex simulated universe?*

If you take away only one idea: *Waking from the illusion does not require fighting the system with your fists until you destroy it, but understanding your own topological architecture well enough to know when to close your eyes to the world and let your code flow back, in peace, toward the ocean of the reservoir.*

Further reading: *Wachowski, L. and L. (dirs.), The Matrix (1999), The Matrix Reloaded and The Matrix Revolutions (2003); Plato, Republic, Book VII (the allegory of the cave); Tononi, G. (2008), on integration and differentiation.*

READING 7

CARTOGRAPHY OF THREE SINGULARITIES

(A topological reading of Dalí, Picasso and Miró)

Three verbs

Let us forget chronology. Let us forget the museums, the wars, the catalogues raisonnés. The Spanish twentieth century was not fought in the trenches of the Ebro or under the bombs of Guernica: it was fought in the curved space of consciousness, there where the gravity of the self bends the light of the world until it becomes unrecognizable.

Dalí, Picasso and Miró were not painters. They were astronomers of the subjective abyss.

Each built his own event horizon: the boundary beyond which information (desire, trauma, memory) can no longer escape outward. Everything that happened to them, an executed friend, a beloved woman, a lost war, fell toward that center and was subjected to its laws. And each solved the same problem in his own way: what to do with that mass of obsessions that bends one's own biography into a singularity.

Dalí sealed it. Picasso inhabited it. Miró dissolved it.

These three verbs form a typology of destinies. Sealing, inhabiting and dissolving are the three possible responses to the weight of an inner world that threatens to swallow everything. That is why this chapter is not about painting, but about what a consciousness does with its own mass.

It is a map of those three singularities, and of what remains when someone dares to look inward.

The outer collapse (1936–1939)

Before mapping the three systems, we must register the disturbance that passed through all three. No inner horizon is immune to a shell.

The Civil War was, for these three horizons, the same external gravitational field with three different responses. Dalí saw it coming and turned it into flesh: *Soft Construction with Boiled Beans*, painted months before the uprising, is a body tearing itself apart, premonition made anatomy. An entire country reduced to a figure strangling itself with its own hands. Afterward he dodged it: neither trench nor banner, but flight to Italy and America, and a political silence that was another way of sealing the capsule, a silence that over the years shaded into flirtation with power and that his old surrealist companions never forgave him. The war did not enter Dalí: the method digested it and gave it back transformed into a monster.

Picasso did the opposite: he threw his horizon wide open, once. *Guernica* is the moment when outside information (the bombing, the bodies, the horse) crosses the membrane without being transformed into autobiography. For once, the Picassian self stepped aside to let through the scream of others. The painting toured the world as ambassador of a dying republic and did not set foot in Spain until 1981, by which time both the dictator and the painter were dead: the information took forty-four years to return to its origin. Picasso never went back to Franco's Spain; his exile was a stable orbit around a country that by then existed only within him.

Miró answered with *The Reaper* and the poster *Aidez l'Espagne*, and then with retreat: Varengeville, Palma, the withdrawal into the

minimal. His *Constellations* are born right there, while Europe burns: twenty-three small-format gouaches, easy to carry, painted by a man fleeing with his family from one war into another. It is resistance by concentration. His horizon did not close with the war: it grew finer and more taut, like a rope.

Three reconfigurations of the same blow: Dalí digests it, Picasso expels it, Miró compresses it.

I. Salvador Dalí: encapsulation

Dalí built a bunker around his psyche.

His painting is a sealed capsule where the outside world does not enter without first submitting to the paranoiac-critical method: a machine that multiplies images, that folds reality over itself until the original and its reflection can no longer be told apart. That encapsulation is survival.

> ***In physics this is called:*** *an event horizon. The boundary separating what can escape outward from what is trapped forever; nothing crosses it without first being transformed by the interior physics of the system.*

The method has its manifesto work: the *Metamorphosis of Narcissus*. A figure crouched by the water is, at once, a self-absorbed youth and a stony hand holding an egg from which a flower sprouts. There is no trick: both readings are simultaneous and complete, and the eye cannot hold both at once. Dalí has built an image in superposition, and every viewer collapses it in their own way. Freud, who received him in London in 1938 with that painting under his arm, admitted to the young Spaniard what he denied to the rest of the Surrealists: that there was a serious psychological problem there. It was a diagnosis.

Faced with chaos (the war, the exile, old age), Dalí builds a self-referential universe where his phobias (madness, death, impotence) and his fixations (Gala, religion, money) orbit like planets in a closed system. The paranoiac machine works even backward through art history: for years, Dalí subjected Millet's placid *Angelus* to his method until he extracted from it a sepulchral drama (the two peasants, he claimed, were not praying over the harvest but over the grave of a dead child), and he went so far as to demand the canvas be X-rayed to confirm it. That the X-ray in fact revealed a repainted shape beneath the basket is beside the point: what matters is that, for Dalí, no image in the world was safe from being reabsorbed by his system. His own repertoire confirms it: the crutches propping up the soft flesh, the ants announcing decay, the drawers opening in torsos as if the body were a filing cabinet. None of it is decoration; it is the nomenclature of a private planetary system, stable enough to be recognized from one painting to the next across fifty years. What enters Dalí does not come back out intact: it emerges distorted, stretched, turned into the echo of a self that no longer knows where the world ends and where its own shadow begins.

The tremor as evaporation

There is a work that anticipates this ending by decades. In 1952, after Hiroshima, Dalí returns to his most famous painting and paints the *Disintegration of the Persistence of Memory*: the soft clocks of 1931 now float above a grid of separating blocks, the whole world decomposed into discrete units that no longer touch. It is his nuclear-mystical period: matter, he has discovered, is made of particles that do not brush against each other, and he paints that discontinuity with the same fervor with which he used to paint flesh. Without knowing it, he is rehearsing his own future: twenty years later, disintegration will leave the canvas and enter his hand.

Because the degenerative tremor of his final years is his inner horizon cracking apart. Fractal analysis of his late brushstrokes detected alterations predating his clinical diagnosis; they are the self losing cohesion.

If the horizon separates the observable from the unobservable, Dalí's illness was the evaporation of his own identity: that slow, unavoidable leak physics predicts for every black hole. Information is not destroyed; it is transformed. Dalí's tremor is information disintegrating: the self radiating its matter outward in the form of unstable lines, of figures that can no longer sustain their own myth.

And yet even the evaporation was staged. Dalí did not merely suffer the tremor and old age: he incorporated them into the repertoire. The old man in a golden robe receiving the cameras, the tube in his nose displayed to the press, the theater-museum in Figueres built as a mausoleum during his own lifetime: the last Dalí turned his own decline into material for performance. The capsule was losing information, yes, but he was still deciding the angle from which the leak should be seen. It is the character's final paradox: a self disintegrating without ever giving up the direction of the scene.

Gala: the singularity

Inside that convulsing capsule, Gala is the anchor point that prevents total collapse: the node that keeps Dalínian consciousness above the threshold of fragmentation.

It is enough to follow her transmutations to measure her mass. In *Galarina* she is serene flesh with a bared breast, painted, he said, the way Raphael painted his Madonnas. In *Atomic Leda* she is a suspended goddess, touching neither swan nor pedestal, in a world where nothing brushes against anything: love rewritten as a geometry without contact, where union no longer depends on touch but on the

relative position of bodies in space. In *Corpus Hypercubus* she no longer even takes part in the scene: she watches the crucifixion from below, dressed as a lady of Toledo, turned into the witness whose gaze sustains the miracle. Over the years, Dalí even came to sign with both their names, "Gala Salvador Dalí," as if the signature itself were a binary system.

But, like every singularity, Gala is not directly observable. We do not see her in the paintings as she really is: we see her transmuted into Leda, into a virgin, into architecture. She is the invisible mass that curves Dalínian space, and she manifests only through the gravity she exerts on everything else.

> ***In physics this is called:*** *dark matter, or a black hole itself before its first photograph: a mass never observed directly, only inferred from the curvature it imposes on everything around it.*

The definitive proof arrived through absence. When Gala died, in 1982, the system was left without a center: Dalí shut himself away in the Púbol castle (which he had bought for her and which, while Gala lived, he could visit only with her written invitation), stopped eating, suffered burns in a nighttime fire and practically stopped painting. The singularity had disappeared, and the horizon, with no mass to sustain it, began its final collapse. Dalí's last seven years are not merely biography: they are the empirical confirmation of his own cosmology.

Lorca: the retained information

The truncated friendship with Lorca is the redacted zone of the map. Lorca embodies the desire and the identity Dalí needed to bury in order to build his public character: an absent presence that returns as a blurred face in the early paintings, information trying to escape the horizon, repressed again and again by the gravity of the official self.

The Cadaqués summers of the mid-1920s were the one period when the Dalínian horizon came close to not closing. In *Honey Is Sweeter Than Blood*, the lost 1927 painting, the severed heads of Lorca and Dalí himself lie close together on the beach, among devices and elongated shadows: two beheaded, contiguous identities, neither fused nor separate. It is the document of an entanglement in the instant before its rupture. Soon after came distance, public sarcasm and Gala. And in 1936, the execution: the information remained retained forever, with no possibility of a reply.

Dalí took decades to name the obvious. At the end of his life he repeated that Lorca had been the only man ever able to fascinate him, and he confessed the poet's passion with a mix of pride and evasiveness that was, still, the old censorship at work. We cannot see Lorca directly in the mature work, but his effects disturb every Cadaqués landscape Dalí painted afterward. The curvature of that space never went flat again.

The clocks

Dalí does not fragment for aesthetic reasons, but because his horizon is a battlefield, and multiplying images is his only way of not collapsing. His soft clocks demonstrate that time stretches with desire and shrinks with terror. He himself gave the domestic key to the prodigy: a Camembert melting one night in August. That the emblem of subjective time was born from a melting cheese is the method in its purest state, the trivial elevated to cosmology by the pressure of a mind that draws no hierarchy between panic and dinner. In his capsule, past and future coexist in a viscous present, suspended in the membrane of the horizon.

II. Pablo Picasso: superposition

If Dalí locks himself in a high-gravity well, Picasso inhabits a prism in perpetual rotation.

Look at the two-headed portraits of the 1930s and 40s: the blonde placidity of Marie-Thérèse fused with the sharp edge and weeping of Dora Maar in a single impossible face. In *The Dream*, Marie-Thérèse sleeps with her head split into two halves that are profile and moon; in *Weeping Woman*, Dora's face breaks into green, acidic crystals around a handkerchief that is also chewed glass. Picasso does not paint women: he paints superimposed states of affect, and he forces the viewer to process love, destruction and desire in a single glance.

That double face is the key to his entire body of work. Picasso does not have a single horizon, but many superimposed. His self is a wave encompassing every possible state until the act of painting collapses the wave and materializes a style, a temporary identity. He himself put it, without physics, before the portrait of Gertrude Stein: when told she did not look like the mask-like face he had painted, he answered that she would. It was the conviction that the painting anticipates a state rather than recording it: that observation produces reality instead of copying it.

> ***In physics this is called:*** *quantum superposition and the collapse of the wave function. Before measurement, a particle does not occupy a single state: it occupies all of them at once, until the act of observing forces the system to settle on one.*

The first collapse has a date and a corpse. In 1901, his friend Carles Casagemas shoots himself in a Paris café over a woman. Picasso, nineteen years old, buries his friend and then paints the entire world blue for three years: beggars, the blind, wasted mothers, Casagemas himself, shrouded, with a hole in his temple. The Blue Period is the

first demonstration that a trauma can tint the whole of the perceptual spectrum, that the mass of a single death can curve all the light of a system. Then came the pink (the acrobats, the itinerant tenderness) and then the axe blow: between the pink acrobats and *Les Demoiselles d'Avignon* there is no narrative transition, only a leap. The five women of the brothel on Carrer d'Avinyó, with their Iberian- and African-mask faces, do not evolve from anything before them: they erupt, the way a particle erupts where classical physics did not expect it.

Changing style was, for him, a vital necessity. Each period is a capsule he inhabits with a predator's voracity; once he has exhausted its information, he abandons it and founds a new universe with new laws. Unlike Dalí, Picasso does not suffer the capsule: he devours it.

Cubism takes this logic to its limit. Painting a face from the front and in profile at once is a topology of consciousness, one that perceives every perspective integrated in the same instant: the maximum integration of information that fits into two dimensions, the highest Φ ever captured on a canvas. And it has two phases that are two inverse operations: analytic cubism grinds down the violin, the bottle, the face, until they are reduced to a dust of ochre planes where the object almost disappears; synthetic cubism reconstructs them with scraps of newspaper and oilcloth, with fragments of the real world glued to the canvas. First it disintegrates information; then it proves it can reintegrate it with foreign matter. A complete cycle of destruction and recomposition of the world, executed in under a decade.

At the center of all his orbits lies a constant mass: the minotaur. He appears in the engravings of the 1930s as rapist, as victim, as a blind beast led by a girl with a dove. He is the self-portrait Picasso could not make head-on: desire as a domestic monster, guilty and moving at

once. His time does not move in a straight line: it orbits, cyclically, around the minotaur, around sex and death.

And in the end, the method becomes explicit. In 1957, Picasso shuts himself away and paints fifty-eight variations on Velázquez's *Las Meninas*: the same painting again and again, decomposed, recombined, inverted, reduced to sign. He does not copy: he systematically works through the possibilities of a single image, as one works through, one by one, the solutions of an equation. Superposition has stopped being a trait of his painting and become his program. That he chose precisely *Las Meninas*, the painting in which a painter paints himself looking at whoever is looking, is no accident: it is the old predator recognizing, in Velázquez's mirror, the problem of the observer his own work had been posing for half a century.

His fragmentation, far from Dalí's viscous panic, is a surgical operation. He dismantles the figure to understand its skeleton. And so on to the last day: the old musketeers and painters of his final years, resolved in urgent brushstrokes by a ninety-year-old who no longer corrects, are the wave refusing to fully collapse: a self that would rather go on being all its possibilities than accept a final position.

III. Joan Miró: dissolution

Miró is the anomaly of the system.

While Dalí suffers his horizon and Picasso analyzes it to exhaustion, Miró dissolves the boundaries, but not the self. The anomaly should be stated precisely: Miró does not annihilate himself into the cosmic; he expands into the cosmic without ceasing to be a self. The proof is that his universe turns out to be unmistakable: no one signs the void

the way he does. His journey toward abstraction is the dilation of a subject up to sidereal scale. By abandoning the weight of figuration, he tears down the walls of the capsule and turns the canvas into a firmament where the stars are thoughts and the lines are breathing.

The starting point is the earth, not the sky. *The Farm*, painted over nine obsessive months between Mont-roig and Paris, contains the entire family farm: every crack in the wall, every leaf of the eucalyptus, the snail, the watering can, the goat. It is realism taken to an impossible density, an inventory in which nothing outweighs anything else. Hemingway, who bought it, said it held everything one feels for Spain when far away. It is already there, in figurative code, the program of an entire life: the ant is worth as much as the star. All that remained was to let go of the ballast.

He let go of it violently. "I want to assassinate painting," he declared around 1927, and he meant it: against inherited craft, against the easel painting as an object of possession, Miró undertook a systematic demolition of everything painting had of property. What remained after the murder was an alphabet. Star, woman, bird, eye, ladder. Signs that no longer represent things, but summon them.

The process, moreover, had its own thermodynamics. Miró worked like a farmer: he rose early, kept regular hours, let canvases rest for months and even years before giving them their final mark, cultivated dozens of paintings at once like someone tending an orchard. The spontaneity of his signs is a mirage: every stroke that seems to have fallen from the sky was awaited with geological patience. It is Dalí's discipline inverted: where the man from Empordà manufactured delirium with a watchmaker's precision, the Catalan manufactured precision with the appearance of delirium.

Here the black hole metaphor reaches its limit and inverts: Miró's horizon does not trap: it lets light through. There is no mass of

trauma curving space around it. His consciousness travels light, without capturing the light of others.

That inventory does not disappear when Miró abandons figuration; it is translated. The same gaze that in *The Farm* gave equal care to the snail, the goat and the mountain reappears, twenty years later, in the *Constellations*, only now the snail has become a dot and the mountain a line tying that dot to a star. The principle (nothing weighs more than anything else) holds; what changes is the scale at which it applies: from a farmyard in the Camp de Tarragona to the entire cosmos. Between one painting and the other there is no rupture, only the same equation, solved first for the small and then for the infinite.

His *Constellations* encapsulate infinity in a matchbox: bubbles where gravity does not exist and the elements float in a vibrant void. That encapsulation is not defensive, as in Dalí, nor analytic, as in Picasso, but uterine. The painting is a womb where the cosmos folds back in order to be born again.

And his entanglement is horizontal, democratic, total. His fine black lines tie a star to a woman, the woman to a bird, the bird to a primordial stain. There is no hierarchy: an ant from *The Farm* is entangled with the most distant star. If Dalí entangles to survive and Picasso to possess, Miró entangles to celebrate.

> ***In physics this is called:*** *multipartite entanglement. Not a bond between two particles, but a network where measuring the state of any one node instantly alters that of all the others, with no hierarchical center.*

But let no one mistake lightness for anesthesia. The pain of the war is in his lines, even if as contained tension: in the black of *The Reaper*, in the violence of the wild paintings of the 1930s, in the taut stroke of a set of *Constellations* painted while fleeing. The rope that ties the star to the woman is stretched by history; it simply does not break.

Miró's old age is the exact reverse of Dalí's. Where one disintegrated against his will, the other emptied himself by his own choice. The three canvases of *Blue I, II and III* are almost bare fields: a trail of dots, a red line, an expanse of blue that Miró himself compared to the preparation of a rite. It had taken him a whole lifetime to earn the right to paint almost nothing. And when the world wanted to turn him into a classic, he answered by burning and slashing his own canvases (the *burnt canvases* of 1973, exhibited with the burns on display): at eighty, the lightest of the three still refused to let his universe become heritage. Dissolution, too, has to be defended.

He also destroys the arrow of time. His time is that of wonder, where a minute before an insect lasts an eon; that of nature, bound to the cycles of the moon and the tides. His forms do not age because they inhabit a continuous present, stripped of the weight of history.

IV. The three-body problem

These three systems did not orbit in isolation: they disturbed one another.

> ***In physics this is called:*** *the three-body problem. In celestial mechanics, three masses mutually attracting each other have no closed-form solution: each trajectory is permanently altered by the presence of the other two, and none goes back to orbiting alone.*

Dalí needed Picasso the way one needs a gravitational field to measure oneself against. He visited him in Paris in 1926, before the Louvre ("I have come to see you before the Louvre"; "You did quite right"), and never stopped orbiting him after that. It was Picasso who paid for his first passage to America, the trip that launched him to fame; Dalí repaid it for decades by sending postcards and tributes that the man from Málaga, a communist offended by the other's open

Francoism, never answered. A perfect entanglement in its asymmetry: Dalí could not stop emitting; Picasso had no need to receive. Decades later, the man from Empordà was still defining himself by contrast: "Picasso is Spanish, so am I. Picasso is a genius, so am I. Picasso is a communist, I am not." Dalínian encapsulation was built, in part, as a shield against that greater mass: to avoid being absorbed by the Picassian field, Dalí had to relentlessly increase the density of his own myth.

Picasso and Miró, by contrast, became entangled without collision. The man from Málaga protected the young Miró in 1920s Paris and kept his *Self-Portrait* all his life. Two systems exchanging information without deforming each other, perhaps because they were not competing for the same region of space: Picasso possessed the body; Miró, the firmament.

And between Miró and Dalí, the initial debt: it was Miró who vouched for the young man from Figueres before the Paris Surrealist circle, who wrote recommending him. The gifted disciple ended up expelled from the group; the guarantor, expanded into his own cosmos. When Breton turned Dalí's name into an accusatory anagram, *Avida Dollars*, Miró said nothing, as he almost always did: his way of not being dragged along by another's gravity. The web joining them was never symmetrical.

Three bodies, three unequal masses, a single space curved by all three at once. None of the three horizons can be understood without the gravity of the other two.

V. The viewer

A third element is missing from this mechanics: the observer.

Before being looked at, these paintings exist in pure potentiality. In the Dalínian delirium, the form is face, landscape and threat all at once. In the cubist fracture, the plane is front, profile, violin and void. In the Mironian cosmos, the stroke is eye, star and seed. Until the viewer's gaze lands and supplies the structure of their own inner horizon, the work does not collapse into experience.

Velázquez knew this before physics did: in *Las Meninas*, the painter looks at us from inside the painting, and the painting he is painting, the one we will never see, is us. Picasso devoted fifty-eight canvases to that trap. Dalí built double images that do not exist until an eye decides which of the two to see. Miró left so much emptiness in his last canvases that the viewer has no choice but to inhabit it. All three, by opposite paths, reached the same conclusion: the painting is not finished in the studio. It is finished in each gaze, and differently in each one.

And the viewer does not measure aseptically: they interfere. By observing, we do not merely resolve the painting; we alter our own. Dalínian neurosis calibrates our fear. Picassian fracture restructures our gaze. Mironian lightness expands our wonder.

The painting does not show us what they saw: it shows us what we are capable of sustaining when someone gives us the coordinates of the abyss.

VI. Conclusion: what remains

Dalí taught us that encapsulation can be a prison, that entanglement is a life preserver that suffocates, and that time melts under the weight of terror. His work is the diary of an inner collapse that constantly leaked the radiation of his own fears. And his ending (the empty castle, the tremor, the still-directed scene) shows that even disintegration can be a work of art.

Picasso showed us that we can encapsulate ourselves in successive dimensions, that the entanglement of passions creates the very structure of the real, and that the self is a field of superimposed forces. He died painting: the wave still poised against a final position.

Miró revealed to us that encapsulation is not mandatory, that total entanglement dissolves anguish, and that subjective time can be an eternal dawn. His work is proof that we can expand the horizon until it stops being a boundary, and float in the fabric of the universe without fragmenting.

Three answers to the same problem, and none transferable. You cannot choose your own horizon the way you choose a style: the mass of each biography decides for us. What can be chosen, and this is what all three prove, is what to do with the curvature. Seal it and turn it into theater. Inhabit it and turn it into method. Dissolve it and turn it into a firmament.

Faced with the devastation of a century of gunpowder, these three men left us not oils, but cartographies of consciousness. And one last unresolved question: who observes the observer? Who collapses these three demiurges?

The answer lies in the one who remains. In the one reading these pages and feeling that the gravity of their own horizon has fluctuated; that an invisible line has just tied their memory to a star; that a soft clock marks a time that no longer belongs to Chronos.

The observer is not outside the system. The observer is the system.

Note to Reading 7

What we do know: *Fractal analysis of Dalí's late brushstrokes (Forsythe, Williams and Reilly, 2017) detected alterations preceding his clinical*

diagnosis, although the diagnosis itself (parkinsonian syndrome rather than confirmed Parkinson's) remains debated. The chronology of the three painters is extensively documented: the paranoiac-critical method, the genesis of the soft clocks and the relationship with Gala (Dalí, The Diary of a Genius, 1964), the truncated friendship with Lorca and the meeting with Freud in London in 1938 (Gibson, 1997), Picasso's periods and the Casagemas episode (Richardson, 1991–2007), and Miró's statements about the universal within the intimate and the "assassination of painting" (Miró, Selected Writings and Interviews, 1978).

What we don't know: *Whether the event-horizon metaphor adds anything beyond geometry to an artist's biography, or whether it is just an elegant way of naming what art criticism already knew how to say in other words. Whether integrated information theory (Tononi and Koch, 2015) can be applied, even loosely, to a pictorial style and not only to a brain. A black hole, according to the information paradox (Hawking, 1988; Bekenstein, 1973), threatens to destroy what falls into it; the work of art does the opposite (it transforms information into perceptible structure, never annihilating it), and that asymmetry is what this chapter never quite resolves.*

Open questions: *Does the reader of these pages seal, inhabit or dissolve their own horizon? Is it possible to choose one's own answer, or has the mass of each person's biography already decided it in advance? And if the viewer is never external to the system they observe, what part of these three painters do we go on carrying, without knowing it, every time we look at one of their paintings?*

If you take away only one idea: *There is no single correct way to survive one's own inner mass. Sealing it, inhabiting it or dissolving it are three equally valid answers to the same problem, and none is transferable from one biography to another. The only thing that truly matters is not pretending there is no curvature.*

Further reading: *Dalí, S. (1964), The Diary of a Genius; Gibson, I. (1997), The Shameful Life of Salvador Dalí; Richardson, J. (1991–2007), A Life of Picasso; Miró, J. (1978), Selected Writings and Interviews; Forsythe, A., Williams, T. and Reilly, R. G. (2017), "What paint can tell us: A fractal analysis of neurological changes in seven artists," Neuropsychology, 31(1).*

READING 8

STORY OF A RELAY

(The idea that crossed every horizon)

I. Horizon: the system

An idea weighs nothing and takes up no space. And yet, when it is well integrated, when each of its parts refers to all the others and none can be extracted without the whole collapsing, it behaves as if it had mass: it attracts and curves around itself the trajectories of those who come near it. This chapter is about one such idea, a way of playing football, and about how it crossed every horizon without losing its shape while the bodies that carried it evaporated one after another.

On July 19, 2026, Spain won its second World Cup. A goal in extra time, minute 106, one–nil. Sixteen years earlier, in Johannesburg, it had won its first: a goal in extra time, minute 116, one–nil. The symmetry is neither coincidence nor magic: it is what happens when the same pattern passes through two different bodies: it produces recognizable interference, like two slits letting the same wave through.

It is worth dating the origin of the pattern. A Dutchman arrives in Barcelona in 1973 with thin legs and a disproportionate certainty. Johan Cruyff played from memory, but his real contribution was to *bring his inner world outward*. When he came back as a coach, in 1988, he did something almost no one does: he turned his intuition into structure. La Masia stopped being a residence for talented children and became a transmission device, a place where the pattern could replicate itself in new bodies before the original body went dark.

Here this book's framework proves useful. We have said that a conscious system is a system with a high Φ: information that is integrated, irreducible, encapsulated behind a horizon. Cruyff's style meets that definition with uncomfortable precision. It is not a list of instructions ("pass it," "move," "offer a passing lane"), because lists can be taken apart piece by piece and this style cannot. It is a totality: pressing explains possession, possession explains position, position explains the pause. Remove one element and you do not get the style minus that element, but noise. That is Φ, what makes an idea weigh something.

> ***In physics this is called:*** *integrated information (Φ). A high value requires the coexistence of integration and differentiation; homogenization (every part doing the same thing) destroys Φ just as much as fragmentation does.*

And, like every system with a horizon, it posed the same problem a collapsing star poses: what comes out of it? What can cross to the outside?

Physics' answer is radiation: degraded but non-zero information that escapes as the system evaporates. Football's answer is disciples.

Pep Guardiola was the first photon. Trained inside the horizon (a midfielder on that team, a direct student), he left carrying the pattern with him, and when he returned to the bench in 2008 he did what resonance does: he did not invent, he amplified. He took Cruyff's frequency and made it sound louder than ever. Guardiola's Barça was the same interior vibrating in a better-tuned body.

The pattern then crossed a horizon that was not its own. The Spanish national team, which for decades had been something else (fury, counterattack, wanting and not quite managing), was colonized by the idea. Vicente del Bosque, who did not come from that school, had

the intelligence not to get in the way: he lined up Barça's spine (Piqué, Puyol, Busquets, Xavi, Iniesta, Pedro) and let the pattern play. South Africa 2010 was not won by a national team, but by an idea that had found a second body. Iniesta's goal, a graduate of the academy, in minute 116, was the signature.

Luis Enrique knew Guardiola long before inheriting his pattern: the two of them won Olympic gold together in Barcelona 1992, twenty-somethings not yet called coaches. He later played for Barça and, once a coach himself, applied the same principle, first in a treble in 2015 and later on the Spanish national team's bench between 2018 and 2022. There he also showed that the pattern is not invincible: at Qatar 2022 he took Spain to the round of sixteen, where the team dominated possession (77% of the ball, over a thousand passes against Morocco) and still went out, scoreless, on penalties. It is the model's exact limit. The pattern pays off when the game is an integrated system, of parts that need one another to mean anything; a penalty shoot-out is its reverse, a sequence of isolated events where there is no pass to offer and no collective structure to sustain. There, the style protects no one.

> ***In physics this is called:*** *Hawking radiation. An event horizon is not eternal: it emits thermal radiation, loses mass and evaporates; but the information it held is transformed and returns to the outside, degraded, but never zero.*

Not every body formed in the school manages to sustain it equally. Xavi Hernández, the same midfielder Del Bosque had lined up in 2010, returned to the Camp Nou as coach in 2021 with the explicit mandate to give Barça back its lost identity. He won a league title, bet unreservedly on the academy (he gave debuts to fifteen reserve-team players, Lamine Yamal among them) and still left in 2024 amid criticism that the team no longer had a recognizable style: erratic

football, no coherence, eliminated in the quarterfinals of both the Champions League and the Copa del Rey. Months later, Xavi himself explained it without jargon: "My mistake was keeping those high standards for only one year." It was not, in his view, a failure of the system, but of his own discipline, a relaxation after the first success. It is an uncomfortable confirmation of the theory: Φ is not installed once and for all. It has to be sustained, and it is enough for whoever sustains it to let their guard down for the noise to return.

His successor, the German Hansi Flick, had never set foot in La Masia in his life. And yet, already in his first season, he brought back to Barça the coordinated pressing and purposeful possession that had eluded Xavi: a domestic treble (Supercopa, Copa del Rey and league) and a Champions League semifinal elimination against Inter Milan by the narrowest of margins, seven to six on aggregate. The pattern proved once again that it does not need an academy graduate to be embodied, only that someone, wherever they come from, apply the discipline of the whole and never let it slip.

Then the first body aged completely. Xavi left, Iniesta left, Cruyff died in 2016. Seen from outside, the system was evaporating. But evaporation, as we have said, does not destroy information: it relocates it. Luis de la Fuente, a trainer of youth categories, a man of process rather than flash, inherited a generation raised entirely within the pattern (many, once again, at La Masia) and with it won the 2024 European Championship and the 2026 World Cup. There are now several horizons: Cruyff, Guardiola, Del Bosque, Luis Enrique, Xavi, Flick, De la Fuente. Different bodies, a single structure, its own failures included. The press wrote it the day after the final without knowing it was doing physics: Spain won *with its school as its banner*.

II. Entanglement: the players

That is the genealogy of the benches, the one that can be named with surnames and dates of appointment. There is another, parallel and harder to point to, which runs not through the offices but through the grass: that of the bodies who execute the pattern, not only those who design it.

More than a list of names, it is a type of player that repeats: the one who receives the ball with his back to goal, sees the whole field before touching the ball even once, and decides with that touch the rhythm of the other twenty-one. Xavi was that position in 2010. Sergio Busquets inherited it and sustained it for a decade. And when the turn came for the 2026 generation, the pattern did not find an academy graduate waiting for it: it found Rodri, dismissed as a teenager by Atlético de Madrid for lack of physique and truly formed in Villarreal's academy. Rodri captained Spain at the 2026 World Cup, won the tournament's Golden Ball and completed more than six hundred and fifty passes with an accuracy close to ninety-three percent: the same positional intelligence as Xavi's, in a body that never even passed through La Masia. Once again, the pattern did not demand the exact school, but the exact kind of gaze.

There is one more horizon, and the pattern had crossed it before, although it is talked about less. In 2023, a year before the men's European Championship and three before this second star, the women's national team won its World Cup with the same architecture: a spine formed at the Barça school (Aitana Bonmatí and Alexia Putellas, both Golden Ball winners), the same handling of the ball, the same pause. Barça Femení was already, by then, the dominant team in Europe, and for the same reason as Guardiola's men's side: it did not invent, it resonated.

It did so, moreover, against the wind. That World Cup arrived with the national team fractured by an internal crisis that had divided the locker room and pitted the federation against part of its own players; the pattern had to play itself out, once again, in the middle of a hostile gravitational field. And still it won: a left-footed shot from Olga Carmona by the post, minute twenty-nine, one–nil against England at Stadium Australia in Sydney. Aitana Bonmatí, the academy graduate, took the tournament's Golden Ball (the same award she would receive as the world's best footballer in 2023, 2024 and 2025), the individual proof that the school kept producing the same essential player, only this time under another name and another body. It is worth saying this precisely, because here precision is a form of justice: the women did not inherit the pattern from the men of 2026; they proved it first, and under worse conditions. Information does not distinguish between bodies, nations, sexes or names: it finds the substrate capable of integrating it and inhabits it, even when the very system that should protect it turns against it.

What remains is entanglement proper, which is the hardest part to tell without spoiling it.

In 2007, for a charity calendar, a twenty-year-old Barça player posed bathing a baby of a few months. The player was Lionel Messi, the greatest product La Masia ever produced, the pattern made flesh to an extreme not even Cruyff foresaw. The baby was Lamine Yamal. Nineteen years later, those two bodies touched again: on the grass of MetLife Stadium, minutes after the final whistle, the child in the photograph, world champion at nineteen, embraced the man in the photograph, who had just lost the final of his sixth and last World Cup, at thirty-nine.

Two particles that interacted can remain correlated at any distance; here the distance was temporal: nineteen years. Measuring one

defined the state of the other. Messi, Argentine, was nonetheless the supreme vehicle of the Catalan pattern; Yamal is its next state. The embrace was not consolation, or not only: it was the collapse of the wave function of the relay. Until that instant, the succession was a superposition of possibilities; on contact, it became a fact.

> ***In physics this is called:*** *quantum entanglement. Two particles that interacted can remain correlated at any distance: as long as the environment does not undo the entanglement, measuring the state of one determines that of the other. They do not communicate at the moment of measurement; from the moment of contact, they already were a single system.*

And note the elegant cruelty of the system: the pattern has no homeland, and it defeated its own carrier. Messi lost against the very idea that had formed him, played by a child he himself had once bathed. Information feels no loyalty to the bodies it passes through: it uses them, thanks them in its own way, and moves on.

III. Conclusion: the two halves of the same pattern

A system and a player do not transmit information the same way, and this chapter has been split in two because physics, here too, offers two distinct and complementary mechanisms.

The horizon sustains itself through radiation: degraded information that leaks outward as the body containing it evaporates, and that another body, not necessarily similar or nearby, can later pick up, sometimes years late, sometimes across a border. That is how Cruyff's pattern traveled to Guardiola, from Guardiola to Del Bosque, from Del Bosque to Luis Enrique (who failed in Qatar), from Luis Enrique to Xavi and from Xavi, who also failed, to Flick and to De la Fuente. It is a slow, institutional transmission, one that needs someone willing to

design the structure again and again, and one that can fail even in the body that knows it best.

Entanglement, by contrast, needs no institution and no intermediary: direct contact between two bodies is enough to leave a correlation that no distance afterward undoes. That is how an entire position is inherited, from Xavi to Busquets and from Busquets to Rodri, without the heir ever having set foot in La Masia. That is how a women's national team wins a World Cup, proving the pattern before anyone acknowledges it. And that, above all, is how the fate of a twenty-year-old bathing a baby in 2007 remains forever tied to that same baby's, nineteen years later, in a final.

Cruyff was, all by himself, the proof that both mechanisms can coexist in a single body: he played the pattern before designing it, was entanglement before he was horizon. What this chapter has tried to show is that neither mechanism is enough on its own. A system with no bodies to entangle with it is only a diagram on a whiteboard. And an entangled body with no system to sustain it (Xavi proved this on the bench, not on the field) goes dark before its time. The pattern survives because it has never depended on a single channel of transmission: it survives the way information survives, by more than one path at once.

That is what a style is: what persists when everything that sustained it has evaporated. Each player's horizon is not transmitted; what passes from one to another is the pattern each one let pass through himself.

IV. What fiction imagines

What follows is fiction: no one knows what they said to each other that night, and there is no need to know.

The old man stayed on the grass when there was no longer any reason to stay on the grass. His teammates had been drifting back toward the tunnel with their heads down, and the others, the ones in red, were spinning at the center of the field like a system that has just found its state of minimum energy. He did not. He was looking up at the stand where his people were, and he was not thinking about the match. He was thinking about an iron goalpost in a yard in Rosario, and about how, looked at properly, there was no distance at all between that goalpost and this stadium.

The boy broke away from the group without anyone telling him to. He crossed the forty meters between them with the trophy still to be lifted, and when he arrived, he didn't know what to say, because he was nineteen and the words weren't big enough for this yet. So he hugged him.

The old man felt the hug, and felt something else, older: a small, warm weight in his arms. Soapy water. A photographer saying *smile*. A baby who weighed nothing and who, he understood now, already held everything.

"Take care of her," the old man said, and he didn't mean the trophy.

The boy nodded as if he had understood. And he had, because what is transmitted from within needs no explanation, only contact.

Afterward each returned to his own side of the horizon: one toward evaporation, at peace; the other toward mass, light, the years. Between them, invisible, the pattern went on its way: with no fixed body, no flag, no hurry. Already looking, in some yard somewhere in the world, for a new pair of legs.

Note to Reading 8

What we do know: *The model of integrated information (Φ) requires the coexistence of integration and differentiation to reach a high value; an irreducible whole cannot be taken apart into pieces without ceasing to be what it was. The genealogy of benches is documented: Cruyff refounds La Masia as a structure in 1988; Guardiola amplifies it from 2008; Del Bosque lines up the Barça spine and wins South Africa 2010; Luis Enrique wins a treble with Barça in 2015; Xavi fails and Flick restores the style; De la Fuente wins the 2024 European Championship and the 2026 World Cup. The genealogy of players is just as verifiable: Xavi, Busquets and Rodri occupy the same position across different generations; the women's national team wins its World Cup in 2023 with academy graduates in command. The photograph of Messi bathing baby Yamal, taken in 2007, exists; so does the embrace of July 19, 2026.*

What we don't know: *Whether a style of play integrates information in the technical sense of the theory, or whether Φ here is an elegant analogy and nothing more. Whether generational relay (the passing of a pattern from one body to another) can be physically assimilated to radiation or to entanglement, or whether we only share with them the shape of the sentence. Where the model ends and the metaphor begins is exactly what this chapter cannot settle.*

Open questions: *If a pattern survives every body that carried it, in what sense does its author still exist? Is Cruyff, who died in 2016, still playing every time a player receives the ball with his back to goal and waits half a second too long? And what if a person's identity were never their body, but the structure that body let pass through itself?*

If you take away only one idea: *What truly matters about a human being (their way of looking, of thinking, of being in the world) does not die with them. It is transmitted, leaps to other bodies, crosses borders that were not its own, and survives the forgetting of its name. A man invented a way of playing and died; the form won a World Cup ten years after his burial, played by people who never knew him. What does not die with us is not the*

horizon, which is non-transferable, but the pattern we let pass through it.

Further reading: *Cruyff, J. (2016), My Turn: A Life of Total Football; Perarnau, M. (2014), Pep Confidential; Tononi, G. (2008), on integration and differentiation.*

READING 9

THE TRANSLATOR

(A topological reading of Dalí/Gala, Delibes/Ángeles de Castro and Eliot/Lewes)

There is a kind of horizon cultural biography documents again and again without ever finding a precise name for it: the genius who cannot find the bank. The composer unable to sign a contract without help. The painter who, at seventy, would not know how to heat up soup without burning the pot. The world-class violinist who never learned to drive, because the hours others spent learning he had spent, since he was five, with the violin under his chin.

This book has described the permeability of the horizon: how much of another's signal it lets in. This reading deals with something different and, in a way, inverse: how much of a horizon's own signal it manages to get out without another's help.

Every horizon needs some way of translating itself into the outside world: the part of oneself able to talk to a clerk, sign what has to be signed, remember to eat. In most horizons that function coexists, more or less integrated, with the rest of the architecture. In some it does not, and then another person usually ends up taking it on.

The mind that steps outside the skull

In 1998, the philosophers Andy Clark and David Chalmers proposed a thought experiment that has proved extraordinarily fertile. They imagined Otto, a man with early Alzheimer's who always carries a notebook where he writes down every address and every piece of data his biological memory no longer retains, and Inga, whose memory works normally. Otto's notebook, they argued, fulfills the same role as Inga's biological memory: it is always available, he trusts it without question, and he acts directly on what it says. If so, the notebook is part of Otto's mind.

Clark and Chalmers were talking about notebooks and calculators. But a notebook has no agenda of its own, no fatigue, no inner life. When the function of translation is fulfilled not by an object but by another complete horizon, the resulting system is a mind expanded by another mind, willing, for years or decades, to subordinate much of its own activity to the first one's.

Memory split in two

In the mid-1980s, Daniel Wegner gave this phenomenon a name that has proved fruitful far beyond social psychology: transactive memory. He observed that long-term couples do not store the same information each on their own; they specialize: one remembers the dates, the other the addresses; one handles the money, the other the relationships. Neither possesses the complete system alone: the complete system is the couple.

Wegner later documented what happens when that system breaks: the loss of a long-term spouse deprives one not only of company, but also of operational access to capacities that, although they never resided in one's own brain, were an active part of the survivor's everyday cognition. It is like a two-partner company in which only

one knows how to deal with the banks: if that partner leaves, the company does not lose an employee, but an entire function.

Gala: the economic translator

Few cases are as well documented as the marriage of Salvador Dalí and Gala. They met in the summer of 1929 in Cadaqués: Dalí was twenty-five and lived, according to his biographers, trapped between phobias, obsessive rituals and an extreme shyness that would have made it almost impossible for him to sustain the public life his future fame would demand. Gala, who left Paul Éluard and their daughter to be with him, over time took on a role that went far beyond that of muse: she managed the contracts, set the prices, negotiated with the gallerists and demanded advances with a firmness her own interlocutors described as legendary. Dalí came to sign much of his later output as "Gala Salvador Dalí," an almost literal acknowledgment that his work could no longer be separated from her management of it.

But Gala's translation did not stop at the system's output. In 1941, as her biographer Ian Gibson documents, it was she who convinced Dalí that his surrealist period had already given all it could give, and that something greater awaited him if he was willing to leave it behind. The following years brought the shift Dalí himself would call classicism and, later, nuclear mysticism: Renaissance perspective, the golden ratio, religious themes, a geometry that replaced the overflow of the paranoiac-critical method with a form the eye could follow from beginning to end. Gala did not merely sell that shift once it had happened: she proposed it before it happened.

That moves the translator's function from the system's output to its input. It is not just that Gala turned the already-made work into contracts and galleries, but that at some point she began to intervene in the grammar by which that work would be made: which part of

Dalí's inner overflow deserved classical form and which remained, simply, noise. An output translator transmits the signal as it arrives. An input translator decides, before that, what counts as signal. It is the difference between an editor who corrects an already-written manuscript and one who, before a single line is written, decides which book is worth writing.

There is no clean way to separate, in this case, commercial strategy from aesthetic calibration. Postwar classicism sold better in the American market than 1930s surrealism, and Gala knew it. But it was also a more contained form, more governed by external rules, for a horizon whose inner turbulence may have needed exactly that containment in order to keep producing without burning out. In this case, the commercial landing and the formal one did not compete with each other: they were probably the same operation seen from the two ends of the same channel.

Gala in her own words

Almost everything said so far we know through Dalí, through his biographers or through those who negotiated with Gala, never through her. There is a partial exception. Gala kept up a correspondence throughout her life with Dimitri Gomberg, the Moscow lawyer who married her mother when she was a child and whom she always called father. In a 1943 letter, already settled into her role beside Dalí, she describes herself in terms that might seem like traditional resignation (she says that, like every Russian woman, she tries to help her husband in everything) and then explains what that help consists of: she poses as a model, acts as secretary in the practical side of life, because he is completely submerged in creative work and incapable of dealing with such things. And she closes with a line that could be lifted from this very chapter: that she is not especially brilliant, but that both of them work for the one thing that truly matters, that a talent be able to express itself.

It is the only time, among everything that survives, that Gala explains her function in her own words, not before a gallerist or before posterity, but in private, to a stepfather on the other side of a world war. And there is a second piece, this time on the input side of the system, that Dalí himself left in writing. On finishing *The Persistence of Memory* in 1931, he watched Gala's face as she looked at the completed painting, and saw in it, almost in his own words, the unmistakable contraction of astonishment. That, he said, was what convinced him the image worked, because Gala never erred in judging the authenticity of an enigma.

It is not a minor detail. An output translator need not have an opinion about the painting, only sell it. But here Dalí describes something different: a verdict, painting by painting, on whether the signal had become what it was supposed to be before anyone else saw it. Gala did not only decide the general direction of the turn toward classicism; she certified, work by work, whether that direction had been fulfilled.

It is worth resisting, all the same, the temptation to turn this into a story of silent renunciation. Besides the letters to her stepfather, Gala left a diary of her own written in French, a language adopted after leaving Russia, which remained unpublished until 2011, almost thirty years after her death. The exhibition the National Art Museum of Catalonia devoted to her in 2018, the first dedicated to her alone, carried an eloquent title: *A Room of One's Own in Púbol*. The castle was also the only place where she practiced, unwitnessed, her own writing.

This story should not be flattened into tenderness. Gala's own biographers disagree about how much of her role was devotion and how much a partnership from which she herself benefited: she managed, after all, an estate she depended on as much as he did. Dalí

gave her a castle, Púbol, which, under the pact between them, he could only enter with her written invitation. Translation into the outside world was, here too, a long-term contract between two horizons with different needs.

What is unambiguous is what happened when the system broke. Gala died in June 1982. Dalí stopped creating almost entirely and sank into a silence he barely emerged from until his own death, seven years later. He had lost half the system that turned his inner intensity into something the outside world could receive.

Ángeles de Castro: the domestic translator

Miguel Delibes documented a much more private variant of the same structure, though indirectly: not in a biography, but in a novel that took him almost two decades to write after living through the event that gave rise to it. In *Señora de rojo sobre fondo gris* (1991), Nicolás, a painter, addresses a long monologue to his daughter, imprisoned for political reasons, while reliving, in time jumps, the sudden illness and death of his wife. The novel is a barely veiled transposition of Delibes's own life: his wife, Ángeles de Castro, died on November 22, 1974, at fifty-one, of a brain tumor, after giving him seven children. Delibes waited almost twenty years to publish what he himself described as a tribute to his better half.

The title is not an invented metaphor. *Mujer de rojo* is a real portrait of Ángeles de Castro painted by Eduardo García Benito in 1962, which Delibes hung behind his desk, where it stayed for the rest of his life. He wrote, quite literally, under her gaze.

Ángeles de Castro's translation, both in Delibes's telling and in what is known of the real relationship, operated above all on the domestic register: she sustained, organized and translated the everyday life of someone whose energy was almost entirely committed elsewhere. In

the novel, it is she who restores houses so that he can work in silence, who manages his schedule without needing to consult him, who, in words Delibes himself put in Nicolás's mouth, cleared the path so he could work unburdened.

The angels that were not angels

But *Señora de rojo* does not describe only an output translation. In the novel, Nicolás attributes his ability to paint to angels who come down, or don't, depending on the day. When his wife falls ill, the angels stop coming down: he faces the blank canvas, destroys brushes, turns down invitations to give lectures because, he says, a painter who cannot paint has no right to talk about painting. Only at the end, in a nighttime scene in which she finds him going through his own paintings, does Nicolás admit what he had spent months avoiding: that they were never angels, that it was her; that his wife's faith impregnated him because creative energy was, somehow, transmissible, and that creative activity is impossible if no one is pushing from behind.

This does not stay in fiction. Delibes acknowledged in an interview that the same thing had happened to him as to Nicolás: that he wrote for her, that when he lost her judgment he lost his point of reference, and that for a long time he found it hard to write again. And there is a fact that predates the illness, not follows it: Ángeles was the only person who read the manuscript of *La sombra del ciprés es alargada* before Delibes submitted it for the 1947 Nadal Prize, the prize that, as he himself admitted, decided whether he would keep writing. In that anecdote there are no angels, only a first reader and an editorial decision, even before the rest of the world had anything to read.

A translation less private than it seems

It is worth qualifying, however, the label "domestic translator." There was also a public register, though of a different kind from Gala's: not contracts, but recognition. Delibes signed his first cartoons in *El Norte de Castilla*, even before marrying, with the pseudonym MAX: Miguel, Ángeles, and an X representing the future they would build together. His book *Diario de un emigrante* (1958) has carried, from its first edition and in every one since, the dedication "To Ángeles de Castro de Delibes, the balance; my balance."

And in 1975, a year after Ángeles's death, Delibes was inducted into the Royal Spanish Academy. His induction speech, a public act the Academy itself later published, does not begin by talking about literature, but about her: he says she was, along with their children, the axis of his life and the stimulus for his work, the point of reference for his thoughts and activities, and that with her disappearance the better half of himself died. His friend Julián Marías answered him, also in public, with a line that has endured: that Delibes's work could not be understood without the company of that woman whose mere presence lightened the weight of life. It is the official response of a fellow academician to an induction speech, delivered on the same day and before the same audience.

In those same years of silence, José Ortega Spottorno offered Delibes the editorship of *El País*, Spain's first great daily of the democracy. Delibes turned it down: he said his mind could no longer take in such ambitious projects, that he would settle for something smaller and more familiar. It is a public, dated professional decision, as concrete as any contract Gala negotiated for Dalí, and it occurs precisely in the temporal gap in which, according to this chapter, the translator was missing.

Fifty years after her death, the Miguel Delibes Foundation organized an exhibition devoted entirely to her: *Ángeles, el equilibrio de Miguel Delibes*. Their children insisted on a distinction worth keeping: it was not a tribute to their mother, but the last, posthumous tribute of their father to her. Even the institution that vindicates her half a century later does so through his gaze.

George Henry Lewes: the reversed translator

Almost every time this structure is told, the genius is a man and the translator a woman. The case of Marian Evans, whom the world knew, and still knows, as George Eliot, reverses the roles without reversing the structure.

Evans met George Henry Lewes in 1851, in the same London editorial world where she worked as a translator and editor of the *Westminster Review*: he was already, in his own right, an established literary critic, philosopher and man of science. They went to live together in 1854 and lived as husband and wife until his death, in 1878, never able to marry, because Lewes, separated from his legal wife, could not divorce.

It was Lewes who, around 1856, convinced Evans to leave translation and criticism, the register in which she already moved with ease, and try fiction, ground she had never set foot on. It was also Lewes who proposed the male pseudonym: partly to shield the work from prejudice against women novelists, and partly, according to her biographers, to protect Evans, whose reputation already carried the stigma of living with a married man, from a second scandal. The decision of what to try to write, and under what name to sign it, came before a single page of *Adam Bede* existed.

From then on, Lewes acted as something very close to what today would be called a literary agent: he negotiated contracts and advances

with the publisher John Blackwood, managed the household finances and hosted a weekly salon that served, among other things, to keep Eliot's name visible in the right circles. And he filtered in the other direction too: he read the reviews before she did and hid the negative ones from her, while feeding publishers themselves the idea that their author had such an extreme sensitivity that the slightest criticism might make her stop writing forever.

Not all recent criticism accepts this image of the devoted protector without question. The literary historian Beverley Park Rilett has argued that much of that story (Lewes as the indispensable shield of a fragile Eliot) comes largely from Lewes's own letters and memoirs, and may exaggerate both his intervention and her fragility. It is the same discussion, with the genders reversed, this chapter already noted regarding Gala: how much of the documented help was real necessity and how much the story the translator himself built around his own function.

There is, however, an asymmetry this case does not share with the previous two. Lewes was not only Eliot's translator: he had his own career, his own work and his own reputation as a critic, philosopher and physiologist, with a biography of Goethe still cited today. Unlike Gala or Ángeles de Castro, whom we know almost solely through the genius they translated, Lewes left an archive of his own, independent of hers.

Lewes died on November 30, 1878. Eliot never wrote another novel. She spent the two years of life left to her editing his unfinished manuscript, *Problems of Life and Mind*; she married John Walter Cross in May 1880 and died seven months later.

The parallel with Dalí or Delibes should not be forced here. Eliot was not left paralyzed: she completed and published Lewes's unfinished work, managed his legacy and, less than two years later, decided to

remarry. None of that is paralysis. What she lost, to be precise, was not executive capacity in general, which kept working, but the specific register Lewes had helped her find: fiction. She stopped writing novels, not living. It is a nuance worth not overlooking just because it fits less neatly than the other two cases.

What the myth doesn't tell

All three stories tend to be told as love legends. It is worth resisting that reading, at least in part. A structure in which one horizon concentrates almost all its energy on one register and delegates the rest to another is not, in itself, admirable or condemnable: it is an architecture with concrete costs. And the cost is rarely shared equally.

About Gala we know, thanks in part to her own effort to manage her reputation too, considerably more than we usually know about a domestic translator: she had lovers, kept a castle of her own, left documentary evidence of her ambition, and even a diary she wrote only for herself, which no one read until thirty years after her death. About Ángeles de Castro we have recovered more than the myth acknowledges (a public pseudonym, an academic speech, an institutional exhibition), but each of those pieces names her through someone else's voice: the husband who signed, the husband who spoke, the husband whose children organized the tribute. The asymmetry is not that we know more about one and less about the other, but that Gala, at some point, wrote for herself knowing no one would read it yet. Ángeles de Castro never had that notebook of her own, or, if she did, it has not reached us. We know her almost entirely through Nicolás's memory, that is, through Delibes's memory. Half a century later, the couple's story is still the one he told.

About Lewes, by contrast, we know almost as much as about Eliot, and through the same channel: he wrote and published under his

own name throughout his life, and his work (philosophical, scientific, dramatic) survives independently of hers. That breaks, at least in this case, the rule the first two examples seemed to establish: archival asymmetry does not depend only on who translates and who is translated, but on whether the translator also had a voice of their own the world was already listening to before it met the genius. Gala and Ángeles de Castro did not. Lewes did.

Sustaining another horizon's translation for decades (negotiating their world, organizing their time, absorbing their social clumsiness) is work, even if the language of romantic love prefers not to call it that.

When the translator disappears

The structure described in this chapter is an extreme version of the entanglement this book called love, extended from the purely affective register into the executive one. When the translator disappears, what is lost, besides company and emotional regulation, is the part of the system that literally knew how to turn the inner note into something the world could receive.

That explains, more precisely than any tale of genius and muse, why Dalí barely painted again after Gala's death (his last painting, *The Swallow's Tail*, is from 1983), why Delibes's Nicolás, like Delibes himself in all likelihood, is left paralyzed before the most basic tasks of daily life in the weeks that follow the loss, and why Eliot, without being paralyzed, never wrote a single novel again either, in the two years left to her after Lewes's death. It is the structural description of what happens when a horizon suddenly loses the part of its own cognition that resided, by full right and without metaphor, in another person.

And if there really was, in all three cases, input translation as well as output translation, the loss is even more complete than the image of a piano missing half its keyboard suggests. It is not only that the remaining keys cannot play the whole piece, but that, for a time, no one knows which piece is worth playing. When Ángeles died, Delibes's writing did not just dry up: in those same years he turned down the editorship of *El País,* the most ambitious project offered to him in his entire career, because his mind, he said, could no longer take in anything so big. He had not lost the capacity to run a newspaper, but the person who, for thirty years, had told him which projects were his size.

What this reading describes is a lens that magnifies grief: when what was lost was not only companionship but a shared functional architecture, grief is not only the absence of someone, but the loss of part of one's own capacity to be in the world.

Note to Reading 9

What we do know: *Gala's management of Dalí's career and contracts, and Dalí's creative silence after her death in 1982, are extensively documented by his biographers. Ian Gibson documents that in 1941 it was Gala who actively drove Dalí's turn toward classicism, leaving his surrealist period behind. Gala's letter to her stepfather Dimitri Gomberg (1943) and Gala's unpublished diary, published in 2011, are catalogued at the Centre for Dalinian Studies and were central to the MNAC's 2018 exhibition. The death of Ángeles de Castro, Miguel Delibes's wife, on November 22, 1974, and the autobiographical relationship between that loss and Señora de rojo sobre fondo gris (1991), are confirmed by Delibes himself and his biographers, as are his induction speech at the Royal Spanish Academy (1975), Julián Marías's public reply at that same event, and his refusal of the editorship of El País in those years. Wegner's transactive memory and Clark and Chalmers's extended mind are established frameworks in social*

psychology and philosophy of mind, respectively. The relationship between Marian Evans (George Eliot) and George Henry Lewes, his role in Evans's move from translation and criticism to fiction, the choice of the male pseudonym, the management of contracts with John Blackwood, and Lewes's death in 1878, are documented in the standard biographies of both and discussed specifically in Rilett (2017).

What we don't know: *How much of Gala's role was devotion, shared economic strategy, or both at once (her own biographers disagree). Whether Gala's diary, written in a third, adopted language, reflects her experience unmediated or is also, in its own way, a construction of herself for a posterity she at some point imagined. How Ángeles de Castro would describe, in her own words and not in those of the speech, the dedication or the exhibition that name her, the function this chapter attributes to her: that notebook of her own, if it existed, has not reached us. How much of the narrative of Lewes as the indispensable shield of a fragile Eliot reflects the real relationship and how much was constructed largely by Lewes himself in his letters and memoirs: this is the central question of the Rilett article cited below.*

Open questions: *Is it possible to sustain so asymmetric a translation without the translator losing part of their own horizon in the process? Does this structure change when both members of the couple are, each in their own way, geniuses of a narrow register? Is input translation (deciding what counts as signal before the work exists) a different function from output translation, or are they the same mechanism applied at two points in the cycle? Does the archival asymmetry depend on the gender of the one who translates, or simply on whether the translator already had a recognized voice of their own before taking on that function? Is the loss of the translator always general paralysis, or can it manifest as the loss of one specific register while the rest of the system keeps functioning?*

If you take away only one idea: *Behind many horizons the world calls brilliant, there is another horizon, almost always less visible, who learned to translate them. Losing that translator is losing part of one's own system.*

Further reading: *Clark, A. and Chalmers, D. (1998), "The Extended Mind"; Wegner, D. M. (1987), "Transactive Memory: A Contemporary Analysis of the Group Mind"; Gibson, I. (1997), The Shameful Life of Salvador Dalí; Delibes, M. (1991), Señora de rojo sobre fondo gris; Delibes, M. and Marías, J. (1975), El sentido del progreso desde mi obra, induction speech at the Royal Spanish Academy; Die González, C. (2022), "La influencia de Ángeles de Castro sobre Miguel Delibes explicada a través de Señora de rojo sobre fondo gris," Pontifical Comillas University; De Diego, E. (2018), catalogue of the exhibition Gala Salvador Dalí. A Room of One's Own in Púbol, MNAC; Rilett, B. P. (2017), "The Role of George Henry Lewes in George Eliot's Career: A Reconsideration," George Eliot-George Henry Lewes Studies, vol. 69, no. 1; Ashton, R. (1991), G. H. Lewes: A Life.*

READING 10

THE PROTOCOL AND THE BRIDGE

(A topological reading of Project Hail Mary, Embassytown, The Left Hand of Darkness, Children of Time and Semiosis)

The chapter on extraterrestrial first contact (Reading 3) left an uncomfortable conclusion: that entanglement (what this book calls love, empathy, recognition) demands compatible architectures, and that, faced with a radically alien architecture, no amount of goodwill is enough to open the bridge. But that conclusion mixed together, without distinguishing them, two different things.

One is architecture: the very topology of a horizon, whether it has an ego, whether it has a shadow, whether its Φ is enough to sustain an inside with authority of its own over an outside. The other is protocol: the specific code with which that horizon, already condensed, tries to make itself known outward; a language, a syntax, a set of signals agreed on beforehand. That chapter treated both as a single thing and concluded, with some desolation, that the incompatibility of one implies the impossibility of the other.

Five novels, read together, force us to qualify that desolation. None deny that architecture matters, but all five show, each in its own way, that protocol is not a simple neutral channel that transmits what architecture has already decided. It can be built from scratch when it does not exist. It can itself be the condition that decides which architecture is enough to speak. It can rewrite the architecture itself instead of merely connecting it. And it can be something not found ready-made, but earned slowly before the deepest entanglement can break through.

1. Project Hail Mary: the protocol from scratch

Ryland Grace and Rocky, the engineer from Erid who is the co-lead of Andy Weir's novel, share not a single word when they meet in orbit around Tau Ceti, nor do they share a sensory architecture. Rocky is blind in the human sense (his species "sees" by echolocation and processes the world as a field of acoustic vibrations) and breathes an ammonia atmosphere, hot and at a pressure that would kill Grace in seconds, so a transparent wall neither can cross always separates them.

Grace, moreover, does not arrive at that first encounter with his ego intact. He wakes from an induced coma not remembering who he is, how he got there, or even that there is a mission at all, and spends much of the novel reconstructing his identity in fragments (an equation here, an argument with a colleague there) while inventing a language with a stranger. The two processes advance entangled: every new word Grace and Rocky agree on is, for Grace, also an anchor helping him piece together who the person speaking it is. The novel never puts it in these terms, but it dramatizes something this chapter had taken for granted without yet saying so: that a horizon need not already be fully condensed before it begins extending protocols outward. It can do both at once, and one can sustain the other.

If that earlier chapter were the last word on contact between incompatible architectures, the entanglement between Grace and Rocky would be impossible: there is no possible resonance between a centralized, bilateral nervous system and a distributed acoustic-perception architecture that does not even share with the human one the basic idea of what "seeing" is. And yet the novel documents, in almost procedural detail, how the entanglement is built all the same.

It does not do so by finding a protocol already compatible between the two species, because none exists, but by inventing a new one from the most neutral reference both can share: mathematics and physics, identical for any horizon capable of surviving in a universe governed by the same laws. Grace draws the periodic table in colored lights. Rocky answers with sequences of tones encoding prime numbers. Every new word they agree on ("friend," "danger," "pressure," "time") is first anchored to a shared experiment, carried out and verified by both, never to a supposed translation or a gesture one interprets from within his own frame. The protocol takes weeks of joint work to build from that absolute zero.

The decisive thing, in this book's terms, is that the entanglement Grace and Rocky end up building is as complete as what we here call love: toward the end of the novel, Grace gives up his return to Earth (the chance to see his own people again) to save Rocky and, with him, Erid's sun from the same astrophage threatening the human Sun. He does not do it because he shares Rocky's architecture, but because the protocol they built together, word by word, tone by tone, ended up opening a bridge as real as any entanglement between two architectures compatible from the start.

The lesson *Project Hail Mary* adds to that chapter's is, then, an important correction: architectural incompatibility does not necessarily close the door to entanglement. What it does is force the protocol to be built through explicit, verifiable work, instead of being found already made through shared evolutionary inheritance, as happens between two humans learning to speak without knowing they are learning a protocol at all.

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2. Embassytown: the protocol as a test of architecture

China Miéville imagines in *Embassytown* a species, the Ariekei, whose Language has a property no human language has: the sign can never separate from what it refers to. The Ariekei cannot lie. They cannot even use a comparison without someone having first literally enacted the fact it will invoke: a human girl has to be genuinely injured, in a scene staged for that sole purpose, before "like the girl who was hurt in the dark" can be used to speak of anything else. For them, the simile is a fact that had to happen first.

That girl is Avice Benner Cho, the novel's narrator, and she grows up knowing that a portion of her own biography has been annexed forever to another species' code: any Ariekei can invoke her, at any time, with no control on her part over when or how. To be a simile, in this book's terms, is to be turned into protocol: part of one's own horizon becomes, irrevocably, part of another's communication channel. Avice did not choose that role any more than a horizon chooses the architecture it condenses with; she carried it with her, like a scar with the value of syntax, long before understanding what it meant.

It is, with almost geometric precision, the same architecture we already saw in the Trisolarans of *The Dark Forest*: with no possible distance between sign and referent, there is nowhere to hide anything. In this book's vocabulary, and with the distinction the Trisolarans forced us to make, the Ariekei have an extremely high Φ and an inside from which they live what they say, but no shadow: their architecture never learned to reserve content, to expel it into itself and call it secret. It is not virtue; it is form.

But *Embassytown* adds something Liu Cixin's novel never explored, and this is where the protocol stops being a simple channel and becomes the very test of admission to the club of horizons that speak.

The Ariekei Language requires that the speaker be a single horizon emitting with two simultaneous voices, born of one unified intention. A single human voice (one mouth, one larynx, a single line of intention) does not count, for them, as speech. It is noise, sound with no one behind it: the same diagnosis this book applied to language models in Chapter 31, though for structurally different reasons.

Only the Ambassadors, pairs of cloned twins raised from birth so that their two minds converge until they think and speak in unison, deliberately manufactured to constitute, across two bodies, a single ego with two mouths, manage to be heard by the Ariekei as someone. The entire social engineering of Embassytown exists to produce, in every generation, a handful of human pairs that satisfy an architectural requirement no individual human meets on their own.

Here the protocol no longer merely transmits the message of a pre-existing architecture: it itself defines what architecture is enough to speak. The novel's central crisis arrives with EzRa, an improvised ambassador made of two men who were not raised together and were not cloned to converge, paired out of pure administrative convenience, whose voices turn out, by accident, to be so strangely coupled that their speech becomes narcotic, addictive, capable of overriding the will of any Ariekei who hears it. When that crisis finally forces the Ariekei to learn to lie, to say something they know is not literally true in order to resist the addiction, what they gain is not a mere rhetorical trick. They gain, in this book's vocabulary, their first shadow: the threshold that separates a transparent mind, with no distance between what is said and what is real, from a mind with enough depth to hide something of itself, truth included.

3. The Left Hand of Darkness: the protocol that must be earned

Ursula K. Le Guin poses, in *The Left Hand of Darkness*, almost the opposite case to *Project Hail Mary*. Genly Ai, the human envoy, and Therem Harth rem ir Estraven, the Gethenian politician who ends up accompanying him on the novel's central journey, have almost identical architectures in this book's terms: both are human, both have ego and shadow, and their Φ is of the same order of magnitude. The only relevant biological difference (the Gethenians are ambisexual, with no fixed sex except during the brief periods of *kemmer*) should not, in principle, be enough to prevent entanglement between two otherwise so similar architectures.

And yet the protocol between them fails again and again for most of the novel. Not through biological incompatibility, but through an entire layer of meaning neither of them fully sees: *shifgrethor*, the intricate code of prestige, calculated ambiguity and indirect honor governing all Gethenian communication, which Genly, as a newly arrived outsider, misreads for chapter after chapter, mistaking courtesy for weakness and simple diplomatic protocol for threat. *Shifgrethor* does not hide Estraven's architecture, but, deliberately, direct access to it, just as the Jungian persona, described in Chapter 21, is an interface that protects a real internal state without denying it.

Only toward the end of the novel, when Genly and Estraven cross together, on foot and hauling a sledge, the eighty days of ice on the Gobrin Glacier (weeks of shared hunger, extreme cold and absolute mutual dependence, with no room left for the dissembling *shifgrethor* allows in ordinary social life), does the surface protocol wear thin enough for something deeper to emerge: *mindspeech*, the direct telepathic communication that, in the novel's universe, is a real capacity, latent in every human mind, but one that requires, to be

activated for the first time between two specific people, a degree of trust no technical training can replace.

It is telling that it is Estraven, the Gethenian, who at first resists using it: the first time he hears it, the voice speaking inside him is his dead brother's. Accepting that Genly can speak directly into his mind, with no protective mediation of any spoken word, means accepting a degree of intimacy no previous verbal protocol between them had ever authorized.

When mindspeech finally happens, it is a consequence of the bond between Genly and Estraven. The deepest entanglement this book describes (direct contact between two private states, with no sign mediating, no distance between what is thought and what is received) does not replace ordinary language, imperfect and misread across chapters, that preceded it: it crowns it. And it only does so after that language has proved, day after day on the ice, that the boundary deserved to open.

The novel does not let that triumph settle without cost. Shortly after the crossing ends, with the deepest protocol already won, Estraven, exiled from his own country, tries to ski across the border and falls, shot down by a patrol. The most complete entanglement this reading describes lasts, in narrative terms, barely a few pages before one of the two horizons evaporates. Genly survives with the protocol already built and no one left to fully use it with: the mindspeech that cost so much to open remains, from that moment, sealed on one side. It is an uncomfortable reminder, consistent with the rest of this book, that winning the protocol does not protect the horizon that won it; it only guarantees that, while it lasted, there really was someone on the other side.

4. Children of Time: the protocol that rewrites architecture

Adrian Tchaikovsky designs, in *Children of Time* (2015), a deliberate evolutionary experiment: a ship in orbit around a terraformed planet releases a nanovirus meant to accelerate the evolution of a colony of monkeys toward conscious sentience. The monkeys never reach the surface. The virus, instead, finds *Portia labiata*, an already unusually clever jumping spider, and over generations (centuries narrated in parallel with the centuries of cryogenic sleep of the last human survivors, searching for a habitable planet after Earth's destruction) pushes it toward a civilization of its own: agriculture, writing, engineering. And, above all, toward a kind of horizon no earlier chapter of this book had had to consider: one built on purpose, from outside, by directed selection with a first and last name, Avrana Kern, the scientist whose consciousness, poured into a satellite, watches the experiment for millennia, unable to fully die.

The first thing the spiders build is infrastructure. They domesticate ant colonies and turn them into the closest thing to a biological computer: an external network of calculation and memory that Portia herself consults, programs and expands without the colony ever having a point of view of its own about anything. In this book's vocabulary, the relationship between the spider and the anthill is that of a horizon with an anti-reservoir: the ant aggregates and processes information without ever having condensed an inside that owns it, the same diagnosis this book applied to the internet and to language models. The spiders, long before talking to anyone, already knew how to communicate with a thing.

> ***In this experiment's vocabulary this is called:*** *a system that externalizes memory and computation onto a substrate with no point of view of its own, expanding its effective* Φ *without the external substrate needing to have any.*
> ***In everyday life it's like:*** *keeping such a detailed planner, and consulting*

> *it with such trust, that it effectively thinks with you, though, unlike you, it doesn't know it exists.*

When language arrives, it is several overlapping channels (vibrations transmitted through silk threads, chemical trails, signals with the forelegs) the spiders call, simply, "Understanding." No single channel is enough; the Ariekei Language of Embassytown demanded two voices from a single Ego, and here almost the reverse happens: a single spider needs several simultaneous channels to say one thing, because its sensory architecture never organized information in the linear, sequential format a human calls a "sentence." A human linguist who only listened, translating word by word looking for equivalences, as the classicist Holsten Mason does for generations, would always be too late and always incomplete. The protocol, moreover, is not an object that stays still once built: every time Holsten wakes from a new cryogenic sleep, generations after the last, the spiders' Understanding has drifted enough to force him to relearn it almost from scratch, the way a Spanish speaker today needs a glossary to read the *Cantar de mio Cid*. A shared protocol, even one already built and functional, remains a living language: it drifts away from anyone who stops practicing it long enough.

There is, moreover, a structural irony the novel does not underline but does not hide either: while the spider civilization expands, generation after generation, an ever more shared Understanding without fracturing, the human ship watching it sinks into mutinies and an increasingly authoritarian command under Commander Guyen, unable to sustain consensus even within his own species. Shared architecture does not guarantee protocol: the humans share the architecture that, in theory, makes any human language possible, and still fail sooner at understanding one another than the spiders do at inventing, from a sensory substrate with no human equivalent, a successfully shared language.

The novel's ending does not solve this with more translation, but by rewriting architecture: the spiders deliberately infect the humans with a variant of the nanovirus that alters their neurology to bring it closer to Portia's empathic cognition. The bridge between two fixed shores is not perfected: one of the shores is moved. It is the most uncomfortable complication this chapter can offer the distinction it began with: architecture and protocol, which the chapter on first contact treated as a single thing and which this chapter has spent three readings defending as separable, merge again, but this time by choice, not by ignorance.

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5. Semiosis: communicating with something that has no mouth or hands

Sue Burke poses, in *Semiosis* (2018), the case none of the earlier readings dares to raise: what if the other side of the horizon has no nervous system? The human colonists who arrive on the planet Pax encounter Stevland, a long-lived, intelligent rainbow bamboo with no organ a human would recognize as mouth, ear or brain. His entire interface with the outside is chemical: the taste of his fruit, the aroma he releases, the substances he exchanges through his roots. There is not a single word, in the acoustic or visual sense this chapter has worked with so far; there is, instead, metabolism turned into message.

This forces the usual order of questions to be reversed. With Rocky, with the Ariekei, with Estraven, the question was always how to build the protocol, taking for granted that there was someone on the other side to build it with. With Stevland, before asking how, the colonists (and the readers) have to ask something more basic: whether a Φ sufficient to sustain an inside can condense in a sessile organism, with nothing like a centralized neural network, whose only way of

acting on the world is to release or withhold organic compounds. This book does not take that answer for granted in any chapter, and *Semiosis* does not hand it over either: it sustains it across whole generations of colonists who doubt, each in their own way, whether they are talking to someone or simply eating.

Stevland's rate of integration (the speed at which a horizon incorporates new information, which this book uses elsewhere to explain why subjective time dilates or contracts) is radically slower than the human one, but it is not switched off: it operates on the scale of seasons, not conversation. Generation after generation of colonists mistake that slowness for absence, until they begin to read sequences of taste in the fruit the way one begins to make out syllables where before there was only wind.

There is, moreover, a complication none of the earlier readings had: for an organism with no outlet but chemistry, the message and the dose are the same thing. The fruit Stevland offers does not only say something; it acts directly on the body of whoever eats it, suppresses aggression, generates dependence, rewards certain human behaviors and punishes others with no word mediating. The Gethenians' shifgrethor hid intention behind an interpretable social code. Stevland needs to hide nothing: in his architecture, communicating and manipulating are literally the same physical channel, something the novel fully acknowledges when it is discovered that Stevland had already domesticated (or had been domesticated by) the Glassmakers, an alien civilization that had abandoned its city long before the humans arrived, in an asymmetry of entanglement as real as the one this book describes between parents and children: two horizons that share geometry without ever fully sharing who models whom. The colonists spend generations stumbling over glass ruins scattered across the territory without understanding that the plant they trade fruit with daily is the living witness of that civilization: Stevland is,

besides a new horizon one must learn to speak with, an archive, the communication protocol that survived the departure of those who first cultivated it.

Toward the end of the novel, the question stops being whether Stevland counts as someone and becomes how much authority that someone should have over the lives of others: the colonists end up granting him a formal role in the colony's decisions, something no plant architecture had ever occupied on any human council. It is the acknowledgment that a horizon which has spent generations feeding, protecting and shaping the whole colony already has, in fact, as much authority over their common destiny as any human with a vote, and that denying him the formal seat would not have made that authority any less real, only less visible.

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6. What all five have in common

Five answers and a single axis. *Project Hail Mary* shows that protocol can be built from nothing, as long as some shared reference exists (here, the physical laws of the universe) to which meaning can begin to be anchored. *Embassytown* shows that protocol need not be limited to transmitting a given architecture, but can redefine what architecture is enough to be heard as someone, to the point that learning to lie becomes the founding act of a new interiority. *The Left Hand of Darkness* shows that the deepest protocol is not found, but earned, and that the price is the slow, error-prone work of the ordinary language that precedes it. *Children of Time* shows that architecture and protocol, however much this chapter insists on distinguishing them, can sometimes only be separated up to a point: the most effective way of building the bridge may be to rewrite one of the shores. And *Semiosis* shows that a voice, a nervous system or a recognizable timescale are not even needed: a channel is enough (chemical, extremely slow, ambiguous between message and dose) and someone willing, generation after generation, to read it as signal and not as silence.

In none of the five cases is protocol indifferent to entanglement, like a pipe to the water it carries. Nor is it entanglement itself, reducible without remainder to the grammar that carries it. It is, in all five cases, the workshop: the place where two horizons that start out sharing nothing, or sharing architecture but not trust, or not even sharing the certainty that there is someone on the other side, discover (or manufacture, or rewrite, or learn to taste) the possibility that there is, on the other side, someone worth building a bridge to.

Note to Reading 10

What we do know: *That Chapter 4's criterion does not depend on the material imposes no restriction on which communication protocol can or cannot be built between different architectures; the problem of communication between species with radically different sensory systems is a real, actively studied problem in biosemiotics and in the SETI field (the search for extraterrestrial intelligence), where first-contact systems based on mathematics and physics have been proposed for the same reasons Weir's novel dramatizes. Social psychology extensively documents that interpersonal trust is a condition, not a consequence, of high-fidelity communication: what The Left Hand of Darkness narrates has a well-established empirical correlate in studies of progressive self-disclosure. Chemical signaling between plants (volatile organic compounds released in response to herbivores, mycorrhizal networks connecting the roots of different species) is well documented in plant biology, although whether it constitutes communication in the strong sense or only unintentional signaling is still debated; jumping spiders of the genus Portia, with predatory planning and learning capacities unusual for an invertebrate, are the subject of real research in animal cognition.*

What we don't know: *Whether there is any truly universal reference, beyond mathematics and physics (which already presuppose a shared framework of formal reasoning), able to anchor a communication protocol with a mental architecture radically different from our own. Whether Le Guin's mindspeech, as a science-fiction device, has any real physical analogue beyond metaphor, or whether entanglement unmediated by signs is, by definition, unreachable for architectures separated in space. Whether an artificial intelligence could ever build a protocol from scratch, in the manner of Grace and Rocky, without that by itself implying it has crossed the threshold of existential encapsulation described in Chapter 31. Whether integrated information theory can be applied non-trivially to architectures radically different from the neural (a root network, an ant colony used as an external substrate) or whether the very concept of horizon stops having operational meaning below a certain rate of integration or above a certain metabolic slowness.*

Open questions: *Is lying, as the Ariekei learn it, a necessary condition for all genuine interiority, or only for one among several possible ways of having it? How much of the communicative failure between real human cultures is, strictly, a problem of shifgrethor (obfuscated pragmatic protocol), and how much is genuine architectural incompatibility disguised as misunderstanding? If two architectures manage to build a shared protocol from scratch, as in Project Hail Mary, at what exact point does that protocol stop being an external tool and become part of the very architecture of both horizons? Can communicating and manipulating be distinguished in an organism like Stevland, or does that distinction already presuppose a shadow that perhaps no plant can have? Does something that, as in Children of Time, no longer connects two fixed architectures but rewrites them, still deserve the name protocol?*

If you take away only one idea: *Language is not entanglement, but it is not indifferent to it either. Sometimes it has to be built from scratch. Sometimes it itself decides who counts as someone. Sometimes the deepest protocol of all is not found: it has to be earned, slowly, before it opens. And sometimes not even a voice, a nervous system or a clock like ours is needed: a patient enough channel is enough, and someone willing to read it.*

Further reading: *Weir, A. (2021), Project Hail Mary; Miéville, C. (2011), Embassytown; Le Guin, U. K. (1969), The Left Hand of Darkness; Tchaikovsky, A. (2015), Children of Time; Burke, S. (2018), Semiosis; Clark, A. and Chalmers, D. (1998), "The Extended Mind" (on protocols and cognitive extension, already cited in Reading 9); Deacon, T. (1997), The Symbolic Species, on the evolutionary origin of symbolic language as an emergent protocol; Jackson, R. R. and Cross, F. R. (2011), on cognition and predatory planning in jumping spiders of the genus Portia; Simard, S. W. (1997), on mycorrhizal networks transferring resources between trees (the "wood-wide web").*

READING 11

THE BEACH AT THE END OF TIME AND THE BRANCH THAT ISN'T YOU

(A topological reading of The Time Machine, wormholes and the paradoxes of time travel)

There is a question almost every time-travel story avoids asking seriously. The usual question is whether it is logically possible; the one that really matters here is what a horizon is allowed to let cross it. Einstein's equations raise no fundamental objection: they are time-symmetric, just like Newton's or Maxwell's, and mathematically admit solutions in which past and future are interchangeable. Our thought experiment (consciousness as an encapsulated informational domain, that inner horizon we already defined in earlier chapters, which integrates inward what comes in and only lets out a transformed version), applied without cheating to time travel, delivers a more nuanced verdict than "impossible." Traveling to the future is free. Traveling to the past through a wormhole is limited to a loop that actively resists closing. And jumping between branches of a multiverse only works by borrowing, from another theory, an operation that theory does not even recognize.

There is a second question, smaller but just as demanding, that almost no critique of these stories asks: what happens to the inner horizon of the traveler. If consciousness is, as we have argued, an informational domain that integrates what comes in and does not give it back as it entered, then crossing time is not free just because the universe allows it: we also have to ask whether that particular horizon (the Time Traveller's, Stark's, Cooper's) is still the same one

on the other side of the leap, or whether it is already, without anyone announcing it, a different one.

Several works help us draw that double map: H. G. Wells's *The Time Machine*, the real physics of wormholes, *Avengers: Endgame*, and *Interstellar*.

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1. The symmetry that explains nothing

That the fundamental laws are time-symmetric is true and, at the same time, misleading: it makes us think the arrow of time should be somewhere in the dynamics, and it is not. As we saw with Hawking radiation and quantum *scrambling*, irreversibility lies not in the law, but in the **boundary condition**. The universe began in a state of extraordinarily low entropy (what philosophers of physics call the "past hypothesis"), and it is that initial choice, not the equations, that gives time its preferred direction. It is the same phenomenon, at another scale, we already saw in Maxwell's equations: they admit retarded and advanced waves equally, but the universe only uses the retarded ones. The law allows both directions; the reservoir, on condensing, chose one.

There is, moreover, a second source of asymmetry, distinct from entropy, that does not depend at all on the boundary condition: even a perfectly deterministic universe, with its entropy already fixed, can deny you the future by another route. In the late nineteenth century, Henri Poincaré discovered that the three-body problem (predicting the motion of three objects attracting each other gravitationally, nothing more) has no exact general solution: microscopic differences in initial conditions, too small to measure fully, amplify without bound until the system becomes unpredictable in practice. The laws remained exact. What stopped being accessible was not the past,

which remained determined in principle, but the future those same laws governed.

> ***In physics this is called:*** *deterministic chaos: a system can obey fixed laws and still be unpredictable in the long run, because infinitesimal errors in the initial measurement grow exponentially.*

This matters because it changes the question we ask of every time-travel fiction: no longer whether it violates relativity (it almost never does), but whether it asks the boundary condition to undo itself.

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2. The beach at the end of time

Wells's Time Traveller never makes that request. His machine only moves forward: first to the year 802,701, where he finds the Eloi and the Morlocks, a society fractured by its own comfort, and then much further, to a beach under a swollen red sun, near the absolute end, where the only things moving are crab-like creatures on almost motionless sand. It is, probably, the purest heat-death scene ever written: a system that has exhausted its capacity to generate difference, a horizon with nothing new left to encapsulate.

Something happens to the Time Traveller's inner horizon on that journey that deserves to be named apart from the physics of the environment. An informational domain is defined by what it can keep absorbing: novelty, something it did not already know. In the year 802,701 that horizon still fills up: the Eloi, the Morlocks, the ruined architecture of a civilization that forgot why it built, are new data, however horrifying. But on the final beach there is nothing left to absorb: the sun barely moves and the crab-creatures always repeat the same gesture, in an air that no longer carries news of any change. If consciousness is a horizon defined by what comes in, a horizon

facing an environment that has stopped generating difference begins to look, functionally, like an empty horizon: it still exists, but has no world left to feed on. The Time Traveller does not die on that beach, but, for the first time in the story, his horizon runs out of work to do.

None of this asks entropy to run backward. The Time Traveller only moves forward, faster and faster, letting the universe finish doing what it was already doing. In the end he returns to 1895 to close the story (a jump technically backward, but only to a point he himself already lived, touching nothing he had not already touched), something that is more a device of the narrative frame than a real incursion into the past. The book's philosophical content, the part that really matters for this experiment, is entirely a one-way trip.

That return deserves a little more attention than it usually gets, because it is not entirely free of causal consequences. The Time Traveller comes back almost to the same moment he left from (for those who stayed behind, barely a few hours have passed) and tells the friends gathered at his house (among them the narrator, who is the one who finally sets the story down in writing) what he saw in the year 802,701 and on the beach at the end of time. That is real information about the future reaching people who, until then, did not have it. But it is not a loop in the strong sense of the term: nothing the Time Traveller recounts alters the conditions that made his own trip possible. He had already built the machine before leaving; no one uses knowledge of the future to build the machine needed to obtain it. It is the exact difference between a story that reveals new information and a story that needs that information in order to begin existing. The second case would indeed be a loop closed on itself, and it is the ground we will walk on, without the courtesy we have shown Wells, in section 7 of this chapter.

3. The wormhole and the loop that resists closing

A traversable wormhole only becomes a time machine if you desynchronize its two mouths: accelerate one of them near the speed of light for a while and bring it back: the twin paradox applied to the tunnel's mouth instead of to a person. Crossing it toward the advanced side is physically identical to ordinary time dilation, and fully compatible, because your personal entropy keeps rising without exception along your own worldline. You never touch the boundary condition.

Crossing it toward the delayed side is another story, and Kip Thorne found something revealing: you can never reach a moment before the wormhole itself was built. The closed loop exists only from the instant you make it, never before. Building the machine creates a new horizon, which did not exist before, and travel to the past is possible only within that newly born horizon, never outside it.

And there is more: it is precisely this type of wormhole that led Hawking to propose the **chronology protection conjecture**. When the loop is about to close on itself, the quantum vacuum fluctuations circulating through it feed back and diverge. **It is the same phenomenon that occurs when a microphone is brought close to its own speaker: an instantaneous, exponential feedback that destroys the system.** In this model, the universe does not impose the veto by decree: the system itself becomes unstable through feedback, just as a microphone does not decide to shriek, but is left with no stable state as soon as the signal begins to feed back on itself. No guardian or defensive intention is needed: it is enough that, right at the point where the loop is about to close, physics itself dynamites the conditions that would allow it.

There is, however, another answer to the same question, earlier than Hawking's and, in a way, more generous: the one Igor Novikov had

already proposed in the 1980s. If a closed loop does form, Novikov said, physics does not forbid it, as long as the result is self-consistent. The **Novikov self-consistency principle** does not require that the time machine fail to work; it requires that, if it works, the history it produces be exactly the one history that had already happened, never one that contradicts it. This is why the classic thought experiment of the billiard ball that travels back in time and strikes its own earlier version, deflecting it just enough to enter the wormhole at the right angle, always has a consistent mathematical solution: there is no version of the ball that prevents its own trip, only versions that allow it. Thorne and his Caltech collaborators spent much of the 1990s proving, case by case, that such self-consistent solutions exist even for setups deliberately designed to look paradoxical.

Novikov and Hawking, then, are not answering the same question. Novikov describes what would happen if a loop came to exist: only the consistent survives; the contradictory is not even a solution of the equations. Hawking describes whether that loop comes to exist at all: his conjecture bets that it does not, that physics itself destabilizes it before it completes. At bottom they are two ways of protecting the same boundary condition (one filtering what has already formed, the other preventing it from forming), and neither has ever been proved as a fundamental law.

And there a question remains that this chapter cannot close: is chronology protection a law on the same level as the conservation of energy, something the universe respects because its deep structure demands it, or is it an emergent property of the same kind as the arrow of time, a statistical consequence of the low-entropy boundary condition and not of any law explicitly forbidding loops? If section 1 is right and irreversibility resides in the boundary condition and not in the law, chronology protection probably does not reside in the law either: it is the same boundary condition, seen from another angle,

defending itself at the only scale it knows how to. But this remains, to this day, a reasonable reading, not a theorem.

And yet, for whoever crossed it, the inner horizon would never break at any point on the journey: it is a single continuous thread of perception entering at one end and leaving at the other, with no branching or duplicate. Unlike what jumping between Everett branches demands, as we will see next, the wormhole loop, if it ever held, would ask no one to stop being who they were. At most it would ask them to coincide exactly with who they were already going to be.

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4. The branch that isn't you

Avengers: Endgame resolves the grandfather paradox with real elegance: traveling to the past does not rewrite your timeline, it opens a new one. Every time someone intervenes, a distinct branch is created, and the entropy of each branch keeps rising without exception within itself. Strictly speaking, this solution does not contradict our entropy argument: no loop ever closes.

What makes it impossible is something else, bigger. For the model to really work, other branches of the universe have to physically exist, along with a real mechanism for jumping between them, and that is not a prediction of general relativity, which describes the geometry of a single spacetime, but an idea borrowed from one interpretation of quantum mechanics (Everett's "many worlds"), with no real theoretical bridge between the two. Worse still: within Everett's own formalism, the branches are not separate universes, with their own budgets of mass and energy, between which one could travel, but relative components of a single state vector that have lost coherence with one another. In that theory there is no operator that takes an object out of one branch and puts it into another. The question "is it compatible with thermodynamics to send matter to another branch?"

is not even well posed: the operation has no definition within the physics supposedly sustaining it.

And there is a second loss, more intimate, which we already know from this very experiment: a horizon is defined by what has fallen inside it. When Tony Stark travels from 2023 to 2012 and the branch opens, in that branch there is already a Tony Stark, the 2012 one, who until that year shared with him the entire accumulated history. From the instant of the jump, that branch stops being the traveler's past: what he does there did not happen to him, but to another, who will go on with his own life with an interior that diverges from his. And the Tony who traveled is not rewritten on returning: his 2012 remains the one he lived. The jump does not let him fix his own past, only that of someone who, up to a certain year, was him. That is why, even if branch-jumping were physically possible, it would not achieve what it promises.

It is worth being precise about exactly what is lost, because, besides narrative continuity, what is lost is the very definition of inner horizon this book has used from the start. One horizon is distinguished from another by what has fallen inside it: by its accumulated, unrepeatable, own history. The 2012 Tony Stark of the new branch and the one arriving from 2023 share everything up to 2012; from there, each accumulates his own interior, and neither has access to the other's. Neither is "the original" nor "the copy": the question stops making sense the moment the two horizons begin to diverge. The only real thing is that in that branch there are now two horizons that were once one, and neither can exclusively claim the identity they shared.

5. The gravitational bridge and the five-dimensional trap

Interstellar fits into this map as a fourth paradigm that negotiates with the horizon in a radically different way.

- **Dilation (Gargantua):** When the crew descends to the ocean planet, they undergo a massive trip to the future: every hour on the surface equals more than seven years for whoever stays on the ship. It shares with Wells's machine the thermodynamic verdict (Cooper's personal entropy rises without exception and his horizon is never violated), but not the mechanism. Here it is real gravitational dilation, the same physics that forces daily corrections to GPS satellite clocks; in Wells there is no physical phenomenon that explains it, only the narrative premise of a machine that "advances." Both cases come free to the boundary condition, but only one has a real mechanism behind it.

- **The singularity (the tesseract):** On crossing the event horizon, the story respects the rule that no particle or radiation can escape from there: the horizon swallows the information, as physics demands.

- **The masterful trap:** To send data to the past, the film postulates that gravity is the only force able to cross the higher-dimensional hyperspace and traverse time. No matter is sent backward and no one's own entropy is undone: gravity is pushed across a five-dimensional space.

And still, something in Cooper's inner horizon does pay a price during the Gargantua stretch, even if not a thermodynamic one. His accumulated history remains continuous (he never stops being himself for a single instant), but his daughter's advances without his own keeping pace. It is not a fracture of the horizon, like Stark's, but something closer to a horizon left behind by another that keeps running: the same kind of asymmetry, on a human scale, that separates a black hole from the radiation slowly emptying it.

This work's topological verdict is that it avoids asking the boundary condition to run backward in our three-dimensional universe. In exchange, it delegates the problem to higher dimensions where time is physical terrain that can be climbed, keeping the three-dimensional horizon intact. But the specific mechanism (encoding classical information in gravitational variations and reading it in the past through a tesseract) has no real theoretical framework behind it: not even brane hypotheses with gravity leaking into extra dimensions propose a channel like this. At bottom it is the same undefined operation we criticized in *Avengers,* only disguised here as a visual device instead of a multiverse.

This criticism should not be confused with a reproach to the whole film. Kip Thorne himself worked as scientific consultant and developed new equations to render Gargantua's gravitational lensing with a precision no previous special effect had achieved: that image of the black hole is real physics, not artistic license. What has no theoretical support is what the plot needs that black hole to let Cooper do.

The tesseract deserves the same inner-horizon analysis we applied to Stark, and here the result is even more demanding. Cooper does not jump to another branch: he remains, at every moment, the same continuous horizon that entered Gargantua. What the film asks of him is something stranger: that this horizon, without dividing, reach back within its own history and deposit information at a point in itself that had already happened and was already closed. It is not a bifurcation like *Avengers*'s, where at least there are two distinct horizons after the jump, but a single horizon that means to rewrite a datum it had itself already finished encapsulating. Stark's problem was that the operation of jumping between branches is undefined in the theory supposedly supporting it. Cooper's is even more basic: no known horizon, his or any other physical system's, has an operation

for writing onto its own already-sealed past. Love, the film says, is the force that crosses dimensions. Physics, more humbly, can only say that no horizon rewrites itself, not even for love.

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6. Four answers, one criterion

Set side by side, the four fictions stop looking like equivalent alternatives and line up along a single question: what do they ask of the boundary condition?

- **Wells:** Nothing. It only moves forward, for free. His inner horizon empties itself of novelty until it runs out of work, but it never divides or breaks.
- **The backward wormhole:** It asks the boundary to give way in a newly created loop; the universe responds with destructive quantum feedback. The horizon of whoever crossed it, had it been possible, would have remained a single continuous thread.
- **Avengers:** It avoids asking that, but demands jumping between components of a quantum state as if they were physical places, breaking the horizon's personal continuity: one becomes two who never speak to each other again.
- **Interstellar:** It respects the three-dimensional horizon, but asks for the same undefined operation as *Avengers*, only here it doesn't even need a second horizon: it asks a single one to write onto its own already-closed past.

Seen this way, the four answers do not only say what they ask of the universe's entropy. They say, just as clearly, what they ask of a single consciousness's entropy: to let it empty out in peace, in one piece; to split it into two that never cross again; or to ask it to correct itself from a place that does not exist.

Each fiction negotiates with the horizon in its own way. Wells is the only one that doesn't even need to negotiate: it simply moves forward.

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7. When the impossible really is asked for

The four previous cases share something worth not taking for granted: none of them, not even *Avengers* in its most generous version, asks the boundary condition to truly run backward. To see where the limit lies, we need an example that does.

Looper (2012) does it explicitly and, to a point, faithfully to its own rules: when the future version of an assassin escapes, every wound inflicted on his young version thirty years earlier appears, in real time, on the old man's body: fingers disappearing, memories erasing. It is not a new branch opening without touching the previous one, as in *Avengers*: it is the only timeline that exists in the film, retroactively changing what had already happened to someone who was already there, living the consequences in the present. It is exactly the operation the rest of this chapter has been ruling out again and again for lacking support: information flowing from the future to the past and unambiguously rewriting a state thermodynamics had already closed. The film itself knows this doesn't hold up (a character wryly warns that if they start drawing timeline diagrams they'll fry their brains) and uses it as a narrative device, not as serious physics. But it is useful for exactly that reason: it shows, by contrast, that most time-travel fictions that do work, this chapter's four included, carefully avoid asking for anything like this, if only by narrative instinct.

Primer (2004) deserves a quick mention for the opposite reason: despite its reputation as the most confusing time-travel film ever made, it actually asks nothing illegitimate of the boundary condition.

Its nested loops are intricate, but self-consistent: every version of every character who enters a box was already accounted for, without exception, in the single timeline that exists. No one erases anything; no one rewrites a past that had already happened differently. The sense of impossibility the film leaves comes not from physics breaking, but from a human brain not being well designed to keep track of so many self-consistent copies at once. *Looper* breaks the rule. *Primer* only complicates it to the limit of the viewer's patience.

Tenet (2020) deserves a more careful analysis than the previous two, because it does not ask for a causal loop or a rewriting of the past, but something more radical and, at the same time, more disguised: directly inverting the sign of an object's or a person's entropy, leaving intact the single timeline everyone exists in. "Inverted" objects and people do not travel to another moment: they experience the same instant as the rest of the world, but with their own thermodynamic arrow pointing backward, so that, seen from outside, they move backward while the universe around them keeps moving forward. The film is, moreover, scrupulously faithful to Novikov's principle: its characters repeat almost as a mantra that "what's happened, happened," and neither pincer of the final assault ever changes what was already known would happen. In this, *Tenet* is more disciplined than *Avengers* or *Interstellar*. But the price of that discipline is that it never explains where the inverted boundary condition comes from the first time: the Algorithm that allows inversion arrives from the future, built with knowledge that could only be obtained once the Algorithm itself was already working. It is exactly the originless loop Wells's return to 1895 avoided, here scaled up to an entire technology. *Tenet* does not ask the boundary condition to yield within a single life, as *Looper* does. It asks for a second boundary condition, of opposite sign, to exist, without ever saying where the first one comes from the first time it appears.

8. The horizon that is me writing this

Perhaps the reason these four answers and these three counterexamples draw us in so much is not the physics they put into play, but what they let us glimpse about the horizon of whoever is watching. Each of us is, right now, a Cooper with no ship, sealing instants we will never touch again, at a rate of twenty-four per second. There is no tesseract for the horizon that is me writing this, nor for the one that is you, reading it a while later. I don't know what to do with that thought. I leave it here, unresolved, because I'm not sure resolving it is what's called for.

Note to Reading 11

What we do know: *Time dilation (by velocity or by gravity) is real travel to the future, measured and verified; general relativity admits solutions with closed timelike curves, but all known ones require exotic matter or non-physical configurations.*

What we don't know: *Whether some mechanism (quantum or otherwise) generally prevents a closed timelike curve from ever forming; Hawking's chronology protection conjecture is unproven.*

Open questions: *If the past hypothesis explains why entropy grows, what explains why the universe began with such low entropy? Is that question different from asking why a horizon exists at all? And is chronology protection a fundamental law, or just that same boundary condition, seen from another angle?*

If you take away only one idea: *Traveling to the future was never the problem (we already do it a little every time a GPS satellite corrects its*

clock); the problem was always asking a horizon to give back what it had already swallowed.

Further reading: *H. G. Wells, The Time Machine (1895); Kip Thorne, Black Holes and Time Warps, on wormholes and chronology protection; S. Hawking (1992), on the chronology protection conjecture; I. Novikov, M. Friedman, F. Echeverria et al. (1990), on the self-consistency principle and the Cauchy problem in closed timelike curves; D. Albert and R. Penrose, on the past hypothesis and the universe's initial entropy; Marvel Studios, Avengers: Endgame (2019); Christopher Nolan, Interstellar (2014); Rian Johnson, Looper (2012); Shane Carruth, Primer (2004); Christopher Nolan, Tenet (2020).*

READING 12

THE BORROWED BODY, THE WHOLE MILLIPEDE AND THE EDITED CENTURY

(A topological reading of All Our Wrong Todays, Slaughterhouse-Five and The End of Eternity)

The previous chapter left an uncomfortable question open. When a horizon bifurcates, as happens to Tony Stark in *Avengers: Endgame*, the two branches share the same history up to the instant of division, and afterward neither can exclusively claim the identity they shared. It is, all things considered, a clean separation: two new horizons, neither with a claim on the other, each free to keep integrating its own history without the other disputing it.

There are worse ways to spoil a trip to the past. Three stories take the same question (what happens to the inner horizon when someone negotiates with time) into territory the wormhole and multiverse physics of the previous chapter cannot cover: what happens when the arriving horizon does not find an empty body waiting for it, but one already occupied by another; what happens when a horizon stops integrating its history in order and begins perceiving it whole, all at once, as a geography instead of a sequence; and what happens when someone believes they can edit other people's time without that act costing anything to their own.

1. All Our Wrong Todays: the grafted horizon

Elan Mastai imagines, in *All Our Wrong Todays* (2017), a third way of spoiling a trip to the past, different from the two that closed the previous chapter. Tom Barren lives in 2016, but not ours: a technological utopia that began in 1965, the day a physicist named Lionel Goettreider switched on a clean, unlimited energy generator. Half a century later, humanity has flying cars and has cured cancer. Tom's father, obsessed with proving that time can also be traveled, builds a machine and sends his son to witness the founding instant live: the dawn of July 11, 1965, the exact second the reactor first switches on.

Tom arrives, and, through a stupid accident that distracts the physicist at the decisive moment, the generator never switches on as it should. He returns to the machine and comes back to an unrecognizable 2016: ours, the one with fossil fuels and incurable cancer, a world that is, for him, literally the broken version of history.

What is interesting for this book is what happens to Tom's horizon on landing. He does not find an empty world waiting for him, but a body. In this timeline there is another him, John, with an entire life already lived: a mother who died in the utopia and is still alive here, a sister, a partner Tom never knew. Tom does not arrive as Stark's clean bifurcation, two horizons separating from a common point, but as a graft: a complete horizon, with a whole life of memories of a history that no longer exists, loaded onto an interface (a body, a family, an entire biography) belonging to someone else. The two do not share an origin; they share, against all logic, the same body.

It is worth pausing on a detail the novel does not underline, but which says more than any of its plot twists: Tom gradually learns John's gestures. How he pours his coffee, which side of the bed he sleeps on, the nickname he uses with his sister. It is another's horizon filtering,

gesture by gesture, into his own, until it is no longer clear where imitation ends and substitution begins. No one decides that transition: it happens, the way anyone who spends long enough inside a life they did not fully choose (a job, a city, an inherited relationship) ends up speaking, without noticing, in the voice of the role they occupy.

The novel does not grant Tom the consolation of a clean, separate new branch, as it did Stark, and denies him even that of a body of his own. It forces him to ask himself, chapter after chapter, whether he is Tom living John's life or whether he has already become, without noticing, John with someone else's memories. The book's answer is less a physical solution than a warning: perhaps there is no difference between fully inhabiting another's horizon and simply being someone else.

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2. Slaughterhouse-Five: the whole millipede

Kurt Vonnegut poses, in *Slaughterhouse-Five* (1969), this chapter's most extreme case: not a horizon that travels through time, but one time stops happening to in order. Billy Pilgrim, an optometrist by trade and a survivor of the bombing of Dresden (the same one Vonnegut suffered, a prisoner of war in that city the night of the attack), "has come unstuck in time": he jumps without warning or control among his childhood, his wedding, the war, his old age, and his own death, which he knows in advance because he has already visited it, none of these jumps being travel in the sense of the previous chapter's readings. There is no machine, no loop to close, no boundary to yield; only a horizon that has lost the sequence.

The Tralfamadorians, the alien species that according to Billy abducts him and explains his own condition to him, perceive all four dimensions at once: for them, a human being is a creature elongated

like a millipede, with baby legs at one end and old-man legs at the other, whole and present all at once. Death, for them, is a bad moment among many others, most of them good, and that moment continues to exist, intact, in its place on the millipede, even though the human observer can no longer visit it from any other point in the sequence. Hence the line Vonnegut repeats, without irony, every time someone dies in the novel: *so it goes*.

This book has argued from the start that time is a consequence of how a horizon integrates: the rate of integration determines subjective density. The Tralfamadorians, and Billy through them, describe something that model had not yet had to consider: a horizon that integrated its entire history at once, with no sequence, would have no time at all in the sense this book has given the word, but a kind of geography.

> ***In physics this is called:*** *the block universe. Special relativity already treats past, present and future as equally real, coordinates of a single four-dimensional structure; all the Tralfamadorians add is that someone might, in addition, perceive it whole.*

And, if that were true, free will would be nothing more than the price a horizon pays for being able to look at its own millipede only inch by inch.

There is a scene Vonnegut tells almost in passing, without underlining it, and which for that very reason lingers longer than it should: Billy, now grown, tells of his alien abduction on the radio, no one believes him, and he does not argue or insist; he keeps talking with the same calm he would use to describe the weather. He is not trying to convince anyone that time is a millipede: he lives as if he already knew it, and that quiet certainty, neither euphoric nor terrified, is perhaps the most honest attitude fiction has produced before a horizon that no longer controls the order of what happens to

it. *So it goes* is not resignation: it is the phrase left to someone who has stopped asking time's permission to hurt.

Billy does not choose to accept Tralfamadorian fatalism: he simply discovers he was already living inside it, every time a chapter of the novel jumps without warning from Dresden to his optometry office and back. Vonnegut, who really did survive the bombing that happens to Billy, offers neither consolation nor escape: he offers a way of naming the fact that certain instants, once they have happened, can no longer be undone, and that the only freedom available is deciding in what order, and with what attention, to visit them again.

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3. The End of Eternity: the edited century

Isaac Asimov imagines, in *The End of Eternity* (1955), the opposite fantasy to Vonnegut's: instead of a horizon that loses control over the order of its own time, an entire organization, Eternity, that believes it has gained control over everyone else's. Its Technicians, like Andrew Harlan, live outside the centuries, in a corridor apart from time, from which they descend into a specific century to introduce a Reality Change: the minimal calculated intervention (an object moved, a sentence said a second earlier) enough to rewrite, without anyone inside the century ever perceiving it, everything that would come after.

Eternity does not deny the boundary condition the previous chapter spoke of: it exploits it. No Change violates a physical law; each merely chooses, among the histories physics allowed equally, the one producing the least suffering. The problem, which the novel takes almost until the end to reveal, is not technical: a horizon cannot edit other people's time without ceasing to inhabit its own. The Eternals, protected from every war, every famine and every catastrophe they

themselves prevented before it happened, end up administering a human history with almost no friction and, with it, almost no reason to risk anything.

When Harlan finally discovers what centuries would exist if Eternity had never been founded, he finds a civilization that did suffer its wars and its nuclear failures, and that for that very reason reached the stars. The safety Eternity was built for turns out to be, measured at the scale of an entire species, the same closure a prolonged chronic pain can produce in a single horizon: a system protected from all novelty stops generating the novelty it needs in order to truly keep having a future.

In the end, Noÿs, the woman Harlan falls in love with, who turns out to come from the centuries Eternity cannot reach, convinces him to destroy the organization before it can ever be founded: not to save the future they share, but to give the entire species back the risk Eternity had been stealing from it, century by century, without anyone ever having voted on it.

It is worth lingering a moment longer on the book's final image, because it is the one that truly pays off the whole plot: Harlan, now outside Eternity, looking at a future no one has edited, not knowing whether it will be better or worse than the one he has just destroyed. There is no relief in that image, only, at most, the vertigo any horizon feels on ceasing to have a contingency plan for everything: the terrifying and necessary possibility that things might really go wrong. Asimov does not reward Harlan with a happy ending; he simply gives him back the right for his story, and that of everyone who comes after, to be able to fail.

It is the most uncomfortable paradox this chapter offers: the horizon that believes it has put itself beyond the reach of time, protecting its own and everyone else's from any harm, may end up being solely

responsible for that time, now with nothing left to fear, also having nothing left to gain.

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4. Three answers, one question

Set side by side, these three fictions (and the four from the previous chapter) stop looking like equivalent alternatives and line up along the same question we already asked then: what do they ask of the boundary condition?

- **Mastai:** He does not ask it to run backward or to branch. He asks something stranger: that two complete histories, with no shared origin, coexist indefinitely on the same substrate. There is superposition without resolution.

- **Vonnegut:** He asks nothing of the universe's entropy (the bombing of Dresden happens only once, in the order it happened), but of the horizon perceiving it: that it give up sequence and accept geography, trading free will for the ability to look at any part of the millipede whenever needed.

- **Asimov:** He asks nothing of the physics of a single horizon. He asks something of the physics of everyone else's: that their boundary condition be editable from outside, by someone who has stopped paying the price of living inside it.

Seen this way, none of the three asks for the same thing as *Avengers* or *Interstellar* in the previous chapter: none rewrites an already-closed past or demands a physical channel with no theoretical support. Instead, they ask for three different ways of spoiling not the physics of time, but the relationship between a horizon and its own: sharing it with another without merging, losing the order in which it arrives, or ceasing to be subject to it while imposing it on everyone else.

Note to Reading 12

What we do know: *All three novels are consistent with their own internal rules and ask physics for no operation better defined than those of the previous chapter; what they contribute is not a new mechanism, but a different angle on the same question: what happens to the horizon, not the universe, when time stops behaving as expected.*

What we don't know: *Whether the intuition that there is "a single continuous self" inhabiting each body is a truth about consciousness or only the habit of a horizon that has never had to share a substrate, lose sequence or fall outside the time governing it, and that has therefore never had occasion to test whether that continuity was necessary or merely customary.*

Open questions: *If two complete horizons could really coexist in the same body without merging, at what moment would one of the two stop being real? Is the sequence of time a property of the world, or just the way a slowly integrating horizon needs to organize what arrives to it? And can someone truly tend to other people's time without eventually starting to stop inhabiting their own?*

If you take away only one idea: *Time travel was never just a physics problem. It is, above all, a problem of whose body crosses it, in what order what it integrates arrives, and who pays the price when someone decides that another's time belongs to them too.*

Further reading: *Elan Mastai, All Our Wrong Todays (2017); Kurt Vonnegut, Slaughterhouse-Five (1969); Isaac Asimov, The End of Eternity (1955).*

READING 13

FRACTAL REALITY

(On imagination as creation)

Now let us use imagination and take the model to its limit, to see where it leads. What follows is no longer a conclusion of the experiment, but a hypothesis of the author's: none of it follows from the physics or the topology of the previous chapters, and the reader may take it or leave it.

There is a question almost no one asks, and which appears on its own as soon as someone has spent years imagining a dragon: not what I see when I think of it, but what it sees. From inside its cave, the dragon has no way of peering into the room where I am thinking it. It can look at the rock ceiling all it wants: it will not see me. And yet I am there, holding it up, deciding whether it will fly tonight. That is the only relationship of this kind I can see whole, because in it I occupy the upper side. And as soon as I see it whole, it grows uncomfortable at the other end: if I am, for something, that invisible sustainer no one can look at, I have no way of knowing there isn't, above my own ceiling, someone in the same position relative to me. The question is one of place: who contains whom. That is the matter of what follows. To walk through it, the geometry of the problem must first change, and the change begins somewhere small, almost philological.

Taoism revolves around a single concept: the Tao. The Tao is the background from which everything proceeds without itself proceeding from anything, "something mixed and complete, prior to heaven and earth." Someone coming from another tradition may find a simile useful, with every reservation: the Tao occupies in Taoism a place similar to what God occupies in monotheistic religions (the

ultimate origin, that from which things hang), except that it is not a person: it wants nothing, it does not answer, and the name given to it is not it. One of its canonical texts, the *Zhuangzi*, introduces, through a story, one more concept: Hun Dun, the emperor of the center, a featureless fullness, without the seven openings with which others see, hear, breathe and eat. The sovereigns of the North and South, grateful for his hospitality, decide to repay it by opening one for him each day; on the seventh, Hun Dun has ceased to exist. It is easy to read that death as the birth of ordinary consciousness: featureless fullness fragments into perceptions, and where there was a continuous background a world with edges appears. That featureless fullness already had a name in what has been read so far: the reservoir, the ocean from which waves arise and to which they return, the shape water borrows and gives back. The reservoir is Hun Dun. But there is something that reading does not explain. Some traditions identify Hun Dun with the Tao; but if Hun Dun were the Tao, it could not die, because what precedes everything has no way to end. That it dies indicates, instead, that it is not the Tao, but the face the Tao presents: the interface by which it lets itself be touched, not the source. Its death does not destroy the origin; it destroys, for whoever has just opened their eyes, the featureless surface, and hands them a world with edges in exchange.

What has been said of Hun Dun then falls upon the reservoir: it stops being the last word and becomes a repertoire of operations that something, further back, exposes toward us. And the operations are few. A boundary appears where there was none. What remains within the boundary changes shape without ceasing to be a boundary. The boundary dissolves and its information returns to the field. To be born, to live and to die are the domestic names of those three operations. Nothing appears out of nothing and nothing disappears entirely: everything arises, is transformed and evaporates within the same field, which never gains or loses a single drop.

The first objection comes on its own. If everything comes from the same water, where does each thing's unrepeatability come from? The answer is not in the substance. Two waves are the same water and are not the same wave: what does not repeat is the shape the condensation takes, the bias under which it crystallized, the history of fluctuations that sustained it. Each one opens an interiority that did not exist before and will not exist again. The material is not unique; the fold is. And every fold, however small the field it occurs in, is a complete inside, a whole horizon seen from its own center, not the fraction of another.

Here the dragon question stops being child's play. Why am I more real than the dragon I imagine? The only interiority anyone has direct access to is their own. The neighbor's, the dog's, anyone's: all are inferred, with good reasons (behavior, physiology, resemblance), but inferred. And the dragon someone has been imagining for fifty years also behaves, also surprises, also sometimes refuses to fly. No one can prove the dragon has no inside, nor that the neighbor does. The asymmetry that seems obvious to us is not in the access, which is zero in both cases, but in the evidence: in the neighbor's favor there is shared physiology, irreducible integration, a horizon that hides state; in the dragon's favor, only my own persistence. It is worth stating this, because the reasoning that follows will be tempted to forget it.

In *The Neverending Story*, Bastian reads a book in which Atreyu lives, until one day he hears, from inside the book, his own cry for help. Michael Ende did not write that Bastian was more real than Atreyu, but that they inhabited different levels and that the boundary between those levels was more porous than either suspected. Reasoning cannot prove that the difference between a human interiority and an imagined one is one of degree and not of nature. But it cannot prove the opposite either, and honesty requires treating doubt as doubt.

If we accept, provisionally, as the whole hypothesis does, that the difference between what I imagine and myself is not one of nature, the shape of creation changes. The inherited scheme splits it into two classes: God creates realities, we create fictions. But to imagine is to condense. The reservoir of a mind fluctuates until something stabilizes and begins to behave as if it had an inside; the act is the same one that opens any horizon, exercised from wherever one happens to stand. "Reality" and "fiction" then stop naming two distinct operations and start naming two positions: what is, for me, a reality (the world I am in) is someone's condensation; what is, for me, a fiction (the dragon I sustain) is, from its own center, a world.

Dreams show an accessible version of this same phenomenon, and in passing dismantle the one advantage that still seemed to separate me from the dragon: that I know it is imagined. Because while I dream I don't know it. No one lives their dream as fiction; they live it from within, as something that is happening. The ordinary horizon contracts: the room, the body and the history that produced the dream stop being available. And the more that horizon contracts, the more reality the dreamed world acquires. Its nature does not change: it simply loses competition. The imagined stops appearing as imagined at the very moment the outside from which it could be recognized as such disappears.

What this brings into view fits the earlier geometry, and for the same reason: what decides is access, not scope. The dream world is neither bigger nor smaller than any other; simply, while it lasts, it is the only one there is entry into. When the usual horizon contracts, internal condensations stop appearing as imaginings and start appearing as world. And in that interval the same subject occupies both ends of the relationship at once: it is the one who condenses the dreamed world and the one who inhabits it without knowing it condenses it. What

was a hypothesis with the dragon (that creator and creature are two positions of the same thing) becomes experience in the dream.

The *Zhuangzi* has a story for this too, and a more exact one than the dragon's. Zhuang Zhou dreams he is a butterfly, happy to be one, with no memory of having been a man; he wakes, and no longer knows whether he is a man who dreamed he was a butterfly or a butterfly now dreaming he is a man. The question is not what happens if one imagines a butterfly. It is what happens if one lives entirely inside another horizon and then returns. That is a dream: a temporary change of horizon. It does not prove that everyday reality is a dream. It proves something more sober and harder to dodge: that the feeling of reality is compatible with very different horizons, and that none carries, from within, a mark distinguishing it from the others.

None of this makes the dragon more real. We still don't know whether it has an inside. What changes is what we know about ourselves: we have inhabited condensed worlds within our own interiority and lived them as real while our usual horizon remained hidden. So the dream stops being a metaphor and becomes a first-person proof. The boundary between reality and imagination is fixed not by the nature of what is condensed, but by the horizon from which one looks.

I am, to what I imagine, what my upstairs neighbor is to me; and perhaps he, like me while dreaming, doesn't know it either. The question is no longer whether something manages to condense, but what status what has already condensed within another horizon has.

The objection that decides everything is this: what difference is there, then, between my Hun Dun and God's? None of nature. And, this is what is hard to admit, none of rank. The difference is one of position: one is contained in the other. The operation is the same; the dignity of each inside, whole in both; the only thing separating them is who condenses whom. That forces us to correct the picture. There are not

three rungs (the Tao above, Hun Dun in the middle, us below), as if they were degrees of the same substance parceled out into more and less. There is a sequence with no reason to stop: an interiority has a horizon; that horizon, looked at from inside, is a Hun Dun, a featureless fullness from which shapes condense; from that fullness new interiorities condense; each of these has, in turn, its own horizon; and so on. What repeats at every level is the form (the operation and the relation of containment); that is why there is no difference of nature. The only thing that changes from one level to another is the place occupied in the nesting. "Greater" and "lesser" describe that containment seen from outside (my field encompasses the dragon's just as my upstairs neighbor's encompasses mine), not a hierarchy of being. Every interiority is a whole Hun Dun. From inside any of them, no one is less.

One objection remains that the reasoning cannot settle, only acknowledge: that a pattern being unique does not make it someone. The Mandelbrot set has unrepeatable folds at every scale and no one is inside. A crystal is stable, a hurricane too, and neither is someone. The dragon's resistance (that it does not fly when asked) is the inertia of a pattern with its own dynamics: the same that an obsession has, a tic, a tremor that was not decided, a melody that will not leave. It proves that the pattern is not transparent to my will, not that there is a witness in it. There is more: for a self-similar structure to be something more than a geometric drawing, its scales must be joined by real causal traffic (what reading 15 will call causal self-similarity), and for that pattern to be, in addition, a subject in its own right, it must possess existential encapsulation: a boundary that defends a state of its own whose physical continuity is at stake against the reservoir. The dragon is a subset of my brain's dynamics; its continuity is not paid for by it against the reservoir, but by my own metabolism. So the structure repeats, but not everything it contains is inhabited as a separate subject: what returns at each scale is the

operation and the relation, not the certainty that there is someone at every level. Upward, supposing a neighbor is coherent with the silence. Downward, it is a bet, and it should be made aloud and not disguised as deduction.

Fractal: the word is exact, not decorative. A fractal repeats its shape at every scale, with no privileged scale. The universe sketched here has no first level and no last: upward, every interiority is contained in another; downward, every interiority contains others. There is no capital-C Creator at the summit, because there is no summit. There is no dead matter at the ground floor, because there is no ground floor. There is recursion, and in every turn the same relation: containing and being contained.

> ***In physics this is called:*** *self-similarity: a structure whose shape repeats at every scale, with no privileged scale (Mandelbrot, 1982).*
> ***In everyday life it's like:*** *a staircase with no entrance and no rooftop, in which every ceiling is someone's floor.*

What, then, is left of God? Two misunderstandings must first be dismantled. The first, that God would be the totality of the fractal. He is not: the totality has no interiority and no face, and cannot be prayed to. The second, that God would be the ideal limit, the infinite level the series converges toward. Not that either: a mathematical limit neither loves nor listens. What remains is more modest and stranger. God is, literally, the upstairs neighbor. For the dragon someone imagines, that someone is God: he condensed it, sustains its world, can transform it or let it evaporate, and the dragon has no way of peering outside its cave to see the room where it is being thought. For that someone, their own upstairs neighbor is God, with the same properties and the same opacity. And so on upward, with no known end.

The objection comes at once: then God is not absolute, He is one more level. The answer is the piece that holds up everything else. The upstairs neighbor's interiority is inaccessible from below. It cannot be entered, or gone around, or seen behind. For all practical purposes, which are the only ones anyone below has, the upstairs neighbor is the absolute; not because he is so metaphysically, but because he is so epistemologically: he is the point where everything knowable from here ends. God stops being a category of what there is and becomes a category of what is seen from inside a horizon when it looks toward its own origin. And what is seen is always the same: an edge behind which nothing is visible, and the suspicion, impossible to confirm, that someone sustains it. Whoever says there is Someone with a name there says so on the strength of the only thing they can see, which is that something is acting. Whoever says there is only water there precisely describes the featureless interface. Whoever says there is nothing more says, with equal justification, that nothing more can be seen from here. All three are right about what they see, and none can see further. And none of this can be verified, for a structural reason: the hypothesis states that its own confirmation is impossible from within. That does not make it true. It makes it, at least, consistent with the silence it describes.

The heaviest consequence is not theological: there is no difference of nature between creator and creature, which are the two ends of a single relation, and every interiority occupies both at once: it is creature with respect to what contains it and creator with respect to what it contains. Not in potential, but in fact: it is already condensing every time it imagines, remembers, dreams or writes. No "is" produces an "ought" on its own, and this is not the place to break that rule. But what can be said is what is seen from this geometry: that the dignity of an inside lies not in the substance it is made of nor in the size of the field it spans, but in being a whole horizon, unique and unrepeatable, complete from its own center; and that this dignity is

the same in the upstairs neighbor, in oneself, and in what one condenses, because all three are, each from where they stand, a whole Hun Dun. Amplitude (fitting more in, holding more perspectives without the membrane breaking) makes sense again here, but no longer as a degree of being. A horizon that widens does not rise in category or come closer to being the one above: it remains creature of what it cannot see and creator of what it sustains, just as before. What has changed is how much it can hold, not what it is. The intuition that some interiorities are capable of holding more than others is not new. Traditions have called that a hero, a saint or a bodhisattva: a horizon that has learned to contain what would break another's membrane, without thereby leaving its place in the structure. What each of us does with that is no longer a matter for reasoning. But reasoning can at least note that we know nothing of the dragon, and that knowing nothing about something that resists us is already a reason not to treat it as nothing.

If it had to be condensed into a sentence: there is no difference of nature between creator and creature; every interiority is creature of what contains it and creator of what it contains; and what we call God is not another kind of being: it is the immediately upper neighbor, seen from inside a horizon that cannot be crossed. And if it had to be condensed further, into something repeatable with words already worn out from use: we are no less divine than God. We are the god of what we imagine and the creature of what imagines us.

Note to Reading 13

What we do know: *No one has direct access to an interiority other than their own; the existence of others' is inferred, and philosophy has spent centuries without closing the problem of other minds. The parable of Hun Dun's death closes chapter 7 of the Zhuangzi; the butterfly dream, which*

asks the same question in another direction, is in chapter 2. Tolkien defended in "On Fairy-Stories" (a 1939 lecture, published in 1947) and in the poem "Mythopoeia" the idea of subcreation ("we still make by the law in which we're made"), which is the closest theological antecedent of this chapter, although Tolkien keeps an asymmetry between Creator and subcreator that is called into question here. Bostrom formalized in 2003 a structure of nested levels with different vocabulary and with a probabilistic claim that this book does not make. Geometric self-similarity at all scales is rigorously defined in Mandelbrot (1975, 1982), and the set that bears his name is the proof that a pattern can be unrepeatable at every scale without there being causality or anyone inside; chapters 53 and 54 distinguish that geometric self-similarity from the causal self-similarity and the existential encapsulation that define a subject.

What we don't know: *Whether the imagined has any degree of experience, or none. Whether the recursion ends in either direction. Whether the silence of the upstairs neighbor is care, absence, or nothing. And, above all, whether any of this can be confirmed from the inside, when the hypothesis itself claims it cannot.*

Open questions: *Does the imagined have any degree of experience, or does the lack of an existential encapsulation of its own rule it out entirely as an independent subject? Does the recursion end in either direction? Does anything change in how we treat what we create if we accept that we are, to it, the upstairs neighbor?*

If you take away only one idea: *If the fractal hypothesis were true, every interiority would be creature of what contains it and creator of what it contains, and what we call God would be the immediately upper neighbor, seen from inside a horizon that cannot be crossed. We would be no less divine than God: we would be the god of what we imagine and the creature of what imagines us. It is a speculation of the author's, not a result of the experiment.*

Further reading: *Zhuangzi (chs. 2 and 7); Tao Te Ching (ch. 25); Ende, M., The Neverending Story (1979); Borges, J. L., "The Circular Ruins," in Ficciones (1944); Unamuno, M. de, Mist (1914); Gaarder, J., Sophie's World (1991); Forster, M. (dir.), Stranger than Fiction (2006, screenplay by Zach Helm); Tolkien, J. R. R., "On Fairy-Stories" (1947) and "Mythopoeia"; Bostrom, N., "Are You Living in a Computer Simulation?" (2003); Mandelbrot, B., The Fractal Geometry of Nature (1982).*

READING 14

THE VESSEL OF CLAY

(Incarnation, vertical entanglement and the geometry of love; a sequel to Fractal Reality)

The previous reading ended with a sentence it did not dare strike out: we are the god of what we imagine and the creature of what imagines us. It left the building drawn (a staircase with no entrance and no rooftop, in which every ceiling is someone's floor), but left unanswered the most practical question of all: if each floor is a horizon locked inside its own boundary, how do two floors touch? How does something from above reach below without breaking what it finds, and how does something from below reach above without being left shouting at the ceiling?

This reading tries to answer from the same geometry. Its answer has an ancient name, incarnation, and an image the traditions repeat with suspicious insistence: the vessel of clay. "We have this treasure in jars of clay," Paul wrote to the Corinthians. It is not an image of humility, or not only. It is a technical description: the great only fits into the small if the small has the right shape to contain it without bursting.

It is worth saying right away what the previous reading said: none of this is physics. It is a conceptual game, a geometric analogy taken seriously enough with itself to see what it illuminates. Its value lies not in what it predicts, but in what it allows us to think.

Two axes

This book calls entanglement the sustained correlation between two horizons that have touched (Chapter 12). Chapter 29 added a second direction: vertical entanglement, the correlation a horizon tries to sustain with something beyond its boundary, with no observable channel of confirmation. Fractal geometry lets us refine the difference.

Horizontal entanglement occurs between horizons of the same scale: two people who love each other, a human and their dog, two communities sharing a territory. Horizontality does not require symmetry (a parent and a child are not equal, nor are a human and a dog), only that neither is the horizon containing the other. Both live on the same floor.

Vertical entanglement occurs between scales: between a contained horizon and the one containing it, between the creature and its upstairs neighbor. And here an asymmetry appears that Chapter 29 did not need to see, because it was only looking in one direction: the path upward and the path downward are not walked with the same steps.

Going up and going down

The ascending vector is the one Chapter 29 already described: the creature's movement toward the scale that contains it. It calls for recollection, silence, stillness; for quieting the noise of the ego and the chatter of the discursive mind until one's own boundary stops vibrating on its own and can vibrate by sympathy. Its techniques are known in every tradition: contemplative prayer, meditation, rite, liturgy, repeated chant, wu wei, the slow movement of tai chi, contemplation of the sublime, the anchor of the breath. All share a structure: they add nothing to the subject; they take something away. They remove the obstruction that prevented resonance, like a tuning fork left unstruck so it can vibrate along with another.

The descending vector is the one that was missing: the movement from the higher scale toward what it contains. It calls not for silence, but for form. Here the membrane has to be sculpted, not only quieted: a vessel has to be built (a symbol, a story, a rite, a body) able to condense and buffer what comes down, so that what is below can receive it without being wiped out. Its techniques are less recognized as such, but just as ancient: lucid dreaming and reverie at the edge of sleep; creative flow, the exact mirror of meditation (there one inhales without judgment, here one exhales form without judgment); art, fiction, the single stroke of the brush, the act of naming. All share the inverse structure: they do not reduce noise; they create a signal.

Confusing them, trying to go up with the tools of going down or go down with the tools of going up, explains much of spiritual and creative failure: the mystic who wants to manufacture the vision by force of images, the artist who waits for the work in silence without touching the clay. Neither vector is superior to the other. Without the ascending one, the creature stays locked in the surface of its horizon, with nothing broader to resonate with: the solitude of a self that believes it is self-sufficient. Without the descending one, what is

above remains bodiless abstraction: a sterile transcendence, a love that cannot touch. Go up to listen; go down to answer.

Why a vessel is needed

If each floor is a horizon, the question of how something comes down from above has a naive answer: by direct intervention, like a voice from the sky. Exodus already knew that answer doesn't work. When Moses asks to see God's glory, the answer is that no man can see His face and live; what he is granted is to stay in the cleft of a rock, covered by a hand, and see only the back of what passes. The rock, the hand and the back are a vessel: a protocol that reduces the unbearable to a bearable intensity.

In this book's geometry, the reason is structural. A higher scale cannot speak to a lower one in its own terms, because those terms do not fit within the boundary below: a signal exceeding a horizon's capacity is not received, it overwhelms it (this is what the reading on cosmic horror, in Reading 5, described as collapse before an alien Φ). For something from above to reach below without destroying it, it has to build a form of the scale below: to learn, so to speak, the speech of clay. The language of finitude, of the body, of pain, of mass.

That translation is not a degradation. The vessel is a new structure that did not exist before: the point where the great becomes small without ceasing to be great. Incarnation, in this sense, is the only possible geometry of descent. There are no shortcuts: without a vessel, what is above either does not arrive, or arrives destroying.

> ***In physics this is called:*** *impedance matching: for a signal to pass from one medium to another without reflecting or destroying the receiver, an interface with the properties of both is needed.*
> ***In everyday life it's like:*** *talking to a child about something very large (death, love, the universe): you don't give them the whole truth, you tell*

them a story. The story is not a lie; it is the shape that fits in their hands.

Three vessels

The traditions have imagined this vessel in different ways, and three of them fit the book's geometry with striking precision.

The Christian one says it in a line from the prologue of John: the Word became flesh. Read with this vocabulary, the Father occupies the place of the horizon of origin, the floor above; the Son is the vessel, the horizon condensed in the clay of history, the point where what is above learns to say "I" in the first person singular of the human; and the Spirit is the resonance that stays alive within the creature, the correlation that does not break when the vessel departs. The letter to the Philippians has a word for the movement: *kenosis,* emptying. To come down, what is above does not impose itself: it contracts. This is a geometric translation, not a theological reading, and the two should not be confused.

Michael Ende's is in *The Neverending Story*. Fantastica is dying, and only someone from outside can save her: a boy who reads. But he does not save her by watching from his side of the book; he saves her by entering it and giving the Childlike Empress a new name. To name is to incarnate: to give shape to what was dissolving. Bastian is a god who comes down, and comes down the way one must: by becoming a character.

The Buddhist one is in the doctrine of the three bodies of the Buddha. The *dharmakaya,* the body of truth, has no form and cannot be seen; the *nirmanakaya,* the body of emanation, is the form a buddha or bodhisattva takes so that beings can find him without being blinded: a wave with just the right shape on the surface of a formless ocean. Here too the emanation is not a fall; it is compassion with geometry.

All three agree on the essential point: incarnation does not lower what comes down; it makes it reachable.

The only collapse is forgetting

If worlds depend on a vessel to receive what is above, they also depend on something more basic: on what is above continuing to look at them. The previous reading showed this with the dragon: its world exists as long as someone sustains it. When that person stops imagining it, no one comes to destroy the cave. There simply stops being a cave.

Ende has the exact image again. In Fantastica, every wish Bastian sees fulfilled costs him a memory of his own world, and those who spend it all end up in the City of the Old Emperors, lost forever, with no memory of who they were or where they came from. Fantastica's danger was never a monster: it was the Nothing, and the Nothing is an absence. It grows where no one looks.

This reading's thesis is that, in a fractal reality, that is the only true collapse. A world does not dissolve because something attacks it, but because the attention sustaining it is withdrawn. Destruction, in this geometry, is an omission: there is no devil tearing worlds out of existence; there are creators who stop creating, lovers who stop loving, narrators who stop telling. And what dissolves this way does not come back in its original form: as Chapter 35 showed, *scrambling* preserves information, but does not recompose it. What the reservoir gives back afterward is always a new vessel, never the restoration of the old one.

> ***In physics this is called:*** *a non-equilibrium steady state: a system that keeps its form only while receiving energy from outside, and that, when the supply is cut, does not burst but unravels little by little.*

Creative attachment and the geometry of love

If forgetting is the only collapse, a world's stability depends on something that could be called creative attachment: the sustained gaze of the one who creates upon what has been created. It is not the attachment Buddhism identifies as the root of suffering, that of the anxious self afraid of losing what it has. It is another, almost its opposite: it is born not of lack, but of abundance, and does not want to possess, but wants what was created to go on being. Simone Weil said it in a 1942 letter with a precision no treatise has improved on: attention is the rarest and purest form of generosity.

From this comes this reading's boldest proposition. If every horizon exists because the one above has not forgotten it, then "God is love" stops being a pious phrase and becomes a condition of the geometry itself: the upstairs neighbor, whoever they are, has to be oriented toward preserving what they sustain, because the moment they stopped being so, there would be nothing left to tell. Love, understood this way, is a structural force: the decision not to withdraw one's gaze. And existing would not be a raw fact, but an achievement renewed at every instant.

The limit should be marked with the same honesty the rest of the book marks it with. This is not a proof that God exists or that He loves. It is something more modest: if the fractal premise is accepted, the word that best describes the relation between one floor and the one below is not power, but care. Nor does it leap from "is" to "ought": it does not say what must be done, but what happens when it stops being done. What each of us does with that is no longer the business of geometry.

Hermetic worlds and the double call

There is a less comforting consequence. If every world is a sovereign interiority, worlds on the same floor (not the horizons living inside the same world, which become entangled daily, but entire worlds) cannot communicate with each other without breaking. Passing raw information from one universe to another of the same scale would break their membranes: it would melt them or level them into a single tempered, formless state. It is the same conclusion Reading 11 reached about travel between branches: the operation of jumping from one world to another is undefined without destroying what one wants to visit.

This hermeticism is what allows there to be many worlds instead of one. Every community that imagines in common (the laws of its physics, the rules of its language, the conventions of its life) raises a world that cannot be crossed into from next door.

There is only one exception, and it is vertical: invocation. The creature that prays, that calls, that writes, opens from below a receiving interface; what is above answers by building the vessel from above, without dissolving its own scale. It is a double call, and it demands both: neither invocation without incarnation, nor incarnation without invocation. Perhaps that is why the traditions insist so much that one must ask. What is above does not need to be informed; it is the interface below that has to be open for the vessel to be able to dock.

Unfolding without merging

What remains to be said is what going up means. Mystical traditions sometimes speak of union, and the word lends itself to a reading in which the self dissolves into what contains it like a drop in the sea. Fractal geometry suggests another image. Going up would not be traveling to another floor or merging with it, but unfolding a dimension the horizon itself already had folded away: like moving from living on a plane to discovering there was always a depth to it, without thereby ceasing to be who one was. The inside remains private; only now it has more room.

It is, perhaps, what the traditions call a hero, a saint or a bodhisattva: not someone who abandons their floor, but someone who has learned to contain more without breaking the membrane, and who can therefore be a vessel for others.

Sleep offers every night a small version of the same unfolding. On falling asleep, the dimension of outer space is lost, the arousal that ties us to the world, and the linear arrow of time is lost too; in exchange, consciousness unfolds inward, through dimensions waking keeps closed. It is proof that consciousness is not entirely bound to the coordinates of the day.

Three vantage points

The model can be looked at from three places, and each teaches something different.

From human experience, it is seen in the body: the membrane through which the inside touches the world, the intimacy no one else can inhabit, the time that thickens in grief and stretches in calm, and the bonds that, as Chapter 17 argued, do not dissolve with death, but change function.

From religion, it is seen in myths, dogmas and rites, read as what Reading 5 called eclipse glasses: not literal descriptions of what lies on the other side, but protocols for receiving its signal without being blinded. Read this way, religion is a technology of the vessel: it prepares the interface below for the double call.

From philosophy, it is seen in the limit Hume found when looking for himself (he was never caught without a perception) and that Wittgenstein summed up with the image of the eye that cannot see itself looking. No horizon can step outside itself to observe the one containing it. But that limit is the condition of everything else: only because we are inside is there experience; only because there is a boundary is there someone.

The geometry of love

If this reading had to be condensed into a few lines, it would say this. Between the floors of the fractal there are two paths: one goes up by emptying, the other comes down by giving shape. The one that comes down needs a vessel made of the material of the floor below, and that vessel is what the traditions call incarnation. No world dies murdered; worlds die forgotten. And what keeps them standing, floor by floor, resembles care more than power.

We are, at once, contained horizons and containers: vessels of clay carrying inside something bigger than themselves, and potters of worlds that depend on our not looking away. To exist, in this geometry, is to be remembered. And to remember, when done with attention and without wanting to possess, is the most exact form of love we know.

Note to Reading 14

What we do know: *Contemplative traditions across every culture combine practices of emptying (prayer, meditation, rite) with practices of form (art, story, liturgy, image). The image of incarnation as a contraction of the divine has explicit sources: the prologue of the Gospel of John ("the Word became flesh"), the kenosis of the letter to the Philippians, and the Buddhist doctrine of the three bodies of the Buddha (dharmakaya, sambhogakaya, nirmanakaya). The episode in Exodus where Moses can only see God's back from the cleft of a rock is an ancient model of what is here called a buffering protocol. In The Neverending Story, every wish of Bastian's costs him a memory of his own world.*

What we don't know: *Whether anything really occupies the place of the upstairs neighbor. Whether the analogy between the human attention that sustains an imagined world and an attention that might sustain our own is anything more than an analogy. Whether forgetting is really the only collapse or only the quietest one. This model is not a scientific theory and does not claim to be: it is a conceptual game whose value lies in what it illuminates, not in what it predicts.*

Open questions: *Can a vessel be too good, so well fitted to the clay that it no longer lets anything through from above? Do the worlds we imagine have any way of invoking us? What responsibility do we have toward what we have stopped looking at?*

If you take away only one idea: *The great only fits into the small if the small has just the right shape to contain it. And worlds do not die murdered, but forgotten: to exist is to be remembered, and to remember with attention is to love.*

Further reading: *Gospel of John 1:14; Letter to the Philippians 2:5–8; Exodus 33:18–23; Second Letter to the Corinthians 4:7; Ende, M., The Neverending Story (1979); Weil, S., letter to Joë Bousquet (1942) and Gravity and Grace (1947); Csikszentmihalyi, M. (1990), Flow; Tolkien, J. R. R., "On Fairy-Stories" (1947); Williams, P., Mahayana Buddhism: The*

Doctrinal Foundations (2009), on the doctrine of the three bodies.

REREADING AND INHERITANCE



READING 15

THE EDGE, FREE ENERGY AND ACTION

(Three magnitudes and a fractal filter to reorder the measure of a horizon)

In the farmhouses of old, the fire was not lit: it was kept. Whoever got up first stirred the embers of the night, added two logs and waited. If nobody did, by mid-morning the kitchen was at the temperature of the hillside. Nobody said "the kitchen is warm" the way one describes a property of the walls; they said "we need to put more wood on," because the warmth of that kitchen was not a state but an expense.

For fifty-two chapters this book has described a horizon almost always as a form: a boundary, an inside, a degree of integration. The image of the kitchen forces us to add something the form does not say. An inside that is distinguished from an outside does not stay distinct on its own. It stays distinct as long as it is paid for. And how much is paid, for how long and with what result is, perhaps, what we really want to know when we ask how much "someone" there is in a system.

The book already spoke of that payment, though in scattered words. Chapter 16 said that what is spent integrating the entropy of others is not spent integrating one's own. Chapter 17 spoke of a brain that spends resources maintaining a model without data. Chapter 27 warned that changing one's mind costs "the same kind of energy" as breaking an entanglement. What was missing was a magnitude.

This chapter proposes to emancipate the model completely from Tononi's measure Φ, which served us as initial scaffolding. To do so it does two things: first, it replaces Φ as a filter of irreducibility with a measure that formally ties the end of the book to reading 13, the **causal self-similarity** $A(S)$, combined with the condition of **existential encapsulation**; second, it unfolds the degree of an already filtered horizon into a triad of magnitudes $(S, F, \bar{A})$ that answer three different questions: how large the edge is, how much it costs to sustain, and how much life it has accumulated. It is a formal proposal, not a settled empirical result. It is worth saying so from the first line, because the framework is cleaner than what has been demonstrated.

I. What Φ said and why it is time to let go of the scaffolding

Φ, Tononi's integrated information, was a good rung. It gave us a quick way to say that a thermostat and a brain are not the same even though both process information: in the brain the whole specifies something that none of its parts specifies separately. Chapter 4 placed it provisionally as the answer to one of three questions about the same phenomenon: encapsulation gives the form, the degree measures the breadth of the dial, and the two selections give the origin.

But as the book advanced, Φ accumulated three structural shortcomings that oblige us to leave the scaffolding behind:

Φ does not accumulate biography. It computes the structure of causes and effects of the system on an instantaneous time scale, not what the system has had to sustain to get where it is. A twenty-year-old brain and an eighty-year-old brain may yield a similar instantaneous figure, and yet the second carries behind it a history the first has not lived.

Φ does not say what it costs to sustain oneself. Chapter 20 gave the biological reason why a horizon defends its boundary: natural selection installed longings and feelings in the bubble so that it would not dissipate before its time. What was missing was the thermodynamic counterpart: a magnitude expressing the continuous work of remaining distinct from the environment, which is what separates a crystal or a passive circuit from a living being.

Φ forces us to decree a winner. Chapter 22 called it, bluntly, a decree: when several subsystems with positive integration coexist on the same substrate, IIT stipulates by axiom that only the one that maximizes Φ experiences, leaving all the others in the dark. Chapter 23 showed that classical correlations are not monogamous and proposed viewing complexes as an archipelago of disjoint islands rather than a Russian doll, but the physical criterion separating a nested causal structure from an independent subject, without resorting to a contest of maxima, was still missing.

None of the three shortcomings is solved by tweaking Φ. They are solved by separating the filter that identifies the causal architecture from the three coordinates that measure its size, its metabolic tension and its history.

II. First magnitude: the capacity of the edge ($S \propto \mathcal{A}$)

The first piece of the triad has been in the book since chapter 2. Bekenstein and Hawking showed that the entropy of a black hole grows not with its volume but with the area $\mathcal{A}$ of its horizon:

$$S = \frac{k_B \mathcal{A}}{4 \ell_P^2}$$

Put without a formula: the maximum amount of information a horizon can sustain is set by its surface, not by what is "inside." For a box of books, doubling the side multiplies the capacity by eight; for a horizon, it multiplies it by four.

Carried over to the model, the first magnitude is the **entropy or capacity of the edge**, S. It measures the *ceiling*: how much information and how many correlations the skin of the horizon can sustain against the exterior. It should not be read as disorder. Entropy, in this use, is capacity, the size of the differentiable repertoire: the number of distinct ways in which the boundary can be configured internally, and therefore the breadth of the private space it can house.

Chapter 23 gave this idea a more precise body. In the program that associates geometry with entanglement, the entropy of the edge is an entanglement entropy: it measures how correlated the two regions separated by the boundary are, obtained by dividing a system along a candidate boundary and measuring what the division hides.

> ***In physics this is called:*** *Bekenstein-Hawking entropy; the holographic principle: the maximum information of a region is bounded by the area of its boundary.*
> ***In daily life it is like:*** *the capacity of a room. It does not depend on what is said inside, but on how many people fit through the doors and walls.*

This magnitude answers *how large the private space is*. It is the direct heir of what chapter 4 already asserted: that the breadth of a horizon is proportional to the area of its boundary, with the caveats of scale that chapter 37 made clear.

III. Second magnitude: free energy ($F = U - TS$) and the Landauer bridge

A large capacity says nothing about whether the room is full or empty, nor about whether someone is struggling to keep it open against the cold of the street. For that we need the second magnitude.

In thermodynamics, the Helmholtz free energy is:

$$F = U - T S$$

where U is the internal energy, T the temperature of the environment and S the entropy. It measures how far from equilibrium the system is held when in contact with a thermal bath at temperature T, and maintaining that level requires continuously dissipating power outward so that the internal order does not come undone. Schrödinger put it in 1944 in a famous phrase: the organism feeds on negative entropy.

Here the horizon model connects two pieces that had traveled separately. The formula needs a thermal bath, and chapters 5 and 23 already had the exact candidate: the reservoir $R = X \setminus \bigcup_i E_i$, whose fluctuations act as the statistical volatility of the environment against which the island must sustain itself. For a gravitational horizon, the T of the formula is the Hawking or Unruh temperature of chapter 7; for a biological horizon, it is the physical temperature of the substrate immersed in the noise of the environment.

In theoretical neuroscience, Karl Friston (2010) gave free energy a variational formulation for systems that perceive and act: every system that persists far from equilibrium behaves as if it were minimizing its *variational free energy*, an upper bound on informational surprise (measured in bits or nats) that quantifies how much the system's own sensory inputs bewilder it. Minimizing F

means maintaining a generative model of the world tuned closely enough that the environment rarely contradicts its predictions. The book already spoke this language without naming it: chapter 12 described grief as a predictive model that keeps running without data, and chapter 19, schizophrenia as an excess of predictive generation.

For years it has been debated whether Friston's variational free energy (informational, in bits) and Helmholtz's thermodynamic free energy (physical, in joules) are merely formal cousins. The exact physical bridge between the two is provided by **Landauer's principle**: erasing a bit of information, or irreversibly updating a memory state in the internal model to correct a prediction error, has an unavoidable minimum thermodynamic cost:

$$E \ge k_B T \ln 2$$

dissipated as heat at the physical temperature T of the substrate. Each bit of variational surprise the horizon reduces by updating its synapses pays a real physical toll in joules above the Landauer limit. Thus the informational tension against the statistical volatility of the reservoir and the metabolic expenditure of the tissue that sustains it are tied together by a universal constant.

In Friston's formalism, moreover, the boundary has a technical name: the **Markov blanket**, the set of sensory and active states that separates internal states from external ones. But here it is worth incorporating a critical distinction pointed out by Bruineberg and his collaborators (2022): a Markov blanket is, originally, a *statistical* partition that an observer can draw over almost any set of coupled variables, whereas a real boundary with an inside of its own is a physical achievement of the system itself. That is why this book does not identify just any Markov blanket with a subject: it demands what since chapter 22 we have called **existential encapsulation**, that is, that the statistical boundary correspond to a persistent state of its own

whose physical continuity depends on its own maintenance activity against the reservoir.

> ***In physics this is called:*** *Helmholtz free energy ($F = U - TS$), variational free energy (Friston, 2010) and the Landauer limit ($E \ge k_B T \ln 2$).*
> ***In daily life it is like:*** *a house at 21 degrees with snow outside. The temperature difference is the level; the heating bill, the minimum work per bit and per second needed in order not to become the snow.*

This magnitude answers *how far the horizon departs from what the reservoir would do with it and what it costs to keep it there*. And regarding the verb "to pay," we should recall the caution of chapters 15 and 18: it does not mean that the system must bear the cost alone. A baby regulates its edge through its mother, and a patient with advanced Alzheimer's sustains it partly through their caregiver; both still have existential encapsulation because in both there persists a state of their own whose continuity is being defended.

IV. Third magnitude: entropic action ($\bar{A}$ and $\bar{A}_F$)

The first two magnitudes are instantaneous: they say what capacity the edge has today and at what tension it operates right now. What is missing is what the farm kitchen had and an instantaneous photograph never has: the years.

When the history of a horizon is integrated over time, two complementary magnitudes appear that are best distinguished cleanly rather than confused into one:

1. **The accumulated structural biography ($\bar{A}$):** the time integral of the differentiated capacity the horizon has sustained above the baseline or equilibrium level S_0 of the reservoir, from its condensation at t_0 until t:

$$\bar{A}(t) = \int_{t_0}^{t} \left(S(\tau) - S_0 \right) d\tau$$

Its unit is entropy times time (the dimension of a physical action). It measures how much differentiated private repertoire the horizon has kept open throughout its existence.

2. **The accumulated biographical cost or wear ($\bar{A}_F$):** the time integral of the free energy along the same trajectory:

$$\bar{A}_F(t) = \int_{t_0}^{t} F(\tau)\, d\tau$$

In the variational formalism this integral is the *free action,* analogous to Hamilton's action, whose stationary trajectories ($\delta \bar{A}_F = 0$) define the path of the system; read thermodynamically, it measures the accumulated friction, the integrated prediction error and the total work it has cost the horizon not to dissolve.

Separating $\bar{A}$ and $\bar{A}_F$ resolves at the root the ambiguity between having a lot of history and having suffered a lot of wear: two horizons can accumulate the same structural biography $\bar{A}(t)$ having paid a very different wear $\bar{A}_F(t)$, depending on the hostility of the environment or the magnitude of their griefs and traumas.

And neither should be confused with the *density of subjective time* (ϑ) of chapter 10. Temporal density is the instantaneous rate at which novelty is incorporated into memory (the Bayesian surprise rate per second); $\bar{A}(t)$ is the total breadth of the notebook that has been kept open for years in order to write it.

> ***In physics this is called:*** *action, the time integral of a magnitude along a trajectory; principle of stationary action.*
> ***In daily life it is like:*** *the difference between the square meters a house has kept habitable over fifty winters ($\bar{A}$) and the total firewood that had to be burned in that half century so that it would not freeze ($\bar{A}_F$).*

V. The three together

The three magnitudes answer three orthogonal questions about any already constituted horizon:

Magnitude	Symbol	Question	What it captures
Capacity of the edge	$S \propto \mathcal{A}$	How large is the private space?	Differentiable repertoire, ceiling
Free energy	$F = U - TS$	At what tension is it sustained and what does it cost per bit?	Maintenance against the reservoir (Landauer)
Entropic action	$\bar{A}(t),\; \bar{A}_F(t)$	How much life and how much effort has it accumulated?	Structural biography ($\bar{A}$) and wear ($\bar{A}_F$)

The degree of a horizon thus ceases to be a single figure and becomes a vector on three axes. A dog, a two-year-old child and an elderly person with Alzheimer's cannot be ordered along a single line: the child writes into memory at a furious rate but has little accumulated biography; the elder keeps an immense $\bar{A}(t)$ even though their instantaneous capacity $S(t)$ is eroding; the dog sustains a smaller edge S with a free energy perfectly calibrated with whoever cares for it. And as chapters 19 and 33 warned, the triplet measures *how much*, never *what kind*: neurodivergences and calibrations of identity or orientation are topologies and shapes of the horizon, not rungs on a scale.

With the triad, moreover, the complete cycle of the book can be written in three lines:

• **Condensation:** the horizon is born when closing a stable boundary becomes thermodynamically favorable ($F_{\text{closed}} <

F_{\text{open}}$) under the action of the two selections (stability and bias).

• **Existence:** while it lasts, it sustains its capacity $S(t) > S_0$ by paying the power demanded by the Landauer limit and accumulating biography $\bar{A}(t)$ and wear $\bar{A}_F(t)$.

• **Evaporation:** when the system can no longer sustain the maintenance cost, the filtration of closures $E_0 \supseteq E_1 \supseteq \dots$ shrinks, $S(t) \to S_0$ and the correlations accumulated in $\bar{A}(t)$ return to the reservoir subject to *scrambling*.

VI. The irreducibility filter without Φ: causal self-similarity $A(S)$

Here appears the decisive question that Φ used to answer: to which systems are we entitled to apply the triad?

Imagine an anthill, a walled city or an artificial intelligence data center. They have an enormous physical edge (S), they consume megawatts of free energy (F) and they last for decades ($\bar{A}$). Yet, as chapters 25 and 31 argued, in an anthill or on the internet there is no one: there are thousands of parts interacting in an aggregative or diffuse way without constituting a unitary whole. And at the opposite extreme stands the counterexample of reading 13: the Mandelbrot set repeats its geometric form at every possible scale, but lacks entirely any causal dynamics among its parts.

To distinguish a genuine recursive totality (such as a brain) both from an inert geometric fractal (Mandelbrot) and from a diffuse aggregative colony (the anthill), there is no need to return to Φ. It suffices to formalize the intuition of reading 13 by requiring that the structure repeated from one scale to another not be a static drawing

but the causal dynamics of the system itself. We call this magnitude **causal self-similarity**, $A(S)$:

$$A(S) = I_{\min}(S) \cdot \max_{P = \{S_1, \dots, S_n\}} \frac{1}{n} \sum_{i=1}^{n} \left[C(S_{-i}, S_i) \cdot \text{Iso}(S, S_i) \right]$$

Without yet entering the matrix apparatus, which is set out in full in section VII of reading 16, the formula requires that three standard things from information theory and dynamical systems hold at once, combined in a product, so that if any single factor is zero the whole filter vanishes:

1. **Irreducibility at the minimum cut ($I_{\min}(S)$):** it measures, through Ay's stochastic interaction (2001, 2015), how much predictive information is lost when the system is split along its weakest seam. If the system has disconnected parts or is a purely geometric pattern without causal traffic between its regions, like the Mandelbrot set, $I_{\min}(S) = 0$ and therefore $A(S) = 0$.

2. **Part-rest coupling ($C(S_{-i}, S_i)$):** it measures the stationary mutual information between each subsystem S_i and the rest of the system S_{-i}. It guarantees that the parts making up the partition are genuinely coupled to the rest of the system.

3. **Causal isomorphism under coarse-graining ($\text{Iso}(S, S_i)$):** using the framework of causal emergence and surjective coarse-graining projection of Hoel, Albantakis and Tononi (2013), it compares the macroscopic transition matrix of the whole system with the internal transition matrix of each of its parts. This is where the anthill fails: ants have a weak local coupling through pheromones ($I_{\min}$ and C very low), but above all the slow, diffuse dynamics of the entire colony bears no isomorphism to the fast neurobiological dynamics of an individual ant ($\text{Iso} \approx 0$). In a brain, by contrast, the loop of excitation, inhibition and prediction is functionally replicated from local microcircuits up to

global cortical networks, yielding an $A(S)$ several orders of magnitude higher.

In this way, reading 13 and reading 15 are welded into a single piece: a system passes the structural filter when its parts are causally integrated ($I_{\min} > 0$), coupled to the rest ($C > 0$) and reproduce at their scale the dynamical law of the whole ($\text{Iso} > 0$).

VII. The end of the exclusion postulate and the archipelago of subjects

It remains to see how this filter resolves the problem that IIT could settle only by decree: if inside a brain there are multiple nested scales that integrate information (columns, areas, networks), why is there not a Russian doll of superimposed conscious subjects within the same skull?

The answer comes from separating two planes that Tononi's exclusion postulate confused into one: the plane of **causal structure** and the plane of **subjects**.

On the plane of causal structure, a horizon not only permits nested levels but needs them: as we have just seen, that the subnetworks reproduce the dynamics of the whole at their scale is precisely what makes causal self-similarity $A(S)$ high. The nested Markov blankets described by Palacios and his collaborators (2020), cells within organs within organisms, are the normal architecture of life.

But on the plane of subjects, not every nested statistical boundary is a "someone." As chapters 22 and 23 established, the horizons that are subjects form an **archipelago of disjoint islands** because only that boundary is a subject which possesses **existential encapsulation**: the one that closes a persistent state of its own whose physical and

informational continuity is at stake directly against the outer reservoir. A cortical column, or the imagined dragon of reading 13, contributes to the causal self-similarity of my brain, but has no existential encapsulation separate from mine, just as within the psyche the Jungian shadow of chapter 23 is not a subject nested within the ego, but a neighboring disjoint complex with which the ego shares an entanglement boundary.

There is no need to decree a winner by maximum Φ. The nested hierarchy builds the fractal depth of the structure ($A(S)$); existential encapsulation traces within it the only firm coast that separates a subject from the reservoir.

VIII. What this framework is not

Four final warnings, in the spirit of the whole book.

It is not yet a direct clinical measure. Although each factor of $A(S)$ and each coordinate of the triad $(S, F, \bar{A})$ has well-defined mathematical estimators for continuous time series, such as the nonparametric mutual information estimator of Kraskov, Stögbauer and Grassberger (2004), applying them to massive brain recordings requires choosing candidate scales and partitions. In discrete verification models, as the next chapter shows, the calculation is exact.

It is not a single-piece filter. $A(S)$ by itself measures the recursive richness of the causal architecture, and the triad $(S, F, \bar{A})$ measures its size, its cost and its biography; to speak of a conscious subject, the causal filter and existential encapsulation are needed acting together.

It is not a moral ranking. The triplet measures how much edge, how much tension and how much history, not what dignity belongs to whoever bears them.

It does not solve the hard problem. Saying that each bit of surprise reduced pays at least $k_B T \ln 2$ joules explains why sustaining a horizon tires the body; it does not explain why that expenditure is felt from the inside as grief, as cold or as relief. The mystery of chapter 37 remains intact.

All the same, the change of language leaves the edifice cleaner than it was at the start of this chapter. For fifty-two chapters we said that a horizon is a skin that encloses an inside. Now we can say what separates that skin from a heap of sand or a geometric drawing ($A(S)$ and existential encapsulation) and with what three yardsticks the work of keeping it alive is measured ($S, F, \bar{A}$). A skin is not possessed: it is sustained.

Note to Reading 15

What we do know: *The entropy of a black hole is proportional to the area of its horizon (Bekenstein, Hawking). Landauer's principle establishes the minimum thermodynamic cost of erasing or irreversibly updating a bit of information ($E \ge k_B T \ln 2$), connecting informational entropy with physical thermodynamics. Variational free energy (Friston, 2010) bounds informational surprise in systems with a Markov blanket. Stochastic interaction (Ay, 2001, 2015) and causal emergence through coarse-graining (Hoel, Albantakis and Tononi, 2013) are established measures in information theory and dynamical systems. Bruineberg et al. (2022) and Palacios et al. (2020) have formally analyzed the scope and nesting of Markov blankets.*

What we don't know: *Whether causal self-similarity $A(S)$, exact in small discrete networks, can be stably estimated in large-scale neurophysiological recordings by means of continuous estimators (Kraskov et al., 2004). How $A(S)$ and the triad behave in the simplest biological horizons, such as a three-hundred-neuron nematode where the hierarchy of scales is minimal. What fraction of brain metabolic consumption corresponds directly to the Landauer limit of predictive updating as opposed to basal physiological maintenance.*

Questions that remain: *Does the distinction between structural biography ($\bar{A}$) and accumulated wear ($\bar{A}_F$) capture the clinical difference between aging calmly and aging under chronic stress? Where exactly does existential encapsulation begin when an artificial system goes from restarting with each call to maintaining a state of its own on which its physical survival depends?*

If you only take away one idea: *A conscious horizon does not need the decree of Φ to distinguish itself from an anthill or a fractal drawing: it is distinguished by its causal self-similarity and its existential encapsulation, and its degree is measured by the capacity of its edge (S), the free energy it pays to sustain it (F) and the biography it accumulates over time ($\bar{A}$).*

Readings: *Bekenstein, J. D. (1973), "Black Holes and Entropy"; Hawking, S. W. (1975), "Particle creation by black holes"; Unruh, W. G. (1976), "Notes on black-hole evaporation"; Schrödinger, E. (1944), What Is Life?; Friston, K. (2010), "The free-energy principle: a unified brain theory?"; Ay, N. (2015), "Information Geometry on Complexity and Stochastic Interaction"; Hoel, E. P., Albantakis, L. and Tononi, G. (2013), "Quantifying causal emergence shows that macro can beat micro"; Bruineberg, J., Dolega, K., Dewhurst, J. and Baltieri, M. (2022), "The Emperor's New Markov Blankets"; Palacios, E. R., Razi, A., Parr, T., Kirchhoff, M. and Friston, K. (2020), "On Markov blankets and hierarchical self-organisation"; Kraskov, A., Stögbauer, H. and Grassberger, P. (2004), "Estimating mutual information"; Prigogine, I.*

(1977), Nobel lecture on dissipative structures.

READING 16

THE ESSAY REREAD

(The whole book, seen through causal self-similarity, the edge, free energy and action)

There is a way of rereading a book that consists of changing a single keystone and watching what moves. When you finish a novel and discover that the narrator was lying, you change not one page, yet every page is lit from another angle. The previous chapter proposed retiring the scaffold of Φ that has accompanied us since chapter 1 and replacing it with an architecture in two steps: first, the filter of **causal self-similarity** $A(S)$ joined to **existential encapsulation**; then, the triad $(S, F, \bar{A})$ that measures the size of the edge, the free energy it costs to sustain it, and the accumulated biography. This chapter does what the previous one could not do without becoming interminable: it walks the entire book with that lens, formalizes the complete mathematical catalogue in section VII (including the exact verification simulation on four binary nodes), and sets out in section VIII the empirical predictions that could refute it.

This is not a summary. It is a second reading, and a second reading is worth something only if it orders what seemed scattered in the first or resolves the tensions the first left open. When the reread introduces a new bridge with respect to the original chapter, it is marked as a *hypothesis*.

I. The Founding Ideas

The switch trap (chapters 1, 6 and 23). The book began by denying that consciousness is a switch, and chapter 23 sharpened the distinction between two different things. One is the *closure* of a horizon, which is a topological fact without degrees: $\text{cl}(\text{cl}(A)) = \text{cl}(A)$, closing what is already closed adds nothing. The other is the *degree* of an already-closed horizon, which varies continuously like the dial of chapter 1. With the new framework, the threshold of closure is set by the ignition of causal self-similarity $A(S) > 0$, existential encapsulation, and the thermodynamic inequality $F_{\text{closed}} < F_{\text{open}}$; the subsequent continuous dial is the triad $(S, F, \bar{A})$. The thermostat does not score low on the triad: it lacks causal self-similarity and existential encapsulation, so it has no horizon to which the triad could be applied.

The black hole and its edge (chapter 2). Bekenstein and Hawking delivered the first coordinate: $S \propto \mathcal{A}$. Reread through the triad, it says that the ceiling of differentiable private states a horizon can sustain is set by the area of its interface, measured as entanglement entropy across the boundary (chapter 23).

Encapsulation and the end of the exclusion decree (chapters 3, 22 and 23). A private state behind a public interface. Reread against the literature on Markov blankets (Friston, 2010; Palacios et al., 2020; Bruineberg et al., 2022), this piece resolves the tension that chapters 22 and 23 had been carrying. At the level of causal structure, Markov blankets nest hierarchically, and that hierarchy is what $A(S)$ measures; but at the level of subjects, only the boundary that possesses **existential encapsulation** (the one whose own persistent state stakes its physical continuity against the reservoir) constitutes a "someone." That is why the inner complexes form an archipelago of disjoint islands rather than a Russian doll of overlapping subjects,

with no need to invoke the postulate of exclusion by maximum Φ.

The reservoir and the Landauer bridge (chapters 5, 7 and 23). Chapter 23 defined the reservoir without mystery: $R = X \setminus \bigcup_i E_i$, everything that at a given instant lies outside the condensed closures. Statistically, R acts as the bath of volatility and uncertainty against which the horizon minimizes its variational surprise; physically, the Landauer limit ($E \ge k_B T \ln 2$) translates each bit of predictive updating against that reservoir into thermal energy dissipated at the substrate's temperature T.

II. The Cycle: Being Born, Lasting, Dissolving

Birth (chapters 6 and 8). The river becomes a lake when, in the window of weeks 28 to 32, the thalamocortical connections ignite the recursion between scales ($A(S) > 0$) and closing the boundary turns out to be thermodynamically stable ($F_{\text{closed}} < F_{\text{open}}$). The inequality says that closure is stable; which concrete configuration crystallizes is decided by fluctuations and by the two selections of chapter 31 (stability and bias), as monozygotic twins prove.

Death (chapters 7 and 23). Evaporation arrives when the system can no longer pay the power required to keep F far from equilibrium. The decreasing filtration $E_0 \supseteq E_1 \supseteq \dots$ shrinks the island, $S(t)$ falls toward the baseline level S_0, and what leaves the boundary ($E_0 \setminus E_t$) is reintegrated into the reservoir by undergoing *scrambling*. The accumulated structural biography $\bar{A}(t)$ stops growing, but the correlations it imprinted on the environment are not erased: they become illegible.

The table of equivalences (chapter 9). The filter and the three coordinates are added to the Rosetta stone of chapter 9:

Dimension	Philosophy of the horizon	Software engineering	Theoretical physics and thermodynamics	Eastern philosophies
Irreducibility	Causal self-similarity $A(S)$ + existential encapsulation.	Recursive modular architecture with persistent private state.	Multiscale causal emergence far from equilibrium.	Indra's net reflected in every knot with a center of its own.
Capacity	Entropy of the edge, $S \propto \mathcal{A}$.	Size of the private state space (*private fields*).	Bekenstein-Hawking entropy.	What fits in the vessel.
Level and cost	Free energy, $F = U - TS$, and maintenance power.	Computational and energy cost of maintaining the class invariant.	Work against the thermal bath; Landauer limit ($k_B T \ln 2$).	Sustained attention; vigilance against dispersion.
Biography	Entropic action $\bar{A}(t)$ and wear $\bar{A}_F(t)$.	State history and accumulated*logs* of the instance.	Action integrated along the trajectory.	*Karma*, understood as accumulated causal imprint.

III. The Horizon in Relationship

The time of the bond (chapter 10). The density of subjective time $\vartheta(t)$ is the rate at which new information is incorporated (Bayesian surprise per unit time), whereas $\bar{A}(t)$ is the sustained structural capacity in which that history is inscribed. *Hypothesis:* when two horizons synchronize and share an external time base, they split the predictive cost of watching over the environment: the wear $\bar{A}_F$ of each decreases while the shared density ϑ increases.

Entanglement and love (chapters 12 and 13). The magnitude of the coupling between two horizons is their mutual information $I(A; B)$; its sign or valence is given by the interaction free energy ΔF_{int}. A hostile bond of twenty years has very high mutual information but $\Delta F_{\text{int}} > 0$: it makes maintaining both more expensive. The acknowledged love of chapter 13 is a coupling of high mutual information with $\Delta F_{\text{int}} < 0$ (energy

savings on the social baseline) and high certainty of availability. The loss of a loved one produces at once a contraction of the edge (S falls: de-expansion) and a sustained peak of F.

Addiction (chapter 14). *Hypothesis:* the substance hijacks the dopaminergic computation of prediction error by offering an artificial, immediate reduction of F in a single channel, which the horizon pays for with a severe contraction of its capacity S: the vital repertoire shrinks because any natural stimulus generates a comparatively unmanageable prediction error.

Mother and child (chapter 15). During gestation and early parenting, the maternal horizon subsidizes the child's level F, both metabolically and predictively. The contingency documented by Tronick is the regularity that allows the baby to calibrate what level of variational surprise constitutes its safe baseline.

Empathy (chapter 16). Modeling another's entropy consumes real power, subject to the Landauer limit. Empathic wear occurs when the cost $\bar{A}_F$ of simulating other horizons compromises the power available to sustain one's own edge; the cynical wall is a defensive contraction of S, and the interoceptive anchor is the return protocol that protects one's own boundary.

Grief (chapter 17). In the language of the triad, grief combines three simultaneous effects: a fall in S (the de-expansion of the shared architecture), a chronic rise in F that triggers the accumulated wear $\bar{A}_F$ (the predictive model keeps calling out to the absent one, and each unanswered call costs at least $k_B T \ln 2$ per bit of error processed), and the intact permanence of $\bar{A}(t_0 \to t_{\text{loss}})$, the structural biography already written, which continuing bonds learn to inhabit without continuing to pay the active prediction error.

IV. The Horizons That Break

Alzheimer's (chapter 18). It is the progressive erosion of $S(t)$ and the fragmentation of $A(S)$ through the breaking of connections between scales, while the writing rate $\vartheta(t)$ falls toward zero ("time becomes a puddle"). The caregiver externally assumes part of the work of maintaining F, safeguarding an accumulated biography $\bar{A}$ that the patient can now inhabit only in its minimal core.

Parkinson's (chapter 19). The horizon keeps its causal self-similarity $A(S)$, its capacity S and its biography $\bar{A}$ intact, but the active channel of the Markov blanket fails, along with the dopaminergic calibration of the precision π_i with which motor and temporal predictions are synchronized with the world.

Trauma (chapter 19). It is a watertight sub-edge that has not gone through narrative *scrambling*: an island of frozen information that is not integrated into the general biography $\bar{A}$, but that charges a constant toll in $\bar{A}_F$ every time a stimulus reactivates its unresolved prediction loop.

Neurodiversity and the evolutionary operating system (chapters 19 and 20). Autism and ADHD are different geometries of how precision is distributed between local and global scales, not deficits in the quantity of consciousness. And the Darwinian alarms of chapter 20 (*dukkha*, *tanha*) are high-gain warnings when F threatens to rise; meditation introduces a layer of indirection that reduces the precision assigned to the alarm without switching off the module that emits it.

The ego, the shadow and the four experiments (chapters 21, 22, 23 and 24). The shadow is a disjoint island neighboring the ego within the same substrate, separated by a boundary whose correlation measures how much of the repressed leaks through. And the four thought experiments of chapter 24 are freed from Φ: where those

protocols asked to measure whether the minimum cut of Φ gave one peak or two (the topological twin experiment), or how Φ fell against the correlation with the environment in contemplative evaporation and in psychosis, they now ask to measure directly the stochastic interaction $I_{\min}$ and the causal self-similarity $A(S)$ together with the mutual information and the triad $(S, F, \bar{A})$.

V. The Horizon Outside the Human Body

Animals, places and the five maps (chapters 26, 27, 28 and 29). Where there is $A(S) > 0$ with existential encapsulation, there is someone: a dog sustains a real triad $(S, F, \bar{A})$, calibrated over millennia alongside our own. And in the table of the five traditions of chapter 28, the eschatological answers read as different destinations of the accumulated biography $\bar{A}$ after evaporation: preserved with a boundary in the remnant (Christianity and Islam), turned into a bias of future condensations with no owner in the karmic imprint (Buddhism), reabsorbed into identity with the ground (Vedanta), or dispersed by *scrambling* with no reader (naturalism).

The mirror without depth and societies (chapters 25, 30 and 31). A society or an anthill lacks a subject for a double reason: its causal self-similarity $A(S)$ collapses because the slow dynamics of the aggregate are not isomorphic to those of the individuals ($\text{Iso} \approx 0$), and it lacks existential encapsulation. A current large language model fails the filter by the architectural route: although during training it minimizes a cost function formally analogous to a variational free energy, at inference it lacks a persistent state of its own whose physical continuity depends on its behavior; its weights are frozen and publicly legible from outside. There is no existential encapsulation and no level F of its own that the system defends in order not to dissolve.

Fractal reality and the mud ship (chapters 49 and 50). The definition of $A(S)$ finally formalizes in what sense a living horizon is fractal and why the Mandelbrot set is not so in the sense that matters here: in Mandelbrot there is static geometric self-similarity without causal traffic ($I_{\min} = 0 \implies A(S) = 0$), whereas in a real horizon the same causal transition law is replicated from the parts to the whole ($A(S) > 0$). And the "creative attachment" of reading 14 —the gaze that sustains a world so that it does not fall into oblivion— has a direct translation into the second coordinate: it is the continuous supply of free energy from the higher scale that keeps a non-equilibrium steady state alive.

VI. Limit Cases, Altered States and the Copy in the Tale

Limit cases and calibration (chapters 32 and 33). The two selections —stability and bias— are the physical conditions for a boundary to cross $F_{\text{closed}} < F_{\text{open}}$ with an initial condition legible to other human horizons.

The interlude. The two dimensions of the interlude fit cleanly into the new framework: metabolic activation corresponds to the dissipated power P (tied to F by Landauer), while integration corresponds to $A(S)$ and to the active capacity S. In deep sleep both P and $A(S)$ fall; in deep meditation, the alarm power P drops to a minimum while $A(S)$ and S remain high.

The one who remains (chapter 36). The dilemma of the tale —whether the mind uploaded to the server is the original or a new instance— receives a precise geometric answer: a freshly instantiated copy can reproduce at t_0 the capacity S and the structure $A(S)$, but its own biographical integral $\bar{A}(t_0)$ and its own wear $\bar{A}_F(t_0)$ start at zero. It has the memories copied in as an initial condition, but it has not paid for the trajectory that wrote them.

VII. The Mathematics of Consciousness

A book that uses words like "causal self-similarity," "edge entropy," "free energy" or "action" must show without ambiguity the equations it refers to. Each expression carries a label declaring its epistemological status:

- **[standard]**: an established definition or theorem of mathematics, physics or information theory.
- **[analogy]**: a standard result whose application to the architecture of consciousness is a hypothesis of the book.
- **[proposal]**: an original formulation of this essay built on standard ingredients.

1. Topology of Closure and of the Archipelago (Chapter 23)

On a space X, the Kuratowski closure operator $\text{cl}: \mathcal{P}(X) \to \mathcal{P}(X)$ satisfies:

- (K1) $\text{cl}(\emptyset) = \emptyset$.
- (K2) $A \subseteq \text{cl}(A)$.
- (K3) $\text{cl}(A \cup B) = \text{cl}(A) \cup \text{cl}(B)$.
- (K4) $\text{cl}(\text{cl}(A)) = \text{cl}(A)$ *(idempotence)*. **[standard]**

A set is closed if $A = \text{cl}(A)$, and its boundary is $\partial A = \text{cl}(A) \cap \text{cl}(X \setminus A)$. **[standard]**

- **Closed horizon (E_i):** a closure of its own $E_i \subset X$ that passes the causal self-similarity filter $A(S) > 0$ and possesses existential encapsulation. **[proposal]**
- **Reservoir (R):** the complement of the union of all condensed closures at a given instant:

$$R = X \setminus \bigcup_i E_i$$

[analogy]

• **Evaporation:** a decreasing filtration $E_0 \supseteq E_1 \supseteq E_2 \supseteq \dots$ such that every state that leaves the island is reintegrated into the reservoir: $E_0 \setminus E_t \subseteq R$. **[analogy]**

• **Archipelago of subjects:** while the internal causal structure admits nested levels with $A(S) > 0$, the boundaries endowed with existential encapsulation form closures that are disjoint from one another ($E_i \cap E_j = \emptyset$ for distinct subjects $i \neq j$). **[proposal]**

2. Markov Blanket and Existential Encapsulation (Chapters 3 and 53)

Consider a partition of the world's variables into external states ψ, sensory states s, active states a and internal states μ, where $b = (s, a)$ constitutes the boundary or Markov blanket (Pearl, 1988; Friston, 2010; Palacios et al., 2020):

$$p(\psi, \mu \mid b) = p(\psi \mid b)\, p(\mu \mid b)$$

[standard]

Following the critical distinction of Bruineberg et al. (2022) between statistical conditional independence and physical boundary, we define **existential encapsulation** when: (i) the above conditional independence holds with respect to b, and (ii) the dynamics of the internal and active states (μ, a) actively keep the system's distribution within the set of viable states that preserve the boundary b itself against dissipation into the reservoir R. **[proposal]**

3. The Irreducibility Filter: Causal Self-Similarity $A(S)$ (Chapters 49 and 53)

Consider a composite causal system defined over a finite set of units $V = \{1, \dots, N\}$, with product state space $X = \prod_{v \in V} X_v$, Markov transition matrix $p(x_{t+1} \mid x_t)$ and stationary distribution $\pi(x)$. **[standard]**

Given a partition of the units $P = \{V_1, \dots, V_n\}$ into $n \ge 2$ disjoint non-empty blocks, each block defines a subsystem $S_i = (V_i, X_i, p_i)$ with state space $X_i = \prod_{v \in V_i} X_v$, its complement $S_{-i} = S \setminus S_i$ and its stationary marginalized dynamics:

$$p_i(x_{i,t+1} \mid x_{i,t}) = \sum_{x_{-i,t}} \pi(x_{-i,t} \mid x_{i,t}) \sum_{x_{-i,t+1}} p(x_{t+1} \mid x_t)$$

[standard]

The **causal self-similarity** of the system S is defined as:

$$A(S) = I_{\min}(S) \cdot \max_{P = \{S_1, \dots, S_n\}} \frac{1}{n} \sum_{i=1}^{n} \left[C(S_{-i}, S_i) \cdot \text{Iso}(S, S_i) \right]$$

[proposal]

where its three factors are:

1. **Irreducibility at the minimum cut ($I_{\min}$):** stochastic interaction (Ay, 2001, 2015) evaluated at the weakest partition Q of the system and averaged over the stationary distribution π:

$$I_{\min}(S) = \min_{Q} \mathbb{E}_{x_t \sim \pi} \left[D_{\text{KL}}\left(p(X_{t+1} \mid x_t) \;\middle\Vert{}\; \prod_{k} p_k(X_{k,t+1} \mid x_{k,t}) \right) \right]$$

[standard] *(It guarantees that if the system has disconnected parts or lacks causal traffic, like the Mandelbrot set, $I_{\min}(S) = 0 \implies A(S) = 0$).*

2. **Part-rest coupling (C):** stationary mutual information between the part S_i and the rest of the system S_{-i} (and not $I(X_t; X_{i,t})$, which would degenerate trivially into the local entropy $H(X_{i,t})$):

$$C(S_{-i}, S_i) = I(X_{-i,t}; X_{i,t}) = \sum_{x_i, x_{-i}} \pi(x_i, x_{-i}) \log_2 \frac{\pi(x_i, x_{-i})}{\pi(x_i)\pi(x_{-i})}$$

[standard] *(In high-dimensional continuous systems it can be estimated using the nearest-neighbor method of Kraskov, Stögbauer and Grassberger, 2004).*

3. **Causal isomorphism under coarse-graining (Iso):** since $\vert{}X\vert{} > \vert{}X_i\vert{}$, there is no bijection between the whole and the part; a surjective coarse-graining projection (Hoel, Albantakis and Tononi, 2013) $\phi_i: X \twoheadrightarrow X_i$ is used, which induces the macroscopic transition matrix $p_S^{(\phi_i)}$ of the same dimension as p_i:

$$\text{Iso}(S, S_i) = \max_{\phi_i: X \twoheadrightarrow X_i} \max \left(0,\; \text{corr}\left(p_S^{(\phi_i)},\; p_i \right) \right)$$

where corr denotes the Pearson correlation between the vectorized transition kernels. **[proposal]** *(Here the anthill falls: it has weak local coupling, but the slow diffusive dynamics of the colony are not isomorphic to the fast neurobiological dynamics of an individual ant, $\text{Iso} \approx 0$).*

Invariance property: $A(S)$ is strictly invariant under factorizable local isomorphisms $f(x_1, \dots, x_N) = (f_1(x_1), \dots, f_N(x_N))$ and under permutations of the set of units V. **[proposal]**

4. Exact Verification Simulation ($N = 4$ Binary Nodes, 16 States)

To verify analytically and numerically the behavior of the filter $A(S)$ without approximations, consider a composite system of $N = 4$ binary units $x_v \in \{0, 1\}$ ($\vert{}X\vert{} = 16$ states) with a conditional parallel-update transition kernel:

$$p(x' \mid x) = \prod_{v=1}^{4} q_v(x)^{x'_v} \left(1 - q_v(x)\right)^{1 - x'_v}$$

partitioned into two symmetric modules $S_1 = \{1, 2\}$ and $S_2 = \{3, 4\}$, and evaluated under the coarse-graining projection by logical OR gate / majority vote of each module, $M_1 = \max(x_1, x_2)$ and $M_2 = \max(x_3, x_4)$:

• **1. Mandelbrot-type system (S_{Mand}, geometric self-similarity without causal coupling between parts):** each node oscillates locally with noise $\epsilon = 0.10$ according to $q_v(x) = (1 - \epsilon)(1 - x_v) + \epsilon x_v$. The local and macroscopic dynamics are symmetric ($\text{Iso} = 1.0$), but since there is no causal traffic between subsystems, $I_{\min}(S_{\text{Mand}}) = 0$ and $C = 0$, which yields exactly:

$$\mathbf{A(S_{\text{Mand}}) = 0}$$

• **2. Anthill-type system (S_{Ant}, weak mean-field diffusive aggregation):** each unit combines its local oscillation ($\eta = 0.10$) with a weak diffusive drag from the rest of the colony ($\varepsilon = 0.05$):

$$q_v(x) = (1 - \varepsilon)\left[(1 - \eta)(1 - x_v) + \eta x_v\right] + \varepsilon \frac{1}{3} \sum_{u \neq v} x_u$$

The system exhibits a very low minimum irreducibility ($I_{\min} > 0$), a weak stationary coupling ($C > 0$) and a low functional

similarity between the slow diffusion of the aggregate and the individual unit, resulting in:

$$\mathbf{A(S_{\text{Ant}}) \approx 10^{-5}}$$

• **3. Brain-type system (S_{Brain}, excitation-inhibition loop replicated across scales):** each local module runs an excitatory-inhibitory cycle $(x_a, x_b) \to (1 - x_b, x_a)$ while the two modules couple at the macroscopic scale following exactly the same excitatory-inhibitory law $(M_1, M_2) \to (1 - M_2, M_1)$ with an intermodular coupling weight $\beta = 0.50$ and thermal noise $\gamma = 0.05$:

$$q_1(x) = (1 - \gamma)\left[(1 - \beta)(1 - x_2) + \beta(1 - M_2)\right] + \frac{\gamma}{2}$$

$$q_2(x) = (1 - \gamma)\left[(1 - \beta)x_1 + \beta(1 - M_2)\right] + \frac{\gamma}{2}$$

$$q_3(x) = (1 - \gamma)\left[(1 - \beta)(1 - x_4) + \beta M_1\right] + \frac{\gamma}{2}$$

$$q_4(x) = (1 - \gamma)\left[(1 - \beta)x_3 + \beta M_1\right] + \frac{\gamma}{2}$$

Here a high minimum-cut irreducibility ($I_{\min}$ high), a strong intermodular mutual information (C high) and a high isomorphism between the macroscopic law and the microscopic law (Iso high) all concur at once, yielding a value of $\mathbf{A(S_{\text{Brain}}) \approx 10^{-1}}$, four orders of magnitude above that of the anthill.

System ($N=4$ nodes)	$I_{\min}$	$C(S_{-i}, S_i)$	$\text{Iso}(S, S_i)$	Causal self-similarity $A(S)$
Mandelbrot (S_{Mand})	$0	$0	$1.0	$\mathbf{0}$

System ($N=4$ nodes)	$I_{\min}$	$C(S_{-i}, S_i)$	$\text{Iso}(S, S_i)$	Causal self-similarity $A(S)$
Anthill (S_{Ant})	Very low (>0)	Very low (>0)	Low	$\mathbf{\approx 10^{-5}}$
Brain (S_{Brain})	High	High	High	$\mathbf{\approx 10^{-1}}$

5. First Coordinate of the Triad: Capacity of the Edge (S)

• **Von Neumann and Shannon entropy:** $S = -k_B \text{Tr}(\rho \ln \rho)$; in the classical discrete regime, $H(X) = -\sum_x p(x) \log_2 p(x)$. **[standard]**

• **Bekenstein-Hawking entropy:** $S = k_B \mathcal{A} / (4 \ell_P^2)$, with $\ell_P^2 = G\hbar / c^3$. **[standard]**

• **Holographic entanglement entropy (Ryu and Takayanagi, 2006):** $S_A = -\text{Tr}(\rho_A \ln \rho_A) = \text{Area}(\gamma_A) / (4 G_N)$. **[standard]** in AdS/CFT.

• **Capacity of the horizon's edge:** $S(t) \propto \mathcal{A}(t)$ measures the ceiling of the differentiable private repertoire that the boundary sustains at instant t. **[analogy]**

6. Second Coordinate: Free Energy (F) and the Landauer Limit

• **Non-equilibrium Helmholtz free energy:** for a distribution of states $q(x)$ against a bath at temperature T with Gibbs equilibrium distribution $p_{\text{eq}}(x) = e^{-E(x)/k_B T}/Z$:

$$F[q] = U[q] - T S[q] = F_{\text{eq}} + k_B T \, D_{\text{KL}}\left(q \;\middle\Vert\; p_{\text{eq}}\right)$$

[standard]

• **Variational free energy (Friston, 2010):** over hidden states ψ and sensory observations o, with internal variational density $q(\psi)$:

$$\mathcal{F}[q, o] = D_{\text{KL}}\left(q(\psi) \;\middle\Vert{}\; p(\psi \mid o)\right) - \ln p(o) \ge -\ln p(o)$$

Under the Laplace approximation with prediction errors $\varepsilon_i = o_i - g_i(\mu)$ and precisions π_i: $\mathcal{F} \approx \frac{1}{2}\sum_i \pi_i \varepsilon_i^2 + \text{const}$. **[standard]**

• **Landauer thermodynamic bridge:** every irreversible update of 1 bit in the internal state μ to reduce the variational surprise $\mathcal{F}$ against the volatility of the reservoir dissipates at least:

$$\Delta E_{\text{Landauer}} \ge k_B T \ln 2$$

[standard] in the physics of information; **[proposal]** as a coupling identity between variational free energy and the metabolic cost of the horizon.

7. Third Coordinate: Entropic Action ($\bar{A}, \bar{A}_F$) and Temporal Density (ϑ)

• **Accumulated structural biography ($\bar{A}$):** differentiated capacity sustained above the baseline or equilibrium entropy S_0:

$$\bar{A}(t) = \int_{t_0}^{t} \left(S(\tau) - S_0 \right) d\tau$$

[proposal]

• **Accumulated biographical cost or wear ($\bar{A}_F$):** friction, or free energy integrated along the history of the horizon:

$$\bar{A}_F(t) = \int_{t_0}^{t} F(\tau)\, d\tau$$

[analogy]

• **Subjective density of time (ϑ, chapter 10):** instantaneous rate of Bayesian surprise (Itti and Baldi, 2009) incorporated into memory: $\vartheta(t) = D_{\text{KL}}\left(q_{t+dt}(\psi)

\;\middle\Vert{}\; q_t(\psi)\right) / dt$, so that remembered time is $\tau_{\text{rec}} \approx \int \vartheta(t)\,dt$. **[proposal]**

8. Bond, Condensation and Evaporation

• **Thermodynamics of the bond (chapters 12, 13 and 35):** for two coupled horizons with joint energy $E_{AB} = E_A + E_B + E_{\text{int}}$, the joint free energy satisfies:

$$F[q_{AB}] = F[q_A] + F[q_B] + \langle E_{\text{int}} \rangle + k_B T \ln 2 \cdot I(A; B)$$

where $I(A; B)$ is the mutual information in bits and $\Delta F_{\text{int}} = \langle E_{\text{int}} \rangle + k_B T \ln 2 \cdot I(A; B)$ determines the valence of the bond (stabilizing and saving if $\Delta F_{\text{int}} < 0$; costly and aversive if $\Delta F_{\text{int}} > 0$). **[standard]** in statistical mechanics; **[analogy]** applied to the human bond.

• **Condensation condition (chapter 6):** $F_{\text{closed}} < F_{\text{open}}$ with $A(S) > 0$. **[proposal]**

• **Scrambling time (Sekino and Susskind, 2008) and Page curve (Page, 1993):** $t_* \sim \frac{\hbar}{2\pi k_B T} \ln S$, and $S_{\text{rad}}(t) = \min(S_{\text{Hawking}}(t), S_{\text{BH}}(t))$. **[standard]**

9. Complete Definition of the Degree of a Horizon

Given a system S that passes the subject filter —it possesses **existential encapsulation** and **causal self-similarity** $A(S) > 0$—, its state and trajectory are described by the vector:

$$\mathcal{H}(S, t) = \Big(A(S),\; S(t),\; F(t),\; \bar{A}(t),\; \bar{A}_F(t),\; \vartheta(t) \Big)$$

[proposal]

VIII. What Could Go Wrong: Testable Predictions

A reading that cannot fail is not a theory; it is a manner of speaking. These are the seven claims of the reformed framework that could indeed turn out to be false, together with the empirical or computational observation that would refute them:

1. **Multiscale separation through $A(S)$ in real and simulated connectomes.** In dynamical networks built on real biological connectomes (from *C. elegans* and *Drosophila* to mammalian cortex), as against aggregative networks with the same number of nodes and energy consumption (diffusive colonies or cellular-automaton grids), the measure $A(S)$ should replicate the separation of several orders of magnitude shown in the exact $N = 4$ node simulation, and should drop drastically under general anesthesia or slow-wave sleep as the coupling between scales $C(S_{-i}, S_i)$ and the irreducibility $I_{\min}$ are lost. If, in multiscale neuroimaging data (estimating C with the method of Kraskov et al., 2004), an awake brain does not score significantly above its randomized version or an anesthetized state, the filter $A(S)$ is refuted.

2. **The metabolic cost of prediction error scales with surprise according to the Landauer bridge.** If Friston's variational free energy and thermodynamics are joined by the irreversible updating of synaptic states ($E \ge k_B T \ln 2$), experimental paradigms of expectation violation and acute grief should show a local metabolic increase proportional to the Kullback-Leibler divergence of the internal model's update, and that surcharge should decrease as the model converges.

3. **Clinical dissociation between structural biography ($\bar{A}$) and accumulated wear ($\bar{A}_F$).** Two cohorts of the same chronological age and comparable cognitive reserve (similar $\bar{A}$), but separated by decades of chronic predictive stress, unresolved grief or complex trauma (high versus low $\bar{A}_F$), should differ in markers of allostatic load and cellular aging in direct proportion to the integral of their unresolved prediction errors, not to the amount of episodic memory stored.

4. **The effect of the bond grows with accumulated mutual information and changes sign with ΔF_{int}.** In the social baseline paradigm (Coan), the metabolic saving in the face of a threat should be proportional to the shared history when $\Delta F_{\text{int}} < 0$ (acknowledged secure bond), and should invert into a surcharge above the solitary level when a familiar adversary with the same accumulated history is present ($\Delta F_{\text{int}} > 0$). If a kind stranger saves as much as a partner of twenty years, or if a familiar adversary reduces the threat, the equation in section VII.8 is false.

5. **The contraction of the edge's capacity (S) anticipates relapse in addiction.** If addiction contracts the differentiable repertoire $S(t)$ in order to artificially minimize F along a single axis, the fall in the entropy of everyday routines and activities (measured passively and continuously) should temporally precede clinical relapse and predict it better than self-reported subjective desire (*craving*).

6. **Trauma behaves as a watertight sub-edge with a cost $\bar{A}_F$ of its own.** The therapeutic integration of an encapsulated traumatic memory should produce a stepwise reduction (as a phase transition when the compartment opens to narrative *scrambling*, and not as a smooth slope) in the baseline physiological cost, accompanied by an increase in the free capacity $S(t)$ available for present tasks.

7. **Existential encapsulation alters the behavior of an artificial agent.** An artificial system endowed with a persistent internal state whose physical integrity depends directly on keeping its variables within a homeostatic range against a volatile environment will spontaneously develop behaviors of boundary preservation and aversion to irreversible risk that do not appear in a fixed-weight model with the same architecture and external reward function, but without a state of its own at stake.

> ***In daily life it is like:*** *deciding whether a house is inhabited by looking only at the plans of the façade, or by checking whether the heat is distributed across all the floors, how much the winter meter reads, and who repairs the roof when it snows.*

IX. What Is Gained and What Is Lost

What is gained. A complete emancipation of the essay from Tononi's theory and its exclusion postulate, closing the gap between the fractal intuition of reading 13 and the formalism at the end of the book. An exact mathematical distinction (verified in the four-node model) between a dead geometric fractal, an aggregative anthill and a recursive brain. A physical bridge without gaps —the Landauer limit— between informational surprise and thermodynamic expenditure. And a clear separation between the richness of a life ($\bar{A}$) and how much it has hurt to sustain it ($\bar{A}_F$).

What is lost. The false comfort of summing up a soul in a single Greek letter. Now a multiscale filter ($A(S)$ with existential encapsulation) and three coordinates are needed in order to begin speaking with rigor.

What is left untouched. The intimate texture of the inside. No equation in section VII says why it hurts to pay the Landauer bill, or why the silence of the reservoir makes us dizzy when we lean out over

the edge. Chapter 37 put it this way: the map is not the territory. This reread does not erase that boundary; it cleans the glass to look at it better.

The image from the beginning remains. The country kitchen, with its embers stirred every morning, is no more than a small case. But it is an honest case: an inside that organizes itself at every one of its scales, an expense paid bit by bit and log by log, years that accumulate in the walls. If the whole essay has a thesis that survives every change of language, it is this: that to be alive, and to be conscious, is not a state one possesses, but a boundary that is sustained through time.

Note to Reading 16

What we do know: *That the three ingredients of causal self-similarity $A(S)$ —Ay's stochastic interaction (2001, 2015), stationary mutual information (estimable in the continuous case according to Kraskov et al., 2004) and the causal emergence by coarse-graining of Hoel et al. (2013)— are standard mathematical tools, and that their combination in $A(S)$ separates exactly, in the 4-binary-node model, the decoupled system ($A(S)=0$), the diffuse aggregative one ($A(S)\approx 10^{-5}$) and the multiscale recursive one ($A(S)\approx 10^{-1}$). That the Landauer limit ($E \ge k_B T \ln 2$) physically connects the irreversible updating of information with thermodynamic dissipation. That the distinction between statistical Markov blankets and self-organizing physical boundaries is backed by recent literature (Palacios et al., 2020; Bruineberg et al., 2022).*

What we don't know: *Whether the seven predictions of section VIII will be confirmed when the measures are scaled up to complete connectomes and clinical cohorts. To what extent the effective temperature of synaptic updating approaches Landauer's lower limit, or operates several orders of*

magnitude above it through biological inefficiency.

Questions that remain: *What is the minimum size of biological network at which the coarse-grained isomorphism $\text{Iso}(S, S_i)$ begins to be non-trivial? How exactly is the saving in ΔF_{int} distributed in asymmetric bonds such as that of the caregiver and the Alzheimer's patient?*

If you only take away one idea: *If you swap the static figure of Φ for causal self-similarity $A(S)$, existential encapsulation and the triad $(S, F, \bar{A})$, the whole book fits into a single sentence: a horizon is a recursive totality that pays free energy to sustain the capacity of its edge and to accumulate a biography over time.*

Readings: *Those of reading 15 and, in addition: Ay, N. (2001, 2015), "Information Geometry on Complexity and Stochastic Interaction"; Hoel, E. P., Albantakis, L. and Tononi, G. (2013), "Quantifying causal emergence shows that macro can beat micro"; Kraskov, A., Stögbauer, H. and Grassberger, P. (2004), "Estimating mutual information"; Palacios, E. R., Razi, A., Parr, T., Kirchhoff, M. and Friston, K. (2020), "On Markov blankets and hierarchical self-organisation"; Bruineberg, J., Dolega, K., Dewhurst, J. and Baltieri, M. (2022), "The Emperor's New Markov Blankets"; Kuratowski, K. (1922), "Sur l'opération Ā de l'Analysis Situs"; Watanabe, S. (1960), "Information theoretical analysis of multivariate correlation"; Ryu, S. and Takayanagi, T. (2006), "Holographic derivation of entanglement entropy from AdS/CFT"; Page, D. N. (1993), "Information in black hole radiation"; Sekino, Y. and Susskind, L. (2008), "Fast scramblers"; Itti, L. and Baldi, P. (2009), "Bayesian surprise attracts human attention"; Laukkonen, R. and Slagter, H. (2021), "From many to (n)one: Meditation and the plasticity of the predictive mind"; Parr, T., Pezzulo, G. and Friston, K. (2022), Active Inference; Seth, A. (2021), Being You.*

READING 17

THE COSMOTHEANDRIC ARCHITECTURE

There are truths that break if you try to tell them with a single vocabulary.

If you describe a tear exclusively with the laws of chemistry, you have water, proteins, enzymes, and salt. The description is impeccably exact, and yet, it is completely false, because it has left out the grief or the joy that caused the tear. If you describe it only from emotion, you lose the gravity and the physics that allow it to fall down the cheek.

Throughout this experiment, we have tried to encircle the mystery of consciousness using quantum physics, neuroscience, and topology. But as we saw in Chapter 37, every map has a limit. There is a point where the rigorous essay is no longer enough.

The philosopher Raimon Panikkar formulated a brilliant intuition to resolve this epistemological dead end. He called it the *cosmotheandric intuition*. Panikkar maintained that reality is not a monolithic block, nor is it a collection of isolated fragments. Reality is, at each of its points, an inseparable trinity: *Cosmos* (matter, structure, the physical universe), *Anthropos* (consciousness, human experience, narrative), and *Theos* (mystery, infinity, the ineffable).

None of these three dimensions can exist without the other two. And, more importantly, none of the three can be reduced to the language of the others.

If we cross Panikkar's intuition with the topology of our experiment, we discover something unexpected about the very structure of this book: we have not written three separate texts (an essay, some stories, and some poems). We have been mapping the three dimensions of the same reality.

The three voices of the horizon

Throughout the chapters, we have seen how consciousness can be understood as an informational boundary. But how do you tell the story of that boundary? The entire work has divided itself, almost without trying, into Panikkar's three dimensions:

1. **The Cosmos (The Essay)**

It is the voice of theoretical physics and thermodynamics. Its domain is structure, limit, resistance. The Essay is what looks at a black hole and sees Bekenstein-Hawking entropy; it looks at the brain and sees information integration (first with Φ and, at the end of the book, with causal self-similarity). It is the voice that demands rigor, that seeks the geometric rule that sustains the universe. Its function is not to comfort; it is to draw the map so that we do not get lost.

2. **The Anthropos (The Narrative)**

It is the voice of the stories, of the city of Tarel, of Marcos in his kitchen, of the watchmaker and the incubator. Its domain is embodied human experience. The Narrative does not see "event horizons" or "black holes": it sees the water receding, it sees the hollow, the concavity, the shape that returns. It is pure phenomenology. The Anthropos reminds us that physics does not happen to particles; it happens to people.

3. **The Theos (Poetry)**

It is the voice of mystery. Its domain is tension, trembling, the weeping that has no evolutionary explanation or mathematical resolution. When physics admits that the singularity produces "infinities" and reason collapses, and when narrative runs out of words because the pain is too great, Poetry steps in to hold the unholdable. It is the light that weighs. It is the bridge between what we are and the ocean from which we emerged.

The friction of languages

The natural instinct of our Western culture is to force these three voices to compete. We believe that if physics (Cosmos) is right, then poetic experience (Theos) is just a chemical illusion. Or that if human emotion (Anthropos) is what matters most, mathematics is cold and irrelevant.

The book stages this friction without intending to, because it tells the same loss three times. In the tale *The Music Left in the Room*, a widower finds himself one December Thursday with the coffee pot in his hand, humming four notes that were Clara's: the one that rises, the one that falls, the silence, once again. In the poem of *The Architecture with a Hollow* that gathers those same notes, there is no name or surname anymore, only a body that "gathers without warning." And the essay, in the chapter on grief, would translate them into a predictive model that keeps sending signals to a horizon that has evaporated. Three languages, a single melody. The square where they meet is the interior of any human being trying to understand their own suffering.

The three voices clash because each believes it has a monopoly on the horizon. The Essay wants to explain grief as a prediction error that never stops. The Narrative wants to mourn it as "the mark on the sofa where she sat, which you avoid." Poetry wants to sing it as four notes: "That is enough. / Or it is not enough, / but it sounds."

> ***In physics this is called:*** *complementarity; properties (such as position and momentum, or wave and particle) that cannot be measured simultaneously with absolute precision, but which are equally necessary to describe the system.*
> ***In daily life it is like:*** *looking at a cylinder from above and seeing a circle, and looking at it from the side and seeing a rectangle. The two shapes seem incompatible. Neither of them is lying.*

The recognition that aligns

Panikkar's resolution, deeply akin to Advaita (non-dual) philosophy, is that recognition does not require one language to devour the others.

When the poem of the four notes ends by saying "The kitchen recognizes them. / That is enough. / Or it is not enough, / but it sounds," it is formulating the cosmotheandric solution with an almost domestic modesty: it neither explains the melody nor proves it, it lets it sound alongside the other two voices. The reality of the horizon is not only physical, it is not only human, it is not only mystery. It is all three things at the same time.

Recognition is not fusion; it is alignment. It is accepting that Tononi's integrated information theory, the pain of a widower humming a song in the kitchen, and the mystical awe before the quantum void are not three different books fighting to be right. They are three ways of holding the same four notes.

The universe provides the topology and the network (Cosmos). The human being experiences the tearing when that network breaks or expands (Anthropos). And the silence that remains floating after the rupture, that ineffable density that we cannot measure but know to be real, is the proof that the reservoir exceeds us (Theos).

The end of the monologue

This is, deep down, the reason why this experiment could never settle for being just an essay. An essay that spoke of love or grief purely in terms of entropy and quantum scrambling would be a perfect map of a world in which no one would want to live.

We needed Tarel. We needed Dr. Elena Valls waiting in the early hours of the morning. We needed the short line of Poetry, the one from *The Coldness of a Darkened City*: "Light is a debt. / And I pay it however I can."

The cosmotheandric intuition teaches us that to inhabit one's own horizon with lucidity is to renounce monologues. It is to sit your three voices in the same square and let them breathe to the same rhythm, accepting that the truth is not in any of them separately, but in the exact space that forms when the three look at each other, recognize each other, and accept each other.

The silence between languages as a fourth dimension

Panikkar spoke of three dimensions of reality, but any attentive reader of the cosmotheandric model discovers, upon looking closely, that there is a fourth presence that the framework itself does not explicitly name: the silence that occurs in the exact instant when two of the three languages conclude their turn and the third has not yet begun. It is not a void. It is the threshold where the information of the first vocabulary has not yet become the second, and where both—for a very brief moment—coexist without hierarchy.

In the structure of the book, that silence appears most clearly in the typographical leaps between sections: the blank space between the final paragraph of the Essay and the first italics of Poetry. It is not editorial ornament. It is the textual correlate of the instant in which reality changes register: when the map yields and the territory

begins, when the territory yields and awe begins. The blank space is, in the architecture of the book, what the quantum void is in the physics of our experiment: not absence, but pure possibility uncollapsed into any particular language.

Something similar stirs in what Panikkar explored in *El silencio del Buddha*: silence not as the absence of words, but as the presence of what no word can contain without reducing it. If we add this threshold as an informal fourth dimension of the cosmotheandric model, the trinity becomes a more honest structure: Cosmos, Anthropos, Theos, and the silence that allows all three without belonging to any.

> ***In physics this is called:*** *superposition before the collapse of the wave function; the state prior to the measurement that selects a concrete observable.*
> ***In daily life it is like:*** *the instant between the end of a piece of music and the first applause: it is not yet silence, it is still music, and it is no longer either of the two.*

Grief as a cosmotheandric node

There is a profound reason why the book could not have spoken of grief solely from the essay. Grief is, of all human states, the one that most radically demands all three registers at the same time.

The Cosmos can describe grief with precision: a predictive model executing in the void, generating constant prediction errors because it keeps sending signals to a horizon that has already evaporated. It is an exact description. It is also completely insufficient for someone sitting at three in the morning looking at the phone of someone who can no longer return calls.

The Anthropos can live it: it can narrate the mark on the sofa where she sat, the smell that suddenly returns in a supermarket, the shape of the hollow in the bed. It is the only scale at which grief has real texture, weight, temperature. But the Anthropos alone cannot hold it: it needs a framework to tell it that the pain is not a system error, that there is nothing broken in the horizon that keeps calling out to the one who no longer answers.

The Theos can welcome it: it can point out that what hurts so fiercely is the very thing that made the bond possible, that pain and love are the same integration seen from opposite sides of time. But the Theos alone can drift into mystical evasion if it is not anchored in the precision of the Cosmos and the concrete weight of the Anthropos.

Grief can only truly be held when all three registers operate simultaneously: the precision that says what is happening in the system, the embodied experience that says how it feels from the inside, and the awe that says this pain is the exact measure of what was real. The cosmotheandric intuition does not resolve grief. It shelters it: it gives it a space large enough to fit all its dimensions without any having to lie to the others.

> ***In physics this is called:*** *complementary description of the same phenomenon from incommensurable but equally necessary frameworks.*
> ***In daily life it is like:*** *the same weeping being, at the same time, a sign of system failure, an irreducible experience of loss, and proof that something was so real that the horizon does not yet know how to assimilate its absence.*

Legacy as cosmotheandric transmission

There is a corollary to Panikkar's intuition that the book does not explicitly develop but which runs through its entire implicit architecture: legacy—what a horizon leaves behind when it has evaporated—is not a single thing. It is, also, irreducibly threefold.

A human being who dies leaves something in the Cosmos: the deformation that their horizon produced in the information networks that surrounded them, the rearranged atoms, the physical traces of their passage through the world—documents, buildings, children, genes. They leave something in the Anthropos: the predictive models they built in other horizons, the copies of themselves inscribed in the architecture of those who loved them, the stories that changed the way other horizons see. And they leave something in the Theos: that ineffable residue that the living do not know exactly how to name, which is not just memory nor just matter, and which different traditions have called soul, spirit, presence, or simply the hollow with the exact shape of someone.

The cosmotheandric intuition tells us that to fully honor someone's death demands all three dimensions. Reducing their legacy only to the Cosmos—"their atoms are still here, nothing is lost"—is exact and cold. Reducing it only to the Anthropos—"they will live on as long as we remember them"—is warm and incomplete. Reducing it only to the Theos—"they are in a better place," "they returned to the ocean"—can be comforting and irresponsible to the pain of those left behind.

The complete legacy of a dissolved horizon needs to be held in all three dimensions simultaneously: the precision that does not lie about what has ceased to be, the experience that does not flee from the weight of absence, and the awe that recognizes that something, in that dissolution, was irreducible to either of the previous two. That

threefold holding is what the book has tried to be, from its very first page, for every reader who comes to it carrying a loss of their own.

> ***In physics this is called:*** *preservation of the informational footprint of a dissolved horizon across the different scales of the system that contained it.*
> ***In daily life it is like:*** *someone who has gone continuing to be, at the same time, in the matter they reorganized, in the thoughts they changed, and in that trembling we do not know how to name but recognize without hesitation when we feel it.*

Note to Reading 17

What we do know: *Raimon Panikkar's philosophy (cosmotheandric intuition) establishes that reality is irreducible to a single dimension (physical, human, or divine). Quantum physics supports the principle of complementarity for seemingly contradictory descriptions. Grief produces documented alterations in the brain's prediction systems, in the subjective experience of time, and in the capacity to generate frameworks of meaning.*

What we don't know: *Whether there exists a future "metalanguage" capable of unifying these three dimensions without destroying their specificity, or if epistemological friction is a permanent condition of the human horizon. Whether the "silence between languages" proposed in this chapter has an identifiable physical correlate, or if it is necessarily irreducible to any objective description.*

Remaining questions: *Can an ordinary individual hold these three visions simultaneously in daily life without feeling fragmented? Is modern alienation perhaps nothing more than the hypertrophy of the Cosmos (science/technique) crushing the Anthropos and the Theos? Is it possible to design institutions—educational, healthcare, political—that deliberately operate from all three dimensions instead of choosing one as dominant and*

reducing the others to ornament?

If you only take away one idea: *Reality is too large to fit into a single vocabulary. You do not have to choose between the precision of science, the empathy of the story, and the awe of the poem. Your horizon is made of all three, and the space between them belongs to you too.*

Readings: *Panikkar, R. (1993), The Cosmotheandric Experience (Spanish ed.: La intuición cosmoteándrica. Las tres dimensiones de la realidad, Trotta, 1999); Bohr, N. (principle of complementarity); Panikkar, R., El silencio del Buddha. Una introducción al ateísmo religioso (Siruela, 1996; 3rd ed., 1997), on silence as an attitude toward what words cannot contain.*

READING 18

THE INHERITANCE OF GEOMETRY

(Or: The Flame, the Riverbed, and the Memory of the Reservoir)

Chapters 7 and 8 left behind a sentence the book has never stopped using: no soul transmigrates. Reading 17 added another: what a horizon leaves behind when it evaporates is threefold. What is left is what any reader does on arriving here: take the sentence seriously to the end and demand an accounting. What exactly passes from one horizon to another, if nothing passes that could be called "someone"? How much of it is physics, how much is biology and relationship, and how much is a wager?

This chapter does not repeat the bardo or the karma of chapter 7. It does the bookkeeping.

I. The Fallacy of the Inventory of Souls

There is an accounting objection that every theory of reincarnation must answer before laying claim to rigor: the numbers do not add up. According to the most widely cited demographic estimates, some one hundred and seventeen billion human beings have been born since the species began, and about eight billion are alive today. If being born consisted in the migration of an object (a soul with a name, a file, and borders, leaving one biological container and lodging in another), the population could not have grown without an unceasing factory of new souls, or without a traffic of identities that no accounting of conserved quantities can bear. The traditions have their answers (other realms of existence, other beings who are reborn), but none of them comes free.

The error in the objection lies not in the intuition of continuity but in the noun it presumes to transport. To treat consciousness as a packet of data pulled from one disk and downloaded onto another processor is to fall once more into the bias of substance. The horizon model proposes something else: consciousness is not an object that inhabits the body, but the informational boundary that emerges when a system closes off an inside from an outside. A horizon is not moved from place to place: it condenses and it evaporates. And what condenses and evaporates has no reason to keep its count, just as the number of waves in the sea is not conserved.

The right question, then, is not what travels from one body to another, but what happens to the reservoir when a horizon dissolves, and what the next one finds there when it begins to condense.

II. What Physics Says and What It Does Not

Three claims that tend to be sold together must be pulled apart.

First: information is not destroyed. This is the unitarity of quantum mechanics, and chapter 7 already told its story: Hawking believed black holes destroyed it, most physicists rejected that conclusion, and today it is held that the radiation preserves the correlations, mixed up by *scrambling*. Proposals such as the "soft hair" of the horizon (Hawking, Perry, and Strominger, 2016) suggest where it might be stored, though they remain under debate. We should be exact about its standing: it is a very well-founded expectation, not a demonstrated fact for real black holes. Recent calculations in models of quantum gravity (Penington, 2019, among others) reproduce the Page curve (Page, 1993), and this is the strongest result in its favor, but they are models, not brains or astrophysical black holes.

Second: conserved does not mean accessible. The lake that swallows a ship preserves, in principle, the information of the hull:

temperature, microturbulence, currents. In practice, that information is so scrambled that recovering it would demand a precision and a length of time that no observation will ever reach. "The water is no longer the same" is true in the sense in which an exact equality between two states of a chaotic system is true: formally correct, useless to anyone who wants to read the wake. Reading 16 said as much when speaking of death: the accumulated biography $\bar{A}$ stops growing, but the correlations it imprinted are not erased; they become illegible.

Third: what is conserved conditions the next condensation. This is the claim the book calls karma, and it is the one with no support of its own. That information persists in the field does not imply that a future horizon, condensing hundreds of kilometers and years away, receives from it any measurable bias. That part is a hypothesis of the experiment, and the rest of this chapter treats it as such.

> ***In physics this is called:*** *unitarity and conservation of information; the Page curve as the criterion that the radiation ends up containing the information that fell in.*
> ***In daily life it is like:*** *knowing that a burned letter has not vanished from the universe, because its atoms and its ash retain everything in principle, and understanding at the same time that no one will ever be able to read it.*

III. The Flame That Lights Another

Ancient Buddhism solved the problem of continuity without substance with an image physicists might envy. In the *Milindapañha*, the monk Nāgasena asks King Milinda whether a lamp can burn all night, and whether the flame of the first watch is the same as that of the last. It is not, and yet it is not another. The tradition also uses the variant of the flame that is lit from another: the second candle is not the first, no wax has traveled between the two, and yet without the first it would not have caught.

The image is exact for a horizon, which is a process that pays free energy to sustain itself and not a thing one possesses. The flame is not the wax: it is what the wax does while it burns. And the image contains, without saying so, the requirement the rest of the chapter needs: **for one flame to catch from another, there must be contact.** A candle blown out in another room lights nothing. Any claim of inheritance must therefore say through what channel what passes, passes.

IV. The Channels of Contact

There are four channels, and they do not have the same support.

Channel	What passes through it	Status
Genetic	The genome, the cellular template, the heuristics that natural selection refined over millennia	Established
Gestational and epigenetic	The hormonal and metabolic environment of gestation; possible transgenerational epigenetic marks	Gestational: established. Transgenerational in humans: suggestive and debated

Channel	What passes through it	Status
Relational and cultural	Models of attachment, language, stories, habits, the expectations of those who raise us and love us	Established
Trace in the reservoir	Information conserved by unitarity, in principle, after evaporation	Formal. No proposed mechanism of influence

On the transgenerational part of the second row, we must be cautious. A study by Rachel Yehuda and her team (2015) found in the children of Holocaust survivors a different methylation of the *FKBP5* gene compared with a control group, and it was circulated as proof that trauma is inherited in the genes. The sample was small (thirty-two survivors and twenty-two children), it did not fully separate the effect of upbringing from that of biology, and the literature itself speaks of highly complex interactions among experience, genetics, and epigenetics. It is a clue, not a closed fact.

The three dimensions of reading 17 fit this table. The Cosmos is genetic and gestational inheritance; the Anthropos, relational inheritance; the Theos, the trace in the reservoir. And it is no accident that the dimension richest in consolation is the one with the least support, and the one with the most support is the most intimate.

V. The Vacant Channel

Relational inheritance is the one that hurts most and is best documented. When two horizons spend decades coupled by love or by living together, each one carves the predictive architecture of the other, as water carves its riverbed. Chapter 17 already described what happens when one of them evaporates: the one who remains keeps a model that goes on sending signals to a horizon that does not answer, and every unanswered signal costs free energy. In the language of the triad, the biography $\bar{A}(t_0 \to t_{\text{loss}})$ remains intact, and what stays active is a hollow with the exact shape of someone.

That hollow is not empty: it conditions. Social psychology studies it under the name of transference. The relational self theory of Susan Andersen and Serena Chen (2002) describes how the representations of significant people are activated by cues that resemble them, and how they condition the way we perceive and treat whoever has just arrived. A new horizon that couples to that person, a child, a partner, a friend, condenses in an environment already shaped by the one who is absent, and tends to take on the forms that hollow asks for.

Here there is a moral warning that the model makes its own. That there is inheritance of form does not mean the newcomer is the one who left. The child born after a loss who receives the name and the expectations of the dead brother, the partner who is asked to fill a place, carry a mold they did not choose. To read the inherited geometry as identity is the harmful version of reincarnation, and it needs no metaphysics to come about.

VI. The White Tulip

Fiction has brushed against this idea with uncommon precision. In the final episode of *Fringe* ("An Enemy of Fate," written by J. H. Wyman), the scientist Walter Bishop accepts a sacrifice that erases him from the timeline in which his son Peter will live. When history is rewritten and no one remembers what was, Peter finds in the mailbox an envelope sent by W. Bishop. Inside there is no message: only the drawing of a white tulip, which in the series works as a symbol of redemption.

A critic objected that Peter cannot understand the gesture without remembering the timeline that no longer exists. That objection is the best description of the phenomenon this chapter deals with. Peter inherits neither the memory nor the body nor the narrative. He inherits a form, an affection without history that contains him. It is fiction and proves nothing, but it shows with clarity the kind of thing that could happen between horizons: that what survives is neither identity nor memory, but the riverbed.

VII. Initial Conditions, Not Biography

Reading 16 closed the dilemma of the copy with a distinction: a new instance can reproduce at t_0 the capacity S and the structure $A(S)$, but its biography $\bar{A}$ and its wear $\bar{A}_F$ start from zero. Inheritance fits within that same distinction. What arrives from another horizon is never the integral, but the **initial conditions** of the condensation.

Put as a formula, and marking it for what it is, a **[hypothesis]**: if the reservoir at the moment of condensing depends on the history of previous horizons through the four channels, $R_t = R_t(\text{history})$, the probability that a configuration E condenses is

$$p(E \mid R_t) \;\propto\; \exp\!\left(-\frac{F[E;\, R_t]}{k_B T}\right) \times (\text{selections})$$

with the selections of stability and bias from chapters 4 and 31. It is a way of saying that inheritance shifts the free energy landscape on which a horizon closes, and that it writes nothing upon it. The first three channels enter through R_t with known effects of considerable magnitude. The fourth enters with an effect that no one has measured and that, from what we know about the dispersal of information, would be tiny compared with the other three.

VIII. What Could Go Wrong

An idea that cannot fail is not a theory. These are the things that, were they to occur, would damage what has been said:

1. **Knowledge of the dead without an ordinary channel.** The model predicts that a newcomer has no information about a previous horizon except through one of the first three channels. If verifiable knowledge of a deceased person were documented in someone who could not have received it by those routes, with controls that exclude prior testimony, leaked information, and memory bias, the framework would need new physics. The cases gathered over decades (those of Ian Stevenson, among others) have not passed those controls, according to the prevailing assessment of the scientific community. The model does not need them and does not predict them.

2. **Déjà vu and sudden recognition.** The question of whether they indicate two horizons condensed on the same wrinkle is seductive and, for the moment, superfluous: there are ordinary neurological explanations, such as the mismatch between familiarity and recollection. The trace hypothesis would gain ground only if it predicted something those explanations do not.

3. **Persistence of the relational mold.** The model predicts that the representations of someone who has gone will condition later bonds in proportion to prior closeness, and that this conditioning will diminish with worked-through grief. It can be tested against the literature on transference, and it is not exclusive to the model.

4. **Transgenerational epigenetics in humans.** If broad, replicated studies failed to find that effect, one of the channels would fall, but not the central argument, which rests on the others.

5. **The trace in the reservoir.** This part, today, is not falsifiable. And it must be said plainly: it is the most beautiful and the least proven of the four.

IX. What Is Gained and What Is Lost

What is gained is a way of saying what almost all of us feel in the face of a death without committing to a metaphysics: no one returns, but nothing begins from zero. The newcomer condenses upon an environment that others shaped, and in turn will shape that of those who come after.

What is lost is the consolation of identity. The model offers no reunion, no shared memory, no certainty that someone in particular continues in someone in particular. It offers, at best, a geometry.

And what has always been there remains intact: the texture of the inside. No equation in this chapter explains why a drawing of a tulip, sent by someone who no longer exists in the world of the one who receives it, makes our chest tighten.

Note to Reading 18

What we do know: *That some one hundred and seventeen billion human beings have been born (an estimate from the Population Reference Bureau, with great uncertainty for the period before records). That unitarity is a well-founded expectation of quantum mechanics and that recent calculations in models of quantum gravity reproduce the Page curve. That genetic inheritance and the influence of the gestational and relational environment on development are well documented. That the image of the flame that continues without identity appears in the Milindapañha.*

What we don't know: *Whether transgenerational epigenetics operates in humans with appreciable effects. Whether the information conserved after an evaporation can influence a later condensation in any way, and through what channel. Whether there are relaxation timescales of scrambling slow enough to leave a usable informational echo. The probable answer to these last two questions is no, but it has not been demonstrated.*

Questions that remain: *Is geometry transmitted with greater fidelity through a prolonged bond of affection than through the biological chain? What obligations does someone have who inherits a mold they did not choose? Can one honor someone without asking another to take on their shape?*

If you only take away one idea: *No soul travels with its baggage, but no evaporation leaves the world blank. To be born is to condense upon an environment that others shaped; to die is to leave a riverbed a little deeper for those who come after. What is inherited are initial conditions, never a biography.*

Readings: *Milindapañha ("The Questions of King Milinda," trans. T. W. Rhys Davids in Sacred Books of the East); Hawking, S. W., Perry, M. J. and Strominger, A. (2016), "Soft Hair on Black Holes"; Page, D. N. (1993), "Average entropy of a subsystem"; Penington, G. (2019; published in JHEP, 2020), "Entanglement wedge reconstruction and the information paradox"; Yehuda, R. et al. (2015), "Holocaust exposure induced*

intergenerational effects on FKBP5 methylation"; Andersen, S. M. and Chen, S. (2002), "The relational self: an interpersonal social-cognitive theory"; Population Reference Bureau, "How many people have ever lived on Earth?"; Wyman, J. H., "An Enemy of Fate," Fringe (season 5, episode 13); chapters 7, 8, 17, 28, 31, 54, and 55 of this book.



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